# TRICERATOPS Statistical Validation Pipeline

Method of Giacalone et al. (2021) and Complete Technical Description of the Analysis Script

Pipeline documentation for triceratops_validation.py

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# Chapter 1

# Introduction

TESS discovers planet candidates at a high rate. A substantial fraction of transit-like signals are astrophysical false positives: eclipsing binaries on the target star, diluted eclipsing binaries on unresolved companions, or eclipses on resolved nearby stars that contribute flux to the photometric aperture. Statistical validation assigns probabilities to these scenarios given the photometry and the local stellar population, allowing candidates to be classified without a full radial-velocity mass measurement.

**TRICERATOPS** (Tool for Rating Interesting Candidate Exoplanets and Reliability Analysis of Transits Originating from Proximate Stars) is a Bayesian framework designed for TESS. Its defining feature is explicit modelling of transits and eclipses on *resolved nearby stars* as well as on the target and on unresolved companions. The primary outputs are the false-positive probability (FPP) and the nearby false-positive probability (NFPP).

The Python script `triceratops_validation.py` is a complete multi-sector analysis pipeline that:

- builds background-corrected light curves from TESScut full-frame image (FFI) cutouts;
- searches for and characterises the transit with Transit Least Squares (TLS);
- performs diagnostic tests (per-transit depths, centroids, difference imaging, optional neighbour light-curve vetting);
- constructs a realistic scalar flux uncertainty for the TRICERATOPS likelihood;
- runs TRICERATOPS with robust numerics (detection of NaN and degenerate runs, automatic retries);
- reports whether the candidate meets the Giacalone et al. (2021) validation criteria.

This document first summarises the scientific method of the TRICERATOPS paper in detail, then describes every major stage of the script, including algorithms, equations, configuration parameters, failure modes, and design choices.

# Chapter 2

# The TRICERATOPS framework (Giacalone et al. 2021)

## 2.1 Bibliographic reference

Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021,
*Vetting of 384 TESS Objects of Interest with TRICERATOPS and Statistical Validation of 12 Planet Candidates*,
The Astronomical Journal, **161**, 24.
DOI: 10.3847/1538-3881/abc6af
arXiv: 2002.00691

The tool was tested on 68 TOIs with known dispositions (confirmed planets or rejected false positives). Decision thresholds in the FPP–NFPP plane were calibrated on that sample and then applied to 384 unclassified TOIs, yielding 12 statistically validated planets (9 of them newly validated), 125 likely planets, and 52 likely nearby false positives.

## 2.2 Bayesian formulation

For each candidate the code evaluates a discrete set of transit-producing scenarios $S_j$. The posterior probability of scenario $j$ given the data $D$ is

$$P(S_j \mid D) \;=\; \frac{P(S_j)\,P(D \mid S_j)}{\sum_k P(S_k)\,P(D \mid S_k)}.$$

Equivalently, up to normalisation,

$$P(S_j \mid D) \;\propto\; P(S_j)\,P(D \mid S_j).$$

- The **prior** $P(S_j)$ encodes the relative occurrence of planets versus eclipsing binaries and the

number of stars capable of producing the observed depth (through the flux contribution of each star to the aperture).

- The **marginal likelihood** $P(D \mid S_j)$ is estimated by Monte Carlo sampling of the relevant physical parameters (planet radius or EB mass ratio, inclination, eccentricity, argument of periastron, etc.) and comparing simulated light curves to the phase-folded photometry under a Gaussian noise model with a single scalar uncertainty $\sigma$.

In practice TRICERATOPS draws of order $N = 10^6$ samples per scenario (configurable). For each draw a model light curve is generated and the Gaussian log-likelihood is evaluated. The marginal likelihood is approximated via a log-mean-exp of the sample likelihoods.

## 2.3 Scenario taxonomy

TRICERATOPS considers scenarios on several classes of host star:

| **Prefix** | **Host** | **Typical scenarios** |
|---|---|---|
| (none) / T | Target star | TP (transiting planet), EB, EBx2P |
| P | Unresolved bound companion | PTP, PEB, PEBx2P |
| D | Unresolved background / foreground | DTP, DEB, DEBx2P |
| B | Background population (TRILEGAL / Gaia) | BTP, BEB, BEBx2P |
| N | Resolved nearby star (TIC / Gaia) | NTP, NEB, NEBx2P (per star) |
| S | Secondary in a hierarchical triple | STP, SEB, SEBx2P |

**x2P** scenarios place the eclipse at twice the reported orbital period. A secondary eclipse of an EB, when the light curve is folded on half the true period, can mimic a primary transit at period $P$.

**TP / PTP / DTP** are the planet scenarios that contribute to the *planet posterior mass*. All other scenarios are false-positive scenarios.

## 2.4 Summary statistics: FPP and NFPP

After normalising the scenario posteriors,

$$\mathrm{FPP} \;=\; 1 - (P_{\mathrm{TP}} + P_{\mathrm{PTP}} + P_{\mathrm{DTP}}).$$

This is the total posterior mass on all non-planet scenarios that place the signal on the target or on an *unresolved* companion.

$$\mathrm{NFPP} \;=\; \sum (P_{\mathrm{NTP}} + P_{\mathrm{NEB}} + P_{\mathrm{NEBx2P}}),$$

where the sum runs over all resolved nearby stars that were assigned a non-zero required depth. NFPP isolates the posterior mass on scenarios that place the eclipse on a *resolved* neighbour. A low

FPP is not sufficient for validation if a bright neighbour can still produce the observed signal.

## 2.5 Decision criteria

Calibrated in Giacalone et al. (2021):

| **Classification** | **Criteria** |
|---|---|
| Validated planet | FPP $< 0.015$ and NFPP $< 10^{-3}$ |
| Likely planet | FPP $< 0.5$ and NFPP $< 10^{-3}$ |
| Likely nearby false positive | NFPP $> 10^{-1}$ |

The original paper emphasised short-cadence (2-minute) data for formal validation. The present script applies the same numerical thresholds to FFI and mixed-cadence light curves, while making the noise model explicit and conservative.

## 2.6 Field-star population and apertures

- **Resolved nearby stars** come from the TESS Input Catalog (TIC) within a configurable search radius (default 10 TESS pixels, $\approx 3.5$ arcmin).
- **Unresolved field stars** for the D/B scenarios are drawn from a TRILEGAL Galactic simulation, or (in recent package versions and in this script's fallback) from a Gaia DR3 cone search reformatted to a TRILEGAL-compatible table.
- For each TESS sector the user (or pipeline) supplies an aperture: a list of full-FFI pixel coordinates [column, row]. TRICERATOPS computes the flux contribution of every star to that aperture with a circular Gaussian point-spread function of width $\sigma \approx 0.75$ pixels and derives the *required* transit or eclipse depth on that star (`tdepth`). If even a total eclipse of the star cannot reproduce the observed aperture dip, `tdepth` is set to zero and the star drops out of the NFPP calculation.

## 2.7 Contrast curves

Optional high-resolution imaging contrast curves can be passed to `calc_probs` to down-weight unresolved companions inside the achieved contrast limits, reducing the prior weight of PTP/PEB/DTP/DEB-type scenarios.

## 2.8 Likelihood and the role of $\sigma$

The Gaussian likelihood for a phase-folded, binned light curve with $N$ points is proportional to

$$\exp\left(-\frac{1}{2\sigma^2}\sum_{i=1}^{N}(f_i - m_i)^2\right),$$

where $m_i$ is the model flux for a given parameter draw and $\sigma$ is a *single scalar* supplied by the user. If $\sigma$ is far too small (pipeline white-noise errors only), every scenario receives an astronomically poor likelihood, the evidences underflow, and relative probabilities become numerically undefined (NaN). Section~13 describes how this script builds a realistic $\sigma$.

# Chapter 3

# Overview of the validation script

The script implements the following high-level pipeline:

```
TIC ID
  |
  +-- Query TIC stellar parameters and limb darkening
  +-- Find sectors with TESScut coverage
  |
  +-- Per sector: download cutout, aperture, background-corrected LC
  |
  +-- Pass 1 detrend (wide biweight) -> TLS search (or known ephemeris)
  +-- Pass 2 detrend (transits masked) -> TLS again -> refine P, T0, T14, depth
  |
  +-- Per-transit depths (odd/even, chi^2 constancy)
  +-- In-transit centroid offsets
  +-- Fold and bin light curve
  +-- Optional limb-darkened fit (reference model for sigma inflation)
  +-- Realistic sigma for calc_probs
  |
  +-- TRICERATOPS target + calc_depths
  +-- Optional difference imaging -> exclude stars
  +-- Optional neighbour LC vetting -> exclude stars
  +-- remove_star + re-calc_depths
  +-- calc_probs x N_iter (with NaN / degenerate retries)
  |
  +-- FPP / NFPP mean +/- std, Giacalone verdict, plots and tables
```

**Major dependencies:** `lightkurve`, `wotan`, `transitleastsquares`, `triceratops`, `astroquery`, `astropy`, `scipy`, `numpy`, `pandas`, `matplotlib`.

**Entry point:** `main(cfg)` after `argparse` constructs a `Config` instance.

# Chapter 4

# Configuration

All tunable parameters live in the `Config` dataclass. Defaults are chosen for a compromise between sensitivity to short-period planets and stability of the biweight filter.

## 4.1 Cutout and aperture

| Parameter | Default | Meaning |
|---|---|---|
| `cutout_size` | 10 | TESScut cutout size (pixels) |
| `percentile` | 85 | Aperture = pixels above this percentile of the median image |
| `quality_bitmask` | `"default"` | lightkurve quality mask |
| `download_dir` | `outdir/tesscut_cache` | Cache for FITS cutouts |

Lightkurve stores cutouts under `\texttt{<download\_dir>/tesscut/}` and reuses any file already present for the same TIC / sector / size, so subsequent runs skip the MAST download.

## 4.2 Detrending

| Parameter | Default | Meaning |
|---|---|---|
| `window_length_search` | 5.0 d | First-pass biweight window (blind search) |
| `window_factor` | 3.0 | Second-pass window = `factor` $\times T_{14}$ |
| `window_fallback` | 0.5 d | Lower bound on any window |
| `detrend_method` | `"biweight"` | wotan method |
| `outlier_sigma` | 5.0 | Clip only *positive* outliers (robust MAD units) |

**Rule of thumb for the search window.** Long-period candidates ($P \gtrsim 10$ d): 1–2 d is often enough. Short-period / hot Jupiters ($P \lesssim 3$ d): 5–10 d is safer. A window comparable to $P$ can

remove a periodic transit.

## 4.3 TLS search

| Parameter | Default | Meaning |
|---|---|---|
| `P_min`, `P_max` | 0.5, 1.5 d | Period search range (clipped to baseline/2) |
| `n_transits_min` | 2 | Minimum number of transits |
| `SDE_threshold` | 7.0 | Detection threshold |
| `tls_search_bin_minutes` | 10 | Binning for the *search only* |
| `tls_oversampling_factor` | 2 | TLS oversampling (default in TLS is 3) |

## 4.4 TRICERATOPS numerics

| Parameter | Default | Meaning |
|---|---|---|
| `n_iter` | 20 | Independent `calc_probs` runs |
| `bin_minutes` | 10 | Binning of the folded LC for TRICERATOPS |
| `fold_n_dur` | 5.0 | Phase window in units of $T_{14}$ |
| `err_floor_ppm` | 0 | Hard floor on flux uncertainty |
| `tri_inflate_to_model` | True | Inflate $\sigma$ to $\chi^2_{\mathrm{red}} = 1$ of a smooth model |
| `tri_max_inflate` | 50 | Cap on inflation factor |
| `tri_max_retries` | 3 | Retries with larger $\sigma$ if a run is invalid |
| `tri_retry_factor` | 2.0 | Multiplier per retry |
| `tri_min_valid_frac` | 0.8 | Minimum fraction of valid runs for a verdict |
| `dilution_corrected` | None | `False` if LC is uncorrected aperture flux |

## 4.5 Neighbour vetting and difference imaging

| Parameter | Default | Meaning |
|---|---|---|
| `vet_neighbours` | False | Enable neighbour light-curve tests |
| `neighbour_min_sep` | 42 arcsec | $\approx 2$ TESS pixels; closer is not separable |
| `neighbour_max_sep` | 200 arcsec | Outer radius for testing |
| `neighbour_ap_radius` | 1.0 px | Circular aperture on the neighbour |
| `neighbour_sigma` | 3.0 | Exclusion threshold |
| `psf_sigma` | 0.75 px | Gaussian PSF (matches TRICERATOPS) |
| `difference_imaging` | True | Enable difference-image centroids |
| `di_sigma` | 3.0 | Neighbour exclusion threshold in difference images |
| `di_min_snr` | 3.0 | Minimum SNR of the difference image |
| `di_mc` | 300 | Monte Carlo draws for centroid error |

## 4.6 Known ephemeris (optional)

When `use_known_ephemeris` is true, the TLS search is skipped. The pipeline uses `P_orb_known`, `T0_known` (BTJD), `depth_known` (fractional), and `dur_known` (days), and masks transits with that ephemeris in the second detrend pass.

# Chapter 5

# Light-curve extraction from TESScut FFIs

## 5.1 Sector discovery

`find_tess_sectors(tic)` calls `lightkurve.search_tesscut("TIC {tic}")` and returns the sorted unique values of `sequence_number` (sector numbers) with FFI coverage. An empty result raises a runtime error.

## 5.2 Stellar parameters

`stellar_params(tic)` queries the TIC via `astroquery.mast.Catalogs` by column name (not positional index). Returned quantities include effective temperature, surface gravity, metallicity, radius and mass with uncertainties, TESS magnitude, sky coordinates, and quadratic limb-darkening coefficients from the TLS `catalog_info` helper (Claret tables). Missing values fall back to safe defaults (e.g. $u = [0.4, 0.3]$).

## 5.3 Per-sector extraction (`extract_sector`)

For each sector the following steps are performed.

1. **Download or reuse** a `cutout_size` × `cutout_size` TESScut cutout into the cache directory.
2. **Aperture construction.** Compute the median image over time. Threshold at the configured percentile of the pixel values. The boolean mask defines the target aperture. An empty mask raises an error.
3. **Pixel coordinates.** Full-FFI column and row of every aperture pixel, stored as an $(N_{\mathrm{pix}}, 2)$ integer array. These coordinates are required by TRICERATOPS `all_ap_pixels` and must be in the same order as `target.sectors`.
4. **Target photometry.** Aperture sum of the target mask.

5. **Background estimate.** A complementary mask of the faintest pixels (threshold mask with a very low threshold) is used to estimate the background per pixel, scaled to the number of target pixels, and subtracted. Errors are combined in quadrature. If fewer than four background pixels exist, background correction is skipped.
6. **Centroids.** `estimate_centroids` on the same aperture, kept for later in-transit offset tests.
7. **Quality cut.** Require finite time, flux, and error; positive flux; finite centroids. Arrays are made contiguous in memory.
8. **Cube and WCS.** The full flux cube and world-coordinate system are retained for difference imaging.

Diagnostic PDFs (aperture overlaid on the target pixel file, background-corrected light curve) are written per sector.

## 5.4 Data structure

Each sector is stored in a `SectorData` dataclass:

- `sector`, `time`, `flux`, `flux_err`
- `ap_pixels` (full-FFI coordinates)
- `cen_x`, `cen_y` (absolute centroids)
- `norm_flux`, `norm_err`, `trend` (filled after detrending)
- `cube`, `cube_time`, `wcs`, `col0`, `row0` (for difference imaging)

# Chapter 6

# Detrending and outlier rejection

## 6.1 Motivation for two passes

A single detrend with a short window can partially fit into the transit and suppress the depth. A single detrend with a very long window leaves residual stellar activity that inflates the folded scatter. The script therefore uses two passes.

## 6.2 Pass 1 (blind)

A fixed long biweight window (`window_length_search`, default 5 days) removes fast stellar and instrumental variability for the initial TLS search without assuming a transit duration. The window must be long compared with $T_{14}$ so the filter cannot fit into the dip, but not comparable to the orbital period (a window $\approx P$ can remove a periodic transit).

## 6.3 Pass 2 (transit-aware)

After a preliminary ephemeris is known, each sector is re-detrended with the transits *masked*. In wotan, points flagged in the mask are excluded from the trend estimate but are still divided by the trend. The transit is therefore preserved rather than absorbed.

The window length is

$$w_2 = \max\big(\texttt{window_factor} \times T_{14,\mathrm{mask}},\ \texttt{window_fallback}\big).$$

The mask duration $T_{14,\mathrm{mask}}$ is chosen generously:

$$T_{14,\mathrm{mask}} = \max\big(T_{14,\mathrm{data}},\ T_{14,\mathrm{physical}},\ 5 \times T_{14,\mathrm{TLS}},\ 0.05\,\mathrm{d}\big),$$

where $T_{14,\mathrm{data}}$ comes from a trapezoid fit, $T_{14,\mathrm{physical}}$ from the stellar density via wotan's `t14`, and $T_{14,\mathrm{TLS}}$ from the TLS result (often underestimated on binned FFIs).

## 6.4 Positive-only outlier clipping

After each detrend, extreme *positive* outliers are rejected at `outlier_sigma` robust standard deviations:

$$\sigma_{\mathrm{robust}} = 1.4826 \times \mathrm{median}\big(|f - \mathrm{median}(f)|\big),$$

$$\mathrm{keep} \;=\; f < \mathrm{median}(f) + \texttt{outlier_sigma} \times \sigma_{\mathrm{robust}}.$$

Negative outliers are deliberately retained. A real transit is a coherent negative excursion; two-sided clipping can delete in-transit points and leave a residual shallow signal (observed on WASP-18b when two-sided clipping was used).

## 6.5 Stitching

Sectors are concatenated, NaNs dropped, and points sorted by time. Sector identifiers are retained for bookkeeping and for optional sector-coloured diagnostics.

# Chapter 7

# Transit search with TLS

## 7.1 Search setup

`run_tls` constructs a `transitleastsquares` model with:

- limb-darkening coefficients from the stellar parameter dictionary;
- optional stellar radius and mass priors (with clipped minima);
- period grid from `P_min` to min(`P_max`, baseline/$n_{\text{transits,min}}$);
- duration grid step and oversampling factor from the configuration;
- optional multi-threading.

## 7.2 Binning for the search only

Before the period search the light curve may be error-weighted binned to `tls_search_bin_minutes` (default 10 minutes). TLS runtime scales with the number of input points; for multi-sector baselines the gain is modest but useful. **Nothing downstream of the search uses the binned series:** per-transit depths, difference imaging, centroids, and the folded LC for TRICERATOPS all use full-resolution data.

## 7.3 Detection threshold

If the Signal Detection Efficiency (SDE) falls below `SDE_threshold` (default 7), the pipeline logs a non-detection, saves diagnostic plots of the weak peak, and exits without running TRICERATOPS.

## 7.4 Known ephemeris mode

When `use_known_ephemeris` is true, the TLS search is skipped entirely. The second detrend masks transits with the supplied ephemeris, and the adopted depth and duration come from the configuration.

# Chapter 8

# Ephemeris refinement and transit shape

## 8.1 Trapezoid model

A trapezoid transit profile is defined by depth $d$, total duration $T_{14}$, ingress duration, and a small mid-time shift. The shape is piecewise linear between the contact points. It is fitted to the phase-folded light curve with robust least squares (`soft_l1` loss).

## 8.2 Fine grid on period and epoch

`refine_ephemeris` evaluates a dense grid around the TLS period and epoch. The score is the significance of the difference between the weighted in-transit and out-of-transit means. The best $(P, T_0)$ is retained and the trapezoid is re-fitted at that ephemeris, yielding refined $T_{14}$ and depth.

## 8.3 Adopted parameters

The pipeline adopts the refined period, epoch, duration, and depth for all subsequent steps. When per-transit depths are available, the **weighted mean** of those depths replaces the single global depth for `calc_depths` and TRICERATOPS.

# Chapter 9

# Per-transit depth diagnostics

## 9.1 Local linear model

For each integer epoch $e$ with mid-time $T_c = T_0 + e\,P$, a local window

$$|t - T_c| < \tfrac{1}{2} \times \texttt{n_dur_window} \times T_{14}$$

is selected (default half-width $1.5\,T_{14}$). Inside that window the model

$$f(t) = a + b\,(t - T_c) - d \cdot s(t)$$

is solved by weighted least squares with weights $w = 1/\sigma_f^2$. Here:

- $a$ is the local continuum level;
- $b$ is a local linear baseline slope (absorbs residual trend and scattered light);
- $d$ is the transit depth;
- $s(t)$ is a **fixed** unit-depth transit shape.

The shape $s(t)$ is preferably the TLS model light curve, linearly interpolated onto the data times and normalised to unit depth. If the TLS duration disagrees strongly with the adopted duration, a soft-edged box of width $T_{14}$ is used instead.

## 9.2 Quality filters

An epoch is discarded if any of the following hold:

- fewer than `min_pts_per_transit` points (default 6);
- the shape maximum in the window is below `min_coverage` (default 0.8) — the transit bottom is not sampled;
- fewer than three points with $s > 0.1$ or fewer than three with $s < 0.1$ — no baseline anchor.

## 9.3 Uncertainty inflation

The formal covariance of the linear solution is scaled by $\max(\chi^2_{\mathrm{red}}, 1)$ so that under-estimated formal errors do not dominate the weighted mean.

## 9.4 Aggregates

**Weighted mean depth**

$$\bar{d} = \frac{\sum_i w_i d_i}{\sum_i w_i}, \qquad w_i = 1/e_{d,i}^2.$$

**Constant-depth test**

$$\frac{\chi^2}{\mathrm{dof}} = \frac{1}{N-1} \sum_i \frac{(d_i - \bar{d})^2}{e_{d,i}^2}.$$

A value near unity indicates consistency with a single depth.

**Odd/even test**

Epochs are split by `epoch mod 2`. Weighted means of the odd and even subsets are compared; the difference in units of the combined uncertainty is reported. A large odd/even difference can indicate a secondary eclipse or a half-period alias. Local visual differences within a single sector, however, often arise from residual detrending or edge effects and should be interpreted together with the global statistic.

# Chapter 10

# Centroid offsets

For each sector the mean in-transit centroid is compared with the out-of-transit median:

$$\Delta x = \langle x_{\rm in} \rangle - \mathrm{median}(x_{\rm out}), \qquad \Delta y = \langle y_{\rm in} \rangle - \mathrm{median}(y_{\rm out}).$$

The offset magnitude and a rough significance (using the out-of-transit scatter) are recorded. Significant offsets can indicate that the transit source is spatially offset from the target. Offsets are computed **per sector** because absolute centroid zero-points differ between sectors; concatenating raw positions produces an artificial step function.

# Chapter 11

# Difference-image analysis

## 11.1 Construction of the difference image

For each sector and each transit epoch with enough frames:

1. **In-transit image:** mean of frames with $|t - T_c| < \texttt{di_in_frac} \times T_{14}/2$.
2. **Out-of-transit image:** mean of frames in an annular time window just outside the transit (`di_out_gap` to `di_out_span` in units of $T_{14}/2$).
3. **Difference:** out − in.

Using a *local* out-of-transit window (rather than the whole sector) cancels slow scattered light and pointing drift. Epochs are combined with weights proportional to the number of in-transit frames. The uncertainty image is estimated from the epoch-to-epoch scatter or, for few epochs, from the out-of-transit frame scatter.

## 11.2 PRF fit

A circular Gaussian PRF plus constant background is fitted to the difference image by bounded nonlinear least squares:

$$I(c, r) = A \exp\left(-\frac{(c - x_0)^2 + (r - y_0)^2}{2\sigma_{\mathrm{PRF}}^2}\right) + B.$$

The fitted $(x_0, y_0)$ is the difference-image centroid in full-FFI coordinates.

## 11.3 Monte Carlo centroid error

Noise realisations are drawn from the difference-error image; each is re-fitted. The scatter of the recovered centroids, scaled by $1/\sqrt{2}$ to account for double-counting of noise already present in the image, provides a conservative positional uncertainty $\sigma_{\mathrm{pix}}$.

## 11.4 Star exclusion

Sky coordinates of all TIC stars are projected into the sector pixel frame. A neighbour is excluded in a sector if its distance from the difference-image centroid exceeds `di_sigma` $\times\sigma_{\mathrm{pix}}$ **and** the difference image SNR is at least `di_min_snr`. Stars excluded in *every* sector where the centroid was measurable are collected for `remove_star`.

Diagnostic three-panel plots (direct image, difference, SNR map) with star markers and centroids are written per sector.

# Chapter 12

# Nearby-star light-curve vetting

Optional; enabled with `--vet-neighbours`.

## 12.1 Motivation

Difference imaging localises the photocentre of the *missing* flux. Neighbour light-curve vetting asks a complementary question: does a resolved neighbour, in its own small aperture, show a transit of the depth required if *it* were the host?

## 12.2 Algorithm (`vet_nearby_stars`)

1. Download a larger TESScut cutout sized to reach `neighbour_max_sep`.
2. For each star in `target.stars` with separation in [`neighbour_min_sep`, `neighbour_max_sep`] and `tdepth > 0`:
   - build a circular aperture of radius `neighbour_ap_radius` centred on the star;
   - extract the aperture light curve from the cube;
   - detrend with the known ephemeris masked;
   - measure depth with a fixed-shape linear model (`_fixed_shape_depth`);
   - compute flux fractions $g_x$ of all stars in that aperture;
   - predict the depth under `signal on neighbour'' and`signal on target";
   - exclude the neighbour if the measurement is inconsistent with '`signal on neighbour'' at more than`neighbour_sigma' standard deviations.

## 12.3 Flux fractions $g_x$

The relative contribution of star $x$ to a given aperture is

$$F_x = 10^{-0.4\,T_{\mathrm{mag},x}},$$

$$f_{\mathrm{ap},x} = \sum_{i \in \mathrm{aperture}} \mathrm{PSF}_x(\mathrm{pixel}_i),$$

$$g_x = \frac{F_x \, f_{\mathrm{ap},x}}{\sum_j F_j \, f_{\mathrm{ap},j}}.$$

The PSF is a circular Gaussian of width $\sigma =$ `psf_sigma` (default 0.75 px). The fraction of the PSF inside a unit pixel is the product of one-dimensional Gaussian cumulative distribution functions (error functions) over the pixel boundaries — the same model TRICERATOPS uses.

Predicted depths in the neighbour aperture:

| **Hypothesis** | **Predicted depth** |
|---|---|
| Signal on the neighbour | $g_{\mathrm{nb}} \times tdepth_{\mathrm{nb}}$ |
| Signal on the target | $g_{\mathrm{tgt}} \times depth_{\mathrm{target}}$ |

## 12.4 Limitations

- Only the first sector in the list is used; there is no multi-sector stacking.
- The neighbour aperture is small (default 1 px), so photon noise is large and exclusion is conservative.
- The test is asymmetric: a neighbour is removed only when inconsistent with hosting the signal, not when merely consistent with the target hypothesis.

# Chapter 13

# Folded light curve and flux uncertainty for TRICERATOPS

## 13.1 Fold and bin

The full-resolution light curve is phase-folded about mid-transit,

$$\phi = (t - T_0 + P/2) \bmod P - P/2,$$

restricted to $|\phi| < \frac{1}{2} \times \texttt{fold_n_dur} \times T_{14}$, and binned to `bin_minutes` with lightkurve. The binned series is the photometric input to `calc_probs`.

## 13.2 Why a custom scalar $\sigma$ is required

`calc_probs` accepts one scalar flux uncertainty for the entire folded series. Formal pipeline errors on binned points are often tens of parts per million and ignore:

- correlated (red) noise;
- residual phase-curve or ellipsoidal variation;
- residual structure after detrending.

At high signal-to-noise every scenario then receives a likelihood so small that $\exp(\ln Z)$ underflows to zero. Relative probabilities become 0/0 and FPP is reported as NaN. This failure mode was observed on WASP-18b (TIC 100100827) before the robust error model was introduced.

## 13.3 Algorithm of `triceratops_flux_error`

1. Restrict to bins with at least three unbinned points and finite flux.
2. Define out-of-transit masks $|\phi| > 0.75\,T_{14}$ on both unbinned and binned series.
3. Estimate:

- $\sigma_{\rm pt}$: MAD of the unbinned OOT flux;
- $\sigma_{\rm white} = \sigma_{\rm pt}/\sqrt{\text{median}(n_{\rm bin})}$: white-noise expectation for a bin;
- $\sigma_{\rm emp}$: MAD of the *binned* OOT flux (includes red noise);
- $\sigma_{\rm floor}$: optional hard floor from `err_floor_ppm`.

4. Set $\sigma_0 = \max(\sigma_{\rm white}, \sigma_{\rm emp}, \sigma_{\rm floor})$.
5. Optionally inflate so a smooth reference model has reduced chi-squared equal to one:

$$\chi^2_{\rm red} = \frac{1}{N - n_{\rm par}} \sum_i \left( \frac{f_i - m_i}{\sigma_0} \right)^2 ,$$

$$\sigma = \sigma_0 \times \text{clip}\left( \sqrt{\chi^2_{\rm red}},\ 1,\ \texttt{tri_max_inflate} \right) .$$

The reference model is the optional limb-darkened fit if available, otherwise a trapezoid fitted on the binned series. Inflation **never decreases** $\sigma$. A diagnostic dictionary (all intermediate values in ppm) is logged.

# Chapter 14

# Optional limb-darkened fit

Enabled with `--ld-fit`, `--ld-free`, or `--ld-coeffs U1 U2`.

## 14.1 Model

Circular-orbit quadratic limb-darkened transit with free parameters:

- radius ratio $R_p/R_\star$;
- impact parameter $b$;
- scaled semi-major axis $a/R_\star$;
- small mid-time shift;
- continuum baseline $f_0$;
- optionally Kipping $(q_1, q_2)$ limb-darkening coefficients when `ld_mode = "free"`.

Otherwise limb darkening is fixed to Claret coefficients from the TIC/TLS catalogue or to user-supplied values.

## 14.2 Exposure smoothing

A single effective integration time is used for the whole folded curve:

$$t_{\rm exp} = \sqrt{(\Delta t_{\rm bin})^2 + \langle \text{cadence}^2 \rangle},$$

where $\langle \text{cadence}^2 \rangle$ is the point-weighted mean square of the median cadences of all sectors that entered the fold. The model is supersampled over this window ($n_{\rm ss} = 5$ sub-samples) so that it matches the binned data.

Two callables are stored:

- **`model` (smoothed):** evaluated with $t_{\rm exp} > 0$; used for residuals and for $\sigma$ inflation.

- **`model_intrinsic`:** evaluated with $t_{\rm exp} = 0$; sharp contact points, representing the instantaneous transit shape.

When sectors mix cadences (for example 30-minute and 200-second FFIs), the single $t_{\rm exp}$ is a compromise: slightly too sharp for long-cadence data and slightly too broad for short-cadence data.

## 14.3 Outputs

Fitted parameters with uncertainties (covariance scaled by the reduced chi-squared of the fit), derived inclination, stellar density from $a/R_\star$ and $P$, transit depth and contact durations, residual RMS, and a two-panel figure (data + models; residuals).

# Chapter 15

# Building the TRICERATOPS target

## 15.1 Instantiation

`make_target` constructs `triceratops.target` with the TIC ID, the array of sectors, and the search radius. Field-star population sources are tried in order:

1. user-supplied path (`--trilegal-fname`);
2. cached local file in the output directory;
3. online TRILEGAL query (with SSL and User-Agent workarounds);
4. Gaia DR3 cone search written as a TRILEGAL-compatible CSV (fallback).

## 15.2 Apertures and depths

The pipeline's per-sector aperture pixel lists are reordered to match `target.sectors`. Then

```
target.calc_depths(tdepth=depth, all_ap_pixels=all_ap)
```

assigns every star a flux ratio in the aperture and a required depth `tdepth`. Stars that cannot produce the observed dip even with a total eclipse receive `tdepth = 0`.

## 15.3 NFPP input summary

`nfpp_input_summary` logs the adopted depth, the implied aperture dip, the minimum neighbour flux ratio needed to explain the signal, and the list of neighbours that still have `tdepth > 0`. This clarifies whether a low NFPP is forced by the flux budget alone or by genuine scenario likelihoods.

# Chapter 16

# Running calc_probs: robust FPP and NFPP

## 16.1 Star removal

Stars excluded by difference imaging and/or neighbour vetting are passed to `target.remove_star`. Depths and flux ratios are recomputed. Stars with `tdepth = 0` are not removed: they already contribute nothing to NFPP, and removing them only clutters the log. The target itself is never removed.

## 16.2 Period type safety

TRICERATOPS decides whether $P_{\rm orb}$ is a fixed period or a range with

```
if type(P_orb) not in [float, int]:
    # interpret as [Pmin, Pmax]
```

A NumPy scalar (`np.float64`), such as `results.period` from TLS, is neither `float` nor `int`, so the range branch runs and indexing `P_orb[0]` raises `IndexError`. The script therefore forces

```
if np.ndim(P_orb) == 0:
    P_orb = float(P_orb)
```

before calling `calc_probs`.

## 16.3 Checked iterations

For each of `n_iter` independent runs:

1. Call `calc_probs` with the robust scalar $\sigma$.
2. Validate the result (`_check_run`):
    - reject if `FPP` is NaN;

- reject if `FPP_degenerate` is true (package $\geq 1.1$);
- reject if `lnZ` contains NaN or $+\infty$;
- reject if all $\ln Z = -\infty$.

3. On failure, multiply $\sigma$ by `tri_retry_factor` and retry (up to `tri_max_retries` times).
4. Store FPP and NFPP only for valid runs; invalid runs remain NaN and are **never** averaged in.

## 16.4 Verdict

If the fraction of valid runs is below `tri_min_valid_frac` (default 0.8), the verdict is **INDETERMINATE**. Otherwise the mean FPP and NFPP of the valid runs are compared with the Giacalone thresholds:

$$\text{FPP} < 0.015 \quad \text{and} \quad \text{NFPP} < 0.001 \quad \Rightarrow \quad \text{PASSED.}$$

Optional sensitivity runs (`--sensitivity`) repeat `calc_probs` at several multiples of $\sigma$ and report how strongly the result depends on the noise model.

## 16.5 Version note

TRICERATOPS versions before 1.1.0 normalise scenario evidences with a direct $\exp(\ln Z)/\sum \exp(\ln Z)$ that underflows for very poor fits. Version 1.1.0 and later use a log-sum-exp form and expose `lnZ` and `FPP_degenerate`. The script detects the installed version and warns if it is older than 1.1.0.

# Chapter 17

# Outputs and products

Typical files written under the output directory (default `TIC{tic}/`):

| File pattern | Content |
|---|---|
| `triceratops_output_TIC*.txt` | Full text log |
| `lc_corrected_*`, `lc_detrended_*`, `lc_stitched_*` | Light curves |
| `tls_diagnostics_*.pdf` | Periodogram and folded transit |
| `per_transit_depths_*.pdf` / `.txt` | Depth versus epoch |
| `phased_transit_*.pdf`, `folded_lightcurve_*.txt` | Folded LC |
| `ld_fit_*.pdf` | Limb-darkened fit (if requested) |
| `difference_image_*`, `difference_imaging_*.csv` | Difference imaging |
| `neighbour_vetting_*.csv` | Neighbour LC tests |
| `field_*.png`, `plot_fits_*.pdf` | TRICERATOPS field and scenario fits |
| `trilegal_TIC*.csv` | Cached field-star table |
| `tesscut_cache/` | Reusable FITS cutouts |

# Chapter 18

# Command-line interface

```
python triceratops_validation.py \
  --tic 100100827 \
  --outdir WASP18b_test \
  --n-iter 20 \
  --vet-neighbours \
  --ld-fit \
  --err-floor-ppm 0 \
  --sensitivity \
  --raw-aperture-flux \
  --trilegal-fname /path/to/trilegal.dat
```

| Flag | Effect |
|---|---|
| `-tic` | TIC identifier |
| `-outdir` | Output directory (default `TIC{tic}/`) |
| `-n-iter` | Number of `calc_probs` runs |
| `-vet-neighbours` | Enable neighbour light-curve vetting |
| `-no-difference-imaging` | Disable difference imaging |
| `-ld-fit` / `-ld-free` / `-ld-coeffs U1 U2` | Limb-darkened reference model |
| `-err-floor-ppm` | Minimum $\sigma$ in ppm |
| `-no-inflate` | Disable $\chi^2_{\rm red} = 1$ inflation |
| `-sensitivity` | Extra FPP/NFPP at scaled $\sigma$ |
| `-raw-aperture-flux` | Pass `dilution_corrected=False` (needs triceratops $\geq 1.1$) |
| `-trilegal-fname` | Pre-downloaded field-star table |
| `-download-dir` | TESScut cache location |
| `-contrast-curve` | High-resolution contrast curve file |

# Chapter 19

# Known limitations and best practices

1. **FFI versus 2-minute data.** The Giacalone validation thresholds were calibrated primarily on short-cadence light curves. FFI analyses remain valuable for triage and prioritisation but should be labelled as such in publications.
2. **Single effective exposure time** in the limb-darkened fit when sectors mix cadences.
3. **Neighbour vetting uses one sector only**; low signal-to-noise can prevent exclusion of true contaminants.
4. **TRILEGAL connectivity.** The Padova web form is often blocked for automated clients (SSL or HTTP 403). Prefer `--trilegal-fname` or rely on the Gaia DR3 fallback.
5. **Always inspect** per-transit depths, difference images, and `plot_fits` before trusting a PASSED verdict.
6. **Prefer triceratops $\geq$ 1.1.0** for log-sum-exp normalisation and the `FPP_degenerate` / `lnZ` attributes used by the robust runner.
7. **Positive-only outlier clipping** is essential for deep transits; do not switch to two-sided clipping without checking the phase fold.
8. **`P_orb` must be a Python float**, not a NumPy scalar, when calling `calc_probs`.
9. **Aperture choice.** The percentile aperture is simple and reproducible but not optimised per sector. Crowded fields may benefit from custom masks.
10. **Statistical validation is not dynamical confirmation.** A PASSED verdict means the data favour a planet on the target over the modelled false-positive scenarios at the stated thresholds; it does not replace radial-velocity or transit-timing mass constraints where those are required.

# Chapter 20

# Function-by-function code walkthrough

This chapter walks through every major function in `triceratops_validation.py` in the order they appear in the source. The goal is to make the control flow, inputs, outputs, and design choices explicit enough that the script can be maintained or extended without reverse-engineering the code.

## 20.1 Configuration and data structures

### 20.1.1 `Config` (dataclass)

Central configuration object. Instantiated from command-line arguments in the `__main__` block, or constructed programmatically for tests. `__post_init__` resolves `outdir` to an absolute path (`TIC{tic}/` by default) and creates the directory. Helper methods:

- `path(name)` — absolute path under `outdir`, creating the directory if needed;
- `cache_dir()` — TESScut cache (`download_dir` or `outdir/tesscut_cache`).

All subsequent functions take `cfg: Config` where they need paths or numerical thresholds.

### 20.1.2 `SectorData` (dataclass)

One object per TESS sector. Fields fall into three groups:

1. **Photometry:** `time`, `flux` (background-corrected, un-normalised), `flux_err`, later `norm_flux`, `norm_err`, `trend`.
2. **Geometry:** `ap_pixels` (full-FFI [col, row]), `cen_x`, `cen_y`, `col0`, `row0`.
3. **Difference imaging:** `cube`, `cube_time`, `wcs` (not printed in `repr` to keep logs readable).

## 20.2 Stellar and sector metadata

### 20.2.1 `stellar_params(tic)`

Queries the TIC with `Catalogs.query_criteria(catalog="Tic", ID=tic)`. Values are read **by column name** (not positional index) to survive TIC schema changes. Returns a dictionary with:

- atmospheric parameters: `Teff`, `logg`, `metal`;
- radius and mass with min/max from reported uncertainties;
- `Tmag`, `ra`, `dec`;
- quadratic limb-darkening `u` from `transitleastsquares.catalog_info`, with fallback `[0.4, 0.3]`.

Raises `ValueError` if the TIC ID is not found.

### 20.2.2 `find_tess_sectors(tic)`

Calls `lk.search_tesscut(f"TIC {tic}")` and returns a sorted NumPy array of unique `sequence_number` values. Raises `RuntimeError` if the search is empty.

## 20.3 Light-curve extraction

### 20.3.1 `extract_sector(tic, sector, cfg)`

1. `search_tesscut` for the given sector; `download(cutout_size=..., download_dir=cache)`.
2. Median image → percentile threshold → `target_mask`.
3. `np.mgrid` over the cutout in full-FFI coordinates; stack columns and rows of masked pixels into `ap_pixels`.
4. Centroids via `tpf.estimate_centroids(aperture_mask=target_mask)`.
5. Target light curve via `tpf.to_lightcurve`.
6. Background: inverted low-threshold mask; if at least four pixels, subtract scaled background and combine errors in quadrature.
7. Common boolean quality mask on time, flux, error, centroids; make arrays contiguous.
8. Store cube and WCS; write diagnostic PDF (TPF with aperture, corrected LC).

Returns a filled `SectorData`. Failures in individual sectors are caught in `main` so one bad sector does not abort the run.

\subsection `detrend_sector(sd, window_length, cfg, mask_transits=None)`

Calls `wotan.flatten` with `return_trend=True` and optional `mask`. Writes `sd.norm_flux`, `sd.norm_err` (= flux error divided by the trend), and `sd.trend`. Window length is clamped to at least `cfg.window_fallback`.

### 20.3.2 `clip_outliers(time, flux, flux_err, sector_ids=None, sigma=5.0)`

Robust median and MAD scale. Keeps only points with $f < \text{median} + \sigma \times \text{MAD}$. Optionally propagates `sector_ids`. Returns filtered arrays. Disabled when `sigma` is `None` or $\leq 0$.

### 20.3.3 `stitch(sectors_data)`

Concatenates `time`, `norm_flux`, `norm_err`, and sector ID arrays; drops non-finite points; sorts by time. Returns `(t, f, e, sector_ids)`.

### 20.3.4 `save_sector_lightcurves`, `save_stitched_lightcurve`, `plot_sector_panels`, `plot_transits_on_sectors`

I/O helpers. Text files carry headers describing columns. Multi-panel PDFs show one sector per row; transit-marked plots overlay predicted transit windows from the adopted ephemeris.

## 20.4 TLS search and shape fitting

### 20.4.1 `bin_for_search(time, flux, flux_err, bin_minutes)`

Error-weighted binning on a regular grid of width `bin_minutes`. Used only to accelerate TLS. Returns binned time, flux, and error.

### 20.4.2 `run_tls(time, flux, flux_err, star, cfg, tag="")`

Builds keyword arguments for `transitleastsquares.power` (limb darkening, period limits clipped by baseline, stellar priors, threads, oversampling). Instantiates the model with `verbose=True`, runs the period search, prints a one-line summary (period, $T_0$, depth, duration, SDE), and returns the TLS results object.

### 20.4.3 `_trapezoid_model(phase, depth, t14, ingress)`

Piecewise-linear trapezoid as a function of time from mid-transit. Used by the independent shape fit and by diagnostic plots.

### 20.4.4 `fit_transit_shape(time, flux, flux_err, P, T0, dur_guess=None, depth_guess=None)`

Robust least-squares trapezoid fit on the phase-folded series. Free parameters: depth, $T_{14}$, ingress, $T_0$ shift. Returns `(t14, depth, t0_shift, model_callable)`.

### 20.4.5 `estimate_duration_depth(...)`

Thin wrapper around `fit_transit_shape` for mask sizing in the second detrend pass.

### 20.4.6 `refine_ephemeris(time, flux, flux_err, P, T0, dur)`

Grid search on $P$ and $T_0$ scored by the significance of the in-transit versus out-of-transit mean difference. Re-fits the trapezoid at the best ephemeris. Returns refined `(P, T0, T14, depth)`.

### 20.4.7 `plot_tls_diagnostics(...)`

Two-panel PDF: SDE periodogram with the best period and threshold marked; phase-folded data with binned points and trapezoid model. Optional overrides for refined ephemeris and depth.

## 20.5 Per-transit depths

### 20.5.1 `per_transit_depths(time, flux, flux_err, results, cfg, P=None, T0=None, dur=None)`

Builds unit-depth shape $s(t)$ from the TLS model or a soft-edged box. For each epoch:

- select local window;
- apply quality filters (point count, coverage, baseline anchors);
- solve $f = a + b(t - T_c) - d\,s$ by weighted least squares;
- inflate depth error by $\sqrt{\max(\chi^2_{\mathrm{red}}, 1)}$.

Returns a structured NumPy array with fields `epoch`, `Tc`, `depth`, `e_depth`, `npts`, `coverage`.

### 20.5.2 `depth_diagnostics(dt)`

Weighted mean depth, constant-depth $\chi^2$/dof, and odd/even significance. Returns a dictionary for logging and plot titles.

### 20.5.3 `plot_per_transit_depths(dt, diag, cfg, tic)`

Errorbar plot of depth versus mid-time, coloured by odd/even epoch, with weighted mean and uncertainty band.

## 20.6 Centroids

### 20.6.1 `centroid_offsets(sectors_data, P, T0, dur)`

Per sector: in-transit mask from `transit_mask`; median out-of-transit centroid as reference; mean in-transit offset and rough significance. Returns a pandas DataFrame.

## 20.7 PSF flux fractions

### 20.7.1 `_gauss_pixel_fraction(xc, yc, cols, rows, sigma)`

For each pixel centre, product of one-dimensional Gaussian CDFs over the pixel edges (error function). Returns the fraction of a circular Gaussian PSF inside each unit pixel.

### 20.7.2 `aperture_flux_fractions(star_x, star_y, star_tmag, ap_pixels, sigma=0.75)`

$$g_x = \frac{F_x \sum_i \mathrm{PSF}_x(\mathrm{pixel}_i)}{\sum_j F_j \sum_i \mathrm{PSF}_j(\mathrm{pixel}_i)}, \qquad F_x = 10^{-0.4 T_{\mathrm{mag},x}}.$$

Returns an array that sums to one over stars.

## 20.8 Nearby-star vetting

### 20.8.1 `vet_nearby_stars(tic, sector, stars, P, T0, dur, depth_target, cfg)`

Downloads a larger cutout; for each eligible neighbour extracts a circular aperture light curve, detrends with masked transits, measures depth via `_fixed_shape_depth`, computes $g_{\mathrm{nb}}$ and $g_{\mathrm{tgt}}$, and flags exclusion when the measured depth is inconsistent with $g_{\mathrm{nb}} \times tdepth_{\mathrm{nb}}$ at more than `neighbour_sigma`. Returns a DataFrame.

### 20.8.2 `_fixed_shape_depth(time, flux, flux_err, P, T0, dur)`

Phase-folded linear model $f = a - d\,s$ with a soft-edged box $s$. Returns `(depth, error)` or `(nan, nan)` if sampling is inadequate.

## 20.9 Difference imaging

### 20.9.1 `_prf_model(p, cols, rows)`

Circular Gaussian PRF plus constant background: parameters $(A, x_c, y_c, \sigma, B)$.

### 20.9.2 `fit_source_position(image, err=None, col0=0, row0=0, cfg=None)`

Bounded least-squares PRF fit. Returns a dictionary with amplitude, position, width, background, and SNR, or `None` on failure.

### 20.9.3 `build_difference_image(sd, P, T0, dur, cfg)`

Per-epoch in-transit and local out-of-transit means; weighted average difference; error from epoch scatter or frame scatter. Returns a dictionary with `diff`, `diff_err`, `direct`, `n_epochs`, `per_epoch`.

#### 20.9.4 `difference_image_analysis(sectors_data, stars, P, T0, dur, cfg, tic)`

Loops over sectors: build difference image, fit PRF, Monte Carlo centroid error, compare to catalogue and direct-image centroids, score neighbours for exclusion, write diagnostic plots. Returns `(summary_df, star_table_df)`.

#### 20.9.5 `_plot_difference_image(...)`

Three-panel figure: direct image, difference, SNR map, with star markers and centroids.

#### 20.9.6 `combine_difference_imaging(star_table, cfg)`

Returns the list of star IDs excluded in every sector where the difference-image centroid was trustworthy.

### 20.10 Limb-darkened transit model

#### 20.10.1 `quad_ld_transit(z, p, u1, u2, n_annuli=150)`

Quadratic limb-darkened transit flux as a function of normalised separation $z$ and radius ratio $p$, via annular integration of the specific intensity.

#### 20.10.2 `limb_darkened_lc(t, P, p, aRs, b, u1, u2, t0s=0.0, f0=1.0, exp_time=0.0, n_ss=5, n_annuli=150)`

Full light curve versus time from mid-transit. If `exp_time > 0`, supersamples $n_{\rm ss}$ points across the exposure and averages. Sets flux to continuum on the far side of the orbit ($\cos\,\text{phase} < 0$).

#### 20.10.3 `_u_from_q(q1, q2)`, `_q_from_u(u1, u2)`

Kipping (2013) bijection between quadratic coefficients and the unit square.

#### 20.10.4 `fit_limb_darkened(t, f, P, dur, depth, cfg, u_claret=None, exp_time=0.0)`

Multi-start least-squares fit (several initial impact parameters). Returns a rich dictionary: parameters with errors, derived inclination and density, depth and contact durations, `model` (smoothed) and `model_intrinsic` callables.

#### 20.10.5 `fit_limb_darkened_lc(tp, lc, P, dur, depth, star, sectors_data, cfg, log=None)`

Estimates effective exposure from bin width and sector cadences, calls `fit_limb_darkened` on well-populated bins, logs results, attaches bin arrays for plotting.

### 20.10.6 `plot_ld_fit(tp, fp, fit, cfg, tic)`

Two-panel figure: unbinned and binned data with smoothed and intrinsic models; residuals of the smoothed model on the bins.

## 20.11 Fold and robust sigma

### 20.11.1 `fold_and_bin(time, flux, flux_err, P, T0, dur, cfg)`

Phase-fold, restrict to the configured duration window, sort, bin with lightkurve. Returns unbinned `(tp, fp, ep)` and the binned `TessLightCurve`.

### 20.11.2 `_mad_sigma(x)`

$1.4826 \times \text{median}|x - \text{median}(x)|$ on finite values; `nan` if fewer than three points.

### 20.11.3 `triceratops_flux_error(tp, fp, lc, dur, P, cfg, ref_model=None, ref_npar=4)`

Implements the robust scalar $\sigma$ of Chapter~13: white and empirical OOT estimates, optional floor, optional inflation to $\chi^2_{\text{red}} = 1$ of a reference model. Returns `(sigma, info_dict)`.

## 20.12 TRICERATOPS helpers

### 20.12.1 `_tri_version()`, `_version_tuple(v)`

Installed package version string and comparable integer tuple.

### 20.12.2 `_dilution_kwargs(func, cfg)`

Returns `{"dilution_corrected": False}` only if the user requested raw aperture flux **and** the installed function signature accepts that keyword (triceratops $\geq 1.1$).

### 20.12.3 `_check_run(target)`

Returns `(ok: bool, reason: str)`. Rejects NaN FPP, `FPP_degenerate`, non-finite $\ln Z$, or all $\ln Z = -\infty$.

### 20.12.4 `_calc_probs_checked(target, lc, P_orb, sigma, cfg, max_retries=None)`

Calls `calc_probs` with the current $\sigma$; on invalid results multiplies $\sigma$ by `tri_retry_factor` and retries. Returns `(ok, fpp, nfpp, sigma_used, reason)`.

### 20.12.5 `_ensure_trilegal_ssl()`

Sets certifi CA paths, optionally disables SSL verification for the Padova host, patches `requests` and `urllib` with a browser-like User-Agent to reduce HTTP 403 responses.

#### 20.12.6 `_gaia_field_star_table(ra, dec, tic, cfg, field_deg2=0.1, mag_limit=21.0)`

ADQL cone search on Gaia DR3; maps columns to a TRILEGAL-like CSV (`Mact`, `logg`, `logTe`, `[M/H]`, `TESS`, `J`, `H`, `Ks`). Used when the online TRILEGAL form is unavailable.

#### 20.12.7 `make_target(tic, sectors, apertures, depth, cfg)`

Constructs `tr.target`, attaches field stars (file / TRILEGAL / Gaia), maps apertures to `target.sectors`, calls `calc_depths`. Returns `(target, all_ap)`.

#### 20.12.8 `nfpp_input_summary(target, depth, cfg, log=None, n_removed=0)`

Logs flux-budget information: which neighbours can still produce the observed dip (`tdepth > 0`).

#### 20.12.9 `run_triceratops(target, all_ap, tic, depth, P_orb, lc, cfg, drop_stars=None, log=None, sigma=None)`

1. Log package version and warn if $< 1.1.0$.
2. Plot fields **before** star removal (plotting breaks after `remove_star`).
3. `remove_star` + re-`calc_depths` if needed.
4. Force `P_orb` to a Python float.
5. Loop `n_iter` checked `calc_probs` runs; optional sensitivity grid.
6. Aggregate valid FPP/NFPP; emit PASSED / NOT passed / INDETERMINATE.
7. Return `(target, FPPs, NFPPs, sigma_used)`.

#### 20.12.10 `_log(fh, text)`

Print to stdout and optionally to an open log file.

### 20.13 Orchestration

#### 20.13.1 `main(cfg)`

End-to-end control flow:

1. Open log file; query star and sectors.
2. Extract all sectors; save corrected LCs and panels.
3. First detrend; stitch; clip outliers.
4. Either known-ephemeris path or TLS pass 1 $\rightarrow$ mask sizing $\rightarrow$ second detrend $\rightarrow$ TLS pass 2 $\rightarrow$ refine ephemeris; plot diagnostics.
5. Per-transit depths and diagnostics.
6. Save detrended products; centroid offsets.
7. Fold and bin; optional LD fit; compute robust $\sigma$; plot phased transit.
8. `make_target`; difference imaging; neighbour vetting; filter `drop_stars`.
9. `run_triceratops`; `plot_fits`; close log.

### 20.13.2 `__main__` argument parser

Defines all CLI flags documented in the command-line chapter, builds `Config`, and calls `main(cfg)`.

# Chapter 21

# Worked notes: high signal-to-noise failure mode

On a deep transit such as WASP-18b ($\approx 1\%$ depth, many sectors, high SDE):

1. Formal binned errors may be $\sim 50$–$100$ ppm.
2. Residual structure after detrending is larger than that.
3. Every scenario model (planet and false positive) fails to fit the residuals at the formal precision.
4. All $\ln Z$ values become large negative numbers; $\exp(\ln Z)$ underflows.
5. Relative probabilities are NaN; FPP is NaN; the validation criteria cannot be evaluated.

The robust $\sigma$ path forces the noise model to absorb residual structure (empirical OOT scatter and optional $\chi^2_{\rm red} = 1$ inflation). At least some scenarios then retain finite evidence, and FPP/NFPP become numerically meaningful. Sensitivity runs at scaled $\sigma$ show whether the classification is stable under reasonable changes of the noise model.

A second numerical failure mode is the NumPy scalar period: `results.period` is `np.float64`, which fails TRICERATOPS's `type(P_orb) not in [float, int]` check and is treated as a period range, causing `IndexError` on `P_orb[0]`. Converting with `float(P_orb)` resolves it.

# Chapter 22

# Pipeline updates (integrated)

This chapter consolidates changes made after the initial documentation release. It replaces earlier ad-hoc changelog appendices.

## 22.1 Sector exclusion

### 22.1.1 Manual

Bad sectors (catastrophic normalisation, extreme scatter) can dominate the fold and bias the noise estimate. Drop them before extraction:

```
python triceratops_validation.py --tic 165501544 --exclude-sectors 56 79
```

Config field: `exclude_sectors` (tuple or None).

### 22.1.2 Automatic (default on)

After each detrend pass, `filter_bad_sectors` drops sectors whose normalised LC fails:

| Cut | Default | Meaning |
|---|---|---|
| `sector_max_rms` | 0.02 (2%) | robust RMS of norm_flux |
| `sector_max_median_offset` | 0.05 | |
| `sector_max_outlier_frac` | 0.05 | fraction with |

Disable with `--no-auto-exclude-sectors`. Tighten/loosen with `--sector-max-rms`.

The log reports the dropped and remaining sector lists. Difference imaging, stitch, fold and TRICERATOPS all use only the remaining sectors.

## 22.2 Cache-first TESScut loading

`_load_tesscut_tpf` opens local FITS when present (filename token or SECTOR header) and never calls `search_tesscut` on a hit. Expected log:

```
TESScut cache index: N FITS in <path>
TESScut sector 14  (cache hit: <file>.fits)
```

On a miss: `not in cache → download`. Prefer a Linux-native `--download-dir` under WSL rather than `/mnt/d/...`.

## 22.3 Out-of-transit renormalisation and outlier handling

- `renormalize_sector_oot`: after each detrend, divide by the OOT median so the continuum sits at 1.0 before stitching.
- `clip_outliers`: upward spikes always; downward spikes only out of transit when an ephemeris is known; in-transit negatives preserved.
- `fold_and_bin`: MAD clip on the fold, OOT re-normalisation to 1.0, optional median binning (`fold_robust_bin`).

## 22.4 Known ephemeris and depth adoption

`--ephemeris P_D T0_BTJD DEPTH_PPM T14_H`

TLS is skipped. Per-transit depths run in known-ephemeris mode. Adoption rules:

1. Weighted per-transit mean if chi-squared reduced is less than 5.
2. Else fold / TLS / known depth.
3. After a successful LD fit with small $T_0$ shift: update depth and $T_{14}$ from the fit.

A large LD $T_0$ shift (fraction of $T_{14}$) means a smeared fold; depth is **not** updated from the LD fit.

## 22.5 Phase 0.5 (secondary / half-period) test

After the primary per-transit analysis the pipeline measures the stacked depth at $T_0 + P/2$. A depth comparable to the primary is incompatible with a single transiting planet at period $P$ (it favours a $P/2$ alias or an equal-depth eclipsing binary). The log reports both depths and their difference in units of combined uncertainty.

## 22.6 Red-noise scramble $p$-value

TRICERATOPS quantifies **astrophysical** false positives (EB, NEB, blends). It does **not** quantify the probability that the stacked dip is a draw from correlated or non-Gaussian noise (Ivashtenko and Zackay 2025).

The pipeline implements a lightweight **phase-randomisation** test:

1. Measure the observed stacked depth at $(P, T_0, T_{14})$.
2. Draw $N$ random epochs $T_0'$ uniformly in $[T_0, T_0 + P)$.
3. Remeasure the stacked depth at each $T_0'$ (light curve fixed, so red-noise structure is preserved).
4. Report

$$p = \frac{1 + n(\text{null} \geq \text{observed})}{1 + n_{\text{valid}}}.$$

```
python triceratops_validation.py ... --red-noise-scrambles 2000 --red-noise-seed 42
```

Interpretation:

- $p \lesssim 0.01$: stacked depth is unlikely under pure phase-random noise at this $P$ (still does not rule out EBs).
- $p \gtrsim 0.01$: do **not** treat a low TRICERATOPS FPP as validation; the signal is consistent with a red-noise fluctuation at fixed $P$.

This is not a full matched-filter plus PSD permutation pipeline; it is a practical diagnostic on the ephemeris under test.

## 22.7 FPP pooling and numerics

- Robust sigma for `calc_probs` (OOT scatter, optional model inflation, retries on NaN).
- Pooled evidences across runs (log-mean-exp) with bootstrap errors; optional `--no-pool`.
- `--n-samples` raises the Monte-Carlo draws per scenario when one run dominates the evidence.

## 22.8 CLI reference (selected)

| Flag | Purpose |
|---|---|
| `--exclude-sectors S ...` | Drop sectors before extraction |
| `--red-noise-scrambles N` | Phase-randomisation trials (0 = off) |
| `--red-noise-seed K` | RNG seed for scrambles |
| `--ephemeris P T0 DEPTH_PPM T14_H` | Known ephemeris (skip TLS) |
| `--ld-fit` / `--ld-free` / `--ld-coeffs U1 U2` | Limb-darkened fold fit |
| `--detrend-windows W ...` | Multi-window blind TLS |
| `--sensitivity` | FPP/NFPP vs sigma scales |
| `--n-samples N` | Draws per scenario in `calc_probs` |

# Chapter 23

# References


1. Giacalone, S., Dressing, C. D., Jensen, E. L. N., et al. 2021, "Vetting of 384 TESS Objects of Interest with TRICERATOPS and Statistical Validation of 12 Planet Candidates," *The Astronomical Journal*, **161**, 24. DOI: 10.3847/1538-3881/abc6af.
2. Ivashtenko, O., and Zackay, B. 2025, "An Independent Search for Small Long-period Planets in Kepler Data. I. Detection Pipeline," *The Astronomical Journal*, **170**, 303. DOI: 10.3847/1538-3881/adfc50.
3. Hippke, M., and Heller, R. 2019, "Optimized transit detection algorithm to search for periodic transits of small planets," *Astronomy and Astrophysics*, **623**, A39 (Transit Least Squares).
4. Hippke, M., David, T. J., Mulders, G. D., and Heller, R. 2019, "Wotan: Comprehensive time-series de-trending in Python," *The Astronomical Journal*, **158**, 143.
5. Lightkurve Collaboration et al. 2018, "Lightkurve: Kepler and TESS time series analysis in Python," ascl:1812.013.
6. Kipping, D. M. 2013, "Efficient, uninformative sampling of limb darkening coefficients for two-parameter laws," *MNRAS*, **435**, 2152.
7. Stassun, K. G., et al. 2019, "The revised TESS Input Catalog and Candidate Target List," *The Astronomical Journal*, **158**, 138.
8. Girardi, L., et al. 2005, "Star counts in the Galaxy... TRILEGAL," *Astronomy and Astrophysics*, **436**, 895.
9. Claret, A. 2017, "Limb and gravity-darkening coefficients for the TESS satellite...," *Astronomy and Astrophysics*, **600**, A30.
10. Ricker, G. R., et al. 2015, "Transiting Exoplanet Survey Satellite (TESS)," *JATIS*, **1**, 014003.
11. Morton, T. D. 2012, background on statistical validation (VESPA).
12. Torres, G., et al. 2004, "Testing blend scenarios for extrasolar transiting planet candidates," *ApJ*, **614**, 979.
13. Thompson, S. E., et al. 2018, "Planetary Candidates Observed by Kepler. VIII...," *ApJS*,

**235**, 38 (reliability of long-period small planets).

14. Burke, C. J., et al. 2019, on systematic false alarms in Kepler.

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# Chapter 24

# Addendum: summary of the version 3 changes

This addendum extends the manual with the changes made to `triceratops_validation.py` after the first edition and after the changelog of Chapter 22. It also corrects statements in Chapters 1–23 that no longer describe the script. All numbers quoted for the default configuration refer to the script as uploaded in October 2026; the `Config` defaults of that file are *problem specific* (for example the period range and the known ephemeris) and should be adapted per target.

## 24.1 What changed

1. **Individual transits are now detected in every mode.** In known-ephemeris mode there is no TLS result; the per-transit block used to be skipped silently, so no per-transit table, no odd/even test and no constancy test were produced (Chapter 25). The mid-time of every transit is now a free parameter, and every unusable epoch is listed with the reason.
2. **The ephemeris is refined from the measured mid-times** before the light curve is folded, and a P/2 test at phase 0.5 is added.
3. **FPP and NFPP are computed from pooled scenario evidences** instead of by averaging per-run FPPs, with a bootstrap error and a convergence check (Chapter 26). New option `--n-samples`.
4. **Star removal is verified.** After `remove_star` the script checks that every requested star is really gone.
5. **NFPP bookkeeping.** Stars with `tdepth = 0` are ignored in vetting and a log section explains which neighbours can produce the signal (Chapter 27).
6. **Limb-darkened model** implemented without external dependency, validated against `batman`; the LD fit may update the adopted depth and duration only when the fold is trustworthy; the fold window uses the larger of the adopted and the measured T14 (Chapter 28).
7. **Detection and detrending.** Optional multi-window blind search, a candidate edge-fraction diagnostic, rejection of cadences with blown-up formal errors, and a repaired folded-light-curve plot (Chapter 29).

8. **Command line.** New options `--ephemeris`, `--blind-search`, `--detrend-windows`, `--n-samples`, `--no-pool`, `--sens-runs`, `--no-refine-ephemeris`, `--no-tc-fit` (Chapter 30).

## 24.2 Corrections to the earlier chapters

| Where | First edition | Current behaviour |
|---|---|---|
| §4.2, §6.2 | `window_length_search` default 5.0 d. | The uploaded configuration uses 1.5 d. A list of windows can be searched (`window_length_grid`, `--detrend-windows`); see Chapter 29. |
| §4.3, §7.3 | `P_min, P_max` = 0.5, 1.5 d; `SDE_threshold` = 7.0. | The uploaded configuration is set for a long-period candidate: 130, 135 d and 3.0. These are per-target settings; the period range must contain the period of the target. |
| §4.4 | Lists the TRICERATOPS numerics. | Adds `tri_N`, `pool_evidences`, `tri_boot`, `tri_sens_runs`, `err_clip_factor` and the per-transit fields (Chapter 30). |
| §4.6, §22.5 | Known-ephemeris fields "not yet exposed as CLI flags". | `--ephemeris P T0 DEPTH_PPM T14_H` sets them per target; `--blind-search` forces the TLS search. The `Config` defaults are global and silently apply to every target, so the log now prints the ephemeris in use. |
| §8.3 | Weighted mean of per-transit depths replaces the global depth when available. | Only when the epochs are mutually consistent (reduced $\chi^2 < 5$); otherwise the fold/TLS/known depth is kept, and the log states which was adopted. (This is the "future improvement" of §22.6; it is implemented.) |
| §9, §20.5 | Local window of $\pm 1.5\,T_{14}$ with fixed $T_c$; requires a TLS result. | Free mid-time within $\pm 0.5\,T_{14}$, window $\geq 1.75\,T_{14}$, works without TLS result, new columns `dTc, e_dTc, snr, dchi2` (Chapter 25). |
| §13.1, §13.3 | Fold window and out-of-transit regions defined with the adopted $T_{14}$. | Defined with $\max(T_{14,\mathrm{adopted}}, T_{14,\mathrm{measured}})$; a catalogue duration shorter than the observed transit put ingress and egress into the "out-of-transit" region. |
| §14 | The LD fit is a reference model for $\sigma$ only. | It may also overwrite the adopted depth and $T_{14}$, but only if the fitted T0 shift is below $0.15\,T_{14}$ (Chapter 28). |
| §16.1 | `remove_star`, then recompute depths. | Additionally verified; missing removals are forced by hand and logged as "removed k of n requested". |

| Where | First edition | Current behaviour |
|---|---|---|
| §16.3–16.4 | Verdict from the mean FPP/NFPP of valid runs. | Verdict from the pooled evidences, with bootstrap error and a dominance check (Chapter 26); `--no-pool` restores the old behaviour. |
| §18 | Ten command-line flags. | Twenty-three; see Table 30.2. |
| §19, item 3 | Neighbour vetting uses one sector. | Unchanged; stars with `tdepth = 0` are now skipped. |
| §20.12.9, §20.13.1 | Description of `run_triceratops` and `main`. | Both changed; see Chapter 31. |

# Chapter 25

# Individual transits and ephemeris refinement

## 25.1 Why the per-transit table was missing

With `use_known_ephemeris = True` the TLS search is skipped and the variable holding the TLS result is `None`. The block of `main()` that computes the per-transit depths was guarded by `if results is not None:`. The consequence was silent: in known-ephemeris runs the log contained no per-transit depths, no odd/even statistic and no constancy statistic. In addition, the original estimator evaluated every epoch at the nominal $T_c = T_0 + eP$; with an ephemeris that is uncertain by a fraction of $T_{14}$ (long period, few transits, small $P$ error multiplied by the epoch number) the transit is fitted too shallow or rejected.

## 25.2 Unit-depth shape

The estimator needs a profile $s(\phi)$ that is 0 outside the transit and 1 at the bottom, as a function of the phase offset $\phi$ in days. The first available of the following is used (`_unit_shape_func`):

1. the supplied `shape_model`: in known-ephemeris mode a trapezoid fitted to the fold, re-centred on its fitted T0 shift, $s(\phi) = \big(1 - m(\phi)\big)/d_{\mathrm{ref}}$ with $d_{\mathrm{ref}} = 1 - \min m$;
2. the TLS model light curve re-expressed as a function of phase (only if the TLS duration is within 30 % of the adopted $T_{14}$);
3. a soft-edged box of width $T_{14}$ with edge length $\max(0.05\,T_{14},\, 1\,\mathrm{min})$.

In known-ephemeris mode the trapezoid duration replaces the adopted $T_{14}$ for the per-transit window when it lies between 0.3 and 3 times it.

## 25.3 Estimator

For epoch $e$ the nominal mid-time is $T_c = T_0 + eP$. Data are selected in $|t - T_c| < h$ with

$$h = T_{14}\,\max\Big(\tfrac{1}{2} n_{\text{dur_window}},\; 0.5 + r + 0.75\Big), \tag{25.1}$$

where $r$ is `per_transit_tc_range` (0.5) if the mid-time is fitted and 0 otherwise; for the defaults $h = 1.75\,T_{14}$. The mid-time shift $\delta$ is scanned on a grid with step $\max(T_{14}/40,\, 2\,\text{min})$ over $[-rT_{14},\, rT_{14}]$. For each $\delta_k$ the linear model

$$f(t) = a + b\,(t - T_c) - d\; s\big(t - T_c - \delta_k\big) \tag{25.2}$$

is solved by weighted least squares (weights $1/\sigma_f^2$), giving $\chi^2_k$ and the covariance $C_k$. The best grid point $k^*$ minimises $\chi^2_k$. With $c_0, c_1, c_2 = \chi^2_{k^*-1}, \chi^2_{k^*}, \chi^2_{k^*+1}$ and curvature $\kappa = c_0 - 2c_1 + c_2 > 0$, the sub-step position is

$$\delta^* = \delta_{k^*} + \Delta\delta\;\text{clip}\Big(\frac{c_0 - c_2}{2\kappa}, -1, 1\Big). \tag{25.3}$$

Uncertainties follow from the fit at $k^*$ with $\chi^2_{\text{red}} = \chi^2/(n-3)$ and inflation factor $\mathcal{I} = \max(\chi^2_{\text{red}}, 1)$:

$$e_d = \sqrt{C_{dd}\,\mathcal{I}}, \qquad \text{SNR} = d/e_d, \tag{25.4}$$

$$\Delta\chi^2 = (\chi^2_{\text{null}} - \chi^2_{k^*})/\mathcal{I}, \qquad e_{T_c} = \Delta\delta\,\sqrt{2\mathcal{I}/\kappa}. \tag{25.5}$$

Here $\chi^2_{\text{null}}$ is the fit without a transit. If the best shift lies at the edge of the search range, or the curvature is not positive, $e_{T_c}$ is set from the interval $\chi^2 \le \chi^2_{k^*} + \mathcal{I}$, or to NaN at the edge; such epochs are marked "at edge of search range" in the plot and are not used for the refinement.

## 25.4 Epoch accounting

Every epoch with $T_c$ in $[t_{\min} - h,\; t_{\max} + h]$ is accounted for. Epochs that cannot be used are listed with the reason:

- *no data*: no cadence within $h$;
- *too few points*: fewer than `min_pts_per_transit` (6);
- *no baseline anchor*: fewer than three points outside $0.5\,T_{14} + rT_{14}$;
- *no data in the transit window*: fewer than three points inside it;
- *bottom not sampled at the best shift*: the profile maximum in the data is below `min_coverage` (0.8);
- *singular fit.*

This turns "the detection did not work" into a concrete list: which transits fall in gaps and which are lost to the quality cuts.

## 25.5 Refinement of $P$ and $T_0$

Epochs with SNR $\geq$ `per_transit_min_snr` (3) and a finite $e_{T_c}$ define the offsets $y_i = \delta_i^*$ at epochs $e_i$. At least three such epochs spanning at least two epochs are required. The weighted line $y = a + b\,e$ with $\sigma_i = \max(e_{T_c,i},\, 0.01\,T_{14})$ gives

$$P' = P + b, \qquad T_0' = T_0 + a, \tag{25.6}$$

with errors scaled by $\sqrt{\max(\chi^2_{\rm red}, 1)}$. If the line fits poorly ($\chi^2_{\rm red} >$ `refine_chi2_max`, default 4), the ephemeris is *not* updated and the log says so; a poor fit indicates transit timing variations or false detections, and a linear ephemeris would then be wrong. After a successful update the epochs are fitted again with the new ephemeris, and the new $P, T_0$ are used for the fold, the difference imaging and TRICERATOPS. Options: `--no-refine-ephemeris`, `--no-tc-fit`.

## 25.6 Odd/even and P/2 tests

The odd/even test of Chapter 9 is unchanged. A new test measures the depth at phase 0.5 (fixed mid-time, no shift search, to avoid a look-elsewhere effect) by running the same estimator with $T_0 + P/2$. A significant depth there means either that the true period is $P/2$ or that the signal is an eclipsing binary with a secondary eclipse; together with the odd/even statistic it addresses the $P$, $P/2$ and $2P$ aliases, which is the central question when the `EBx2P` scenarios dominate.

## 25.7 Adopted depth

After the per-transit stage the depth used downstream is the weighted mean of the individual depths if their constant-depth $\chi^2_{\rm red} < 5$; otherwise the fold/TLS/known value is kept. The log states which one was used and why. At low S/N individual epochs often return depths near zero with underestimated formal errors, and the inverse-variance mean would then understate the depth of the stacked transit.

## 25.8 Outputs

`per_transit_depths_<TIC>.txt` has the columns `epoch Tc dTc e_dTc depth e_depth snr npts`. The figure `per_transit_depths_<TIC>.pdf` has two panels: depth versus nominal mid-time (odd and even epochs distinguished, weighted mean and error band) and the measured minus nominal mid-time in minutes.

## 25.9 Validation on synthetic data

Synthetic light curve: $P = 132.424$ d, $T_{14} = 8.2$ h, depth 1900 ppm, 11 observed transits in sector-like chunks, 600 ppm noise per 10-min point, and a nominal ephemeris that is wrong by $\Delta P = -0.004$ d and $\Delta T_0 = -0.073$ d. For one realization (fixed seed):

- the old estimator (fixed $T_c$, soft box) returned 188–1401 ppm with a median of about 1100 ppm, with mid-time errors growing to $\sim$180 min;
- the new estimator returned 1681–2154 ppm (median about 2000 ppm), and recovered the true mid-times to within 16 min;
- the refinement gave $P = 132.42334 \pm 0.00041$ (truth 132.42400) and $T_0 = 1767.2064$ (truth 1767.2000), with $\chi^2_{\rm red} = 1.65$; after refinement the mean $|\delta^*|$ was 7.6 min and the weighted depth 1997 ppm;
- adding 0.04 d Gaussian scatter to the mid-times raised $\chi^2_{\rm red}$ to 42 and the ephemeris was correctly *not* updated;
- the depth at phase 0.5 was $85 \pm 55$ ppm (consistent with zero).

These results hold for this noise model and a single realization; they demonstrate the logic, not the achievable precision on real FFI data. When the true offset exceeded the search range ($0.5\,T_{14}$) the affected epochs were flagged as at the edge of the range and excluded from the refinement.

# Chapter 26

# Pooled evidences, convergence and the verdict

## 26.1 Monte Carlo evidences

For every scenario $k$ and run $r$, `calc_probs` returns the log-evidence

$$\ln \hat{Z}_k^{(r)} = \ln \frac{1}{N} \sum_{i=1}^{N} \exp\bigl(\ln L_i + \ln \pi_i\bigr), \tag{26.1}$$

where $N$ is the number of Monte Carlo draws ($10^6$ by default), $L_i$ the Gaussian likelihood of draw $i$ and $\pi_i$ the companion prior. Each run is an unbiased estimate of the same $Z_k$. When the likelihood is narrow (high S/N, small $\sigma$) a handful of draws dominates each sum, and $\ln \hat{Z}_k^{(r)}$ scatters by several units between runs. In the TIC 165501544 run of October 2026 the leading scenario evidences were within 3–5 units of each other and the per-run FPP ranged from $\sim 0$ to $\sim 0.95$.

## 26.2 Why averaging per-run FPPs is poor

FPP is a ratio of evidences. The mean of the ratio over runs is neither the ratio of the means nor an efficient estimator: runs in which a false-positive scenario happens to draw a good sample give a large FPP, others a small one, and their average has a large variance and a bias. The mean was also reported with a "standard deviation" that mixes Monte Carlo noise with nothing physical.

## 26.3 Pooling

For runs with the *same* flux uncertainty the pooled log-evidence is

$$\ln \hat{Z}_k = \ln\Bigl[\frac{1}{R} \sum_{r=1}^{R} e^{\ln \hat{Z}_k^{(r)}}\Bigr] = m_k + \ln\Bigl[\frac{1}{R} \sum_{r} e^{\ln \hat{Z}_k^{(r)} - m_k}\Bigr], \quad m_k = \max_r \ln \hat{Z}_k^{(r)}, \tag{26.2}$$

which is the estimate of a single run with $RN$ draws (`_pool_lnZ`). The scenario probabilities are $p_k = \mathrm{softmax}(\ln \hat{Z})$ and, with the indices of triceratops 1.1,

$$\mathrm{FPP} = 1 - (p_0 + p_3 + p_9), \qquad \mathrm{NFPP} = \sum_{k \geq 15} p_k, \tag{26.3}$$

where indices 0, 3, 9 are the TP, PTP and DTP scenarios (`_probs_from_lnZ`). These index conventions follow the installed package; if a future version changes the scenario order, they must be re-checked. Runs with different $\sigma$ (after a retry) are *not* pooled, because they have different likelihoods.

## 26.4 Uncertainty and convergence check

The uncertainty on FPP and NFPP is a bootstrap over runs (`tri_boot`, default 300 resamples with replacement). It describes the Monte Carlo precision of the pooled estimate, not a systematic error. The convergence diagnostic is the largest share one run has in the pooled evidence of the most probable scenario $k^*$:

$$w_{\max} = \max_r \frac{e^{\ln \hat{Z}_{k^*}^{(r)} - \ln \hat{Z}_{k^*}}}{R}. \tag{26.4}$$

For a converged Monte Carlo $w_{\max} \approx 1/R$; for a single run that decides everything $w_{\max} \to 1$. If $w_{\max} > 0.5$ the log states that the Monte Carlo has not converged and recommends raising `--n-samples` and/or `--n-iter`; the verdict is then marked *provisional*.

## 26.5 Verdict

The verdict is built from the pooled values: INDETERMINATE if fewer than `tri_min_valid_frac` (0.8) of the runs are valid or FPP/NFPP are not finite; PASSED if FPP $< 0.015$ and NFPP $< 0.001$; otherwise NOT PASSED. The log also reports the mean of the per-run values and its scatter for comparison. `--no-pool` restores the previous behaviour.

## 26.6 Sensitivity scan

With `--sensitivity` the script repeats `calc_probs` at flux uncertainties $\sigma \times \{0.5, 1, 2, 4\}$. A single run per scale was found to be too noisy to interpret; `--sens-runs` (default 3) runs are now pooled per scale. The run at scale 1 is executed *last* so that the target object keeps the adopted solution for `plot_fits`.

## 26.7 Verification of star removal

`target.remove_star` matches the supplied identifiers as strings. The script now compares the requested identifiers with `target.stars['ID']` (as strings) after the call; any star still present is removed by hand and `calc_depths` is repeated. The log reports `removed k of n requested star(s)`. Previously the number of requested stars was reported

as “removed”, which could not reveal a failed removal. The same check protects against identifier type mismatches (integer versus string columns).

## 26.8 Example log excerpt

```
valid runs: 20/20; final sigma = 600 ppm
FPP (mean of per-run values) = 0.xxxxx +/- 0.xxxxx [run-to-run scatter]
Pooled evidences of 20 runs at sigma = 600 ppm (= one run with 20 x N draws):
FPP = 0.xxxxx +/- 0.xxxxx [bootstrap over runs]
  most probable scenarios (index: prob, lnZ): 0: ..., 2: ...
  largest share of a single run in the leading scenario's evidence: 0.xx
  (1/n_runs = 0.05 when converged)
Giacalone+2021 validation criteria (...; based on pooled evidences): ...
```

The values are placeholders showing the format.

# Chapter 27

# NFPP and the meaning of tdepth = 0

## 27.1 Definition

For the target aperture the observed dip is $d_{\rm ap} = d\,F_0$ when the depth passed to `calc_depths` is dilution corrected ($F_0$ = flux ratio of the target in the aperture), and $d_{\rm ap} = d$ if the flux is uncorrected (`--raw-aperture-flux`). For every star $i$ with flux ratio $F_i$ in the aperture, `calc_depths` defines the depth that star itself would need,

$$\mathrm{tdepth}_i = \frac{d_{\rm ap}}{F_i}, \tag{27.1}$$

and sets it to 0 if $\mathrm{tdepth}_i > 1$. A star with $F_i < d_{\rm ap}$ cannot reproduce the dip, even with a total eclipse. Only stars with $\mathrm{tdepth}_i > 0$ enter the NTP, NEB and NEBx2P scenarios, so

$$\mathrm{NFPP} = \sum_{i:\,\mathrm{tdepth}_i>0} \left(p^i_{\rm NTP} + p^i_{\rm NEB} + p^i_{\rm NEBx2P}\right). \tag{27.2}$$

Consequently NFPP = 0 can follow from the flux budget alone. It does not mean that follow-up was assumed. Difference imaging, neighbour vetting and ground-based photometry can only *lower* the NFPP of stars with tdepth > 0, through `remove_star`.

## 27.2 Log summary

`nfpp_input_summary` writes, after the table of field stars: the adopted depth, the implied dip in the aperture, the minimum neighbour flux ratio needed (= $d_{\rm ap}$), the number of neighbours in the table and how many still have tdepth > 0 (plus the number removed by vetting), the brightest neighbour with the percentage of its own flux it would have to eclipse, and a table of the neighbours that can host the signal.

## 27.3 Examples

**WASP-18b (TIC 100100827)** Dip of 9916 ppm in the aperture. The brightest neighbour (TIC 100100825, $T_{\rm mag}$ = 14.35, 52.5″) has a flux ratio of $2.0 \times 10^{-3}$ and would need an

eclipse of $\sim$499 % of its flux. No neighbour can host the signal; NFPP = 0 is a consequence of the flux budget.

**TIC 165501544** With a dip of $\sim 1.9 \times 10^{-3}$ four neighbours remain able to host it; the brightest of them (TIC 332477346, $T_{\mathrm{mag}} = 15.3$, $110''$, flux ratio $1.7 \times 10^{-2}$) needs an eclipse of about 11 % of its flux. An NFPP below $10^{-3}$ then requires exclusion of these stars by difference imaging or follow-up photometry.

## 27.4 Consequences in the code

`vet_nearby_stars` skips stars with tdepth $\leq 0$: previously the value 0 was read as a predicted neighbour depth of zero, producing meaningless exclusion flags. In `main` only stars with tdepth $> 0$ are passed to `remove_star`; the number of ignored stars is logged.

# Chapter 28

# The limb-darkened model and its use

## 28.1 Dependency-free model

The model is a quadratic limb-darkened occultation of a uniform-disk planet over a star with specific intensity $I(\mu) = 1 - u_1(1-\mu) - u_2(1-\mu)^2$, $\mu = \sqrt{1-r^2}$ (`quad_ld_transit`). The stellar disk is cut into $K$ annuli with edges $r_k = \sqrt{1-\mu_k^2}$ and $\mu_k$ equally spaced between 1 and 0 (dense near the limb), $K = 150$ by default. With the exact overlap area of two discs,

$$A(R;z,p) = R^2 \arccos\frac{z^2+R^2-p^2}{2zR} + p^2 \arccos\frac{z^2+p^2-R^2}{2zp} - \tfrac{1}{2}\sqrt{(-z+R+p)(z+R-p)(z-R+p)(z+R+p)}, \quad (28.1)$$

(with the limiting cases 0 for $z \geq R+p$ and $\pi \min(R,p)^2$ for $z \leq |R-p|$), the flux is

$$F = 1 - \frac{\sum_k I_k \left[A(r_{k+1}) - A(r_k)\right]}{\sum_k I_k \, \pi \left(r_{k+1}^2 - r_k^2\right)}, \quad (28.2)$$

so that the normalisation is consistent with the discretisation. For a circular orbit the separation is $z(t) = a\sqrt{\sin^2\varphi + (\cos i \cos\varphi)^2}$ with $\varphi = 2\pi t/P$ and $b = a\cos i$; points on the far side of the orbit ($\cos\varphi \leq 0$) are set to no transit.

## 28.2 Validation

Against `batman` (Kreidberg 2015) with $a/R_\star = 3.5$, $u = (0.30, 0.25)$ the largest difference over a transit was below 1 ppm for $K \geq 150$ at $b = 0.36$, 0.9, 1.05 and for $p = 0.0975$, 0.12; for $p = 0.3$, $b = 0$ it was 0.7 ppm at $K = 300$. At $K = 100$ the error was 0.2 ppm. Each model evaluation for a few hundred time stamps takes of the order of 10 ms.

## 28.3 Fit

Free parameters: $p = R_p/R_\star$, $b$, $a/R_\star$, a mid-time shift (within $\pm 0.3\,T_{14}$), a baseline $f_0 \in [0.99, 1.01]$, and, in mode `free`, Kipping's $q_1, q_2 \in [0.01, 0.99]$ with $u_1 = 2\sqrt{q_1}\,q_2$ and $u_2 = \sqrt{q_1}(1-2q_2)$.

The period is fixed. Three starting values $b_0 \in \{0.2, 0.55, 0.85\}$ are used, with $a/R_\star$ from $T_{14} \approx \frac{P}{\pi}\frac{\sqrt{(1+p)^2-b^2}}{a/R_\star}$; the best of the three is kept. The covariance of the parameters is $(J^T J)^{-1}$ scaled by the reduced $\chi^2$ with respect to the out-of-transit scatter. Derived quantities are

$$T_{14} = \frac{P}{\pi}\arcsin\frac{\sqrt{(1+p)^2-b^2}}{(a/R_\star)\sin i}, \qquad \rho_\star = \frac{3\pi(a/R_\star)^3}{GP^2}. \tag{28.3}$$

$T_{23}$ uses $1-p$. The density assumes a circular orbit; for eccentric orbits it differs from the true density by the usual factor depending on $e$ and $\omega$, so a $\rho_\star$ different from the TIC value is not by itself a contradiction. The log lists $\rho_\star$ from the fit next to $\rho_\star = 1.41\,M/R^3$ g cm$^{-3}$ from the TIC mass and radius.

## 28.4 Adoption of depth and duration

When `--ld-fit` is active, the LD depth and $T_{14}$ may overwrite the adopted values used by `calc_depths`, the difference imaging and TRICERATOPS. This is guarded: the update happens only if the fitted T0 shift is smaller than `ld_update_max_t0shift` $= 0.15\,T_{14}$. A larger shift indicates that the fold is smeared by an inaccurate $P$ or $T_0$, which makes the profile too shallow and too long; then the log says *adopted depth and T14 are NOT updated*. In the October 2026 run of TIC 165501544 the shift was $0.029 \pm 0.005$ d (41 min, $\approx 6\sigma$) and the fit gave 1903 ppm and 8.7 h, compared with 3228 ppm and 4.8 h adopted; the shift is $0.14\,T_{14}$ for $T_{14} = 0.36$ d, so a refined ephemeris (Chapter 25) should be obtained first.

## 28.5 Fold window and out-of-transit definition

The window of the folded light curve, the baseline normalisation, the clipping and the out-of-transit region of the error estimate use $T_{14}^{\rm fold} = \max(T_{14}^{\rm adopted}, T_{14}^{\rm per\text{-}transit})$. A catalogue duration of 4.8 h while the transit lasts 7.5 h put ingress and egress outside the nominal transit region ($|\phi| > 0.75\,T_{14}$), biased the baseline low and gave a window ($\pm 0.5$ d) that barely contained the transit. The log announces when the larger duration is used.

# Chapter 29

# Detection, detrending and the quality of the folded light curve

## 29.1 Simulation study of the first-pass detrending

When TLS first failed to find the long-period signal, the detrending was suspected. To test this without access to the raw data, a simulated data set was built with a transit of $P = 132.42$ d, $T_{14} = 7.5$ h, depth 1900 ppm; about 13 transits in 14 sector-like data sets with two orbits each; 10-min cadence with 2400 ppm white noise; AR(1) red noise with 1200 ppm rms and a correlation time of 1 d; and ramps of 2500 ppm (start) and −1500 ppm (end) with time constants of 0.5 and 0.4 d at the ends of every orbit. TLS ran on 30-min bins between 100 and 170 d. Detection was counted as successful if the period was within 1 d of the truth. Eight noise realizations:

| Detrending (per sector) | Found | Median SDE | Core depth (ppm) |
|---|---|---|---|
| Biweight 0.75 d | 8/8 | 13.9 | 830 |
| Biweight 1.0 d | 8/8 | 13.8 | 1019 |
| Biweight 1.5 d (previous default) | 6/8 | 12.9 | 1233 |
| Biweight 2.5 d | 7/8 | 9.4 | 1337 |
| 1.5 d with the true transits masked (oracle) | 6/8 | 16.1 | 1614 |
| Two-stage: 1.5 d, events found, 2.0 d with masks | 5/8 | 11.6 | 1346 |
| Biweight 1.5 d with 0.5 d edge cut | 6/8 | 11.8 | 1204 |
| Biweight 1.0 d with 0.5 d edge cut | 8/8 | 12.2 | 1060 |

(The first five rows refer to two series of runs; the median SDE differs slightly between series.)

Findings:

- Shorter windows detect more reliably even though less of the depth is left; the failures are *false peaks* (here ∼106 d, only 3 transits with data, $T_{14} \approx 1.6$ h) produced by residual red noise combined with the sampling.
- The oracle with the true transits masked fails as often as the plain 1.5 d window, so depth suppression by the filter is *not* the cause of the failures; residual correlated noise is.

- Masking events found automatically did not help: the event finder triggers on noise and ramps.
- The wotan `edge_cutoff` had little effect, and the fraction of the in-transit cadences of the false peak near data edges was 0. The initial hypothesis that orbit-edge ramps cause the false peaks is therefore not supported by this simulation.

These are statements about a *simulated* noise model; they motivate the tools below but must be re-checked on the real light curve.

## 29.2 Multi-window blind search

`search_multi_window` detrends all sectors with each window in `window_length_grid`, stitches, clips, bins and runs TLS, keeps the highest-SDE result, and leaves the sectors detrended with the winning window. The log lists for each window: $P$, $T_{14}$, depth, SDE, number of transits, edge fraction and whether $P$ agrees (within 1 %) with the best. If the best period is found with a single window only, the log flags it as a probable red-noise or sampling alias. The second pass (masked, window = `window_factor` $\times T_{14}$) restores the depth afterwards. Cost: one TLS run per window. Usage: `--detrend-windows 0.75 1.0 1.5 2.5`.

In the simulation the single window of 1.5 d found the false peak for seed 2 (SDE 11.0, $P = 105.93$ d, 3 transits); the grid selected $P = 132.409$ d with SDE 14.6 from the 0.75 d window and found the same period with the 1.0 and 2.5 d windows.

## 29.3 Candidate edge fraction

`candidate_edge_fraction` computes the fraction of the in-transit cadences that lie within 0.5 d of the edge of a contiguous data segment (a gap longer than 0.5 d starts a new segment). For orbits of ∼13 d a candidate unrelated to the edges has about $2 \times 0.5/13 \approx 8\,\%$; in the simulation the correct candidate had 7 %. A fraction above 30 % is flagged as suspicious. The diagnostic is logged for the first-pass candidate in every blind search.

## 29.4 Cadences with blown-up errors

In the folded light curve of TIC 165501544 about 30 of ∼2100 unbinned points had formal errors of 10–15 % against a typical 0.2 %, presumably from scattered light or bad pixels. Such points carry no information, and in a plot with error bars they stretched the vertical axis until the real dip of ∼1900 ppm disappeared. `extract_sector` now rejects cadences whose *relative* error exceeds `err_clip_factor` (5) times the sector median and reports the number per sector; set it to 0 to disable.

## 29.5 Folded light-curve plot

`plot_folded_lc` replaces the former figure. The vertical range follows the binned points ($\pm 4$ robust rms outside the extremes), the unbinned points have no error bars, the lines $\pm T_{14}/2$ and the zero phase are marked, the limb-darkened model is overplotted when fitted, and a lower panel gives the residuals. The number of unbinned points outside the range is in the title.

## 29.6 Recommended workflow for a faint, long-period signal

1. Blind search with `--blind-search --detrend-windows 0.75 1.0 1.5 2.5` and a period range that contains the plausible periods; inspect the table of windows and the edge fractions.
2. Rerun with the best candidate as `--ephemeris P T0 DEPTH T14`, so that the second pass uses a mask; inspect the per-transit table, the O$-$C panel, the odd/even and phase-0.5 statistics (Chapter 25).
3. Let the ephemeris be refined; check that the LD-fit T0 shift is small.
4. Run TRICERATOPS with `--ld-fit --sensitivity` and `--n-samples` chosen such that the log reports a dominance well below 0.5.

# Chapter 30

# Configuration and command-line reference

## 30.1 New and changed configuration fields

| Field | Default | Meaning |
|---|---|---|
| `window_length_search` | 1.5 d | First-pass biweight window. |
| `window_length_grid` | None | List of first-pass windows; blind search runs for each (Chapter 29). |
| `err_clip_factor` | 5.0 | Reject cadences with relative flux error above this multiple of the sector median; 0 disables. |
| `per_transit_fit_tc` | True | Fit the mid-time of every transit. |
| `per_transit_tc_range` | 0.5 | Search range of the mid-time, in $T_{14}$. |
| `per_transit_min_snr` | 3.0 | Minimum SNR of transits used for the ephemeris. |
| `refine_from_transits` | True | Refine $P, T_0$ from the measured mid-times. |
| `refine_chi2_max` | 4.0 | Do not update if the linear ephemeris fits worse. |
| `ld_update_max_t0shift` | 0.15 | Largest T0 shift (in $T_{14}$) for which the LD fit may update depth and $T_{14}$. |
| `tri_N` | None | Monte Carlo draws per scenario (None: package default $10^6$). |
| `pool_evidences` | True | Derive FPP/NFPP from pooled evidences. |
| `tri_boot` | 300 | Bootstrap resamples over runs. |
| `tri_sens_runs` | 3 | Runs per scale in the sensitivity scan. |
| `use_known_ephemeris` | True* | Skip TLS; use the next four fields. |
| `P_orb_known, T0_known` | 145.841 d, 1714.174 | Known period and epoch (BTJD).* |
| `depth_known, dur_known` | 863 ppm, 0.5625 d | Known depth and duration.* |
| `P_min, P_max` | 130, 135 d* | TLS period range. |

| Field | Default | Meaning |
|---|---|---|
| SDE_threshold | 3.0* | Detection threshold. |

* Values of the uploaded file; they belong to one candidate and are not suitable as general defaults. Use `--ephemeris` or `--blind-search` to avoid applying them to another target.

## 30.2 Command-line options

Table 30.2: All command-line options of the current script.

| Option | Type, default | Effect |
|---|---|---|
| `--tic` | int | TIC identifier. |
| `--outdir` | path, `TIC<tic>/` | Output directory. |
| `--n-iter` | int, 20 | Number of `calc_probs` runs. |
| `--contrast-curve` | path | High-resolution contrast curve. |
| `--vet-neighbours` | flag | Neighbour light-curve vetting. |
| `--no-difference-imaging` | flag | Disable difference imaging. |
| `--download-dir` | path | TESScut cache. |
| `--trilegal-fname` | path | Pre-downloaded TRILEGAL table. |
| `--err-floor-ppm` | float, 0 | Lower bound on the flux uncertainty. |
| `--no-inflate` | flag | No inflation to $\chi^2_{\rm red} = 1$. |
| `--sensitivity` | flag | Extra runs at scaled $\sigma$ (pooled). |
| `--sens-runs` | int, 3 | Runs per scale for `--sensitivity`. |
| `--raw-aperture-flux` | flag | `dilution_corrected=False` (triceratops $\geq$ 1.1). |
| `--ld-fit` | flag | LD fit, $u_1, u_2$ from Claret values. |
| `--ld-free` | flag | LD fit with fitted $q_1, q_2$. |
| `--ld-coeffs U1 U2` | 2 floats | LD fit with fixed coefficients; excludes `--ld-free`. |
| `--n-samples N` | int | Monte Carlo draws per scenario. |
| `--no-pool` | flag | Mean of per-run FPPs instead of pooled evidences. |
| `--no-refine-ephemeris` | flag | Do not refine $P, T_0$ from the transit times. |
| `--no-tc-fit` | flag | Mid-time of each transit fixed at the ephemeris. |
| `--ephemeris P T0 DEPTH_PPM T14_H` | 4 floats | Known ephemeris for this target (skips TLS); excludes `--blind-search`. |
| `--blind-search` | flag | Run TLS even if `use_known_ephemeris` is set. |
| `--detrend-windows D1 D2 ...` | floats | First-pass windows for the multi-window search. |

## 30.3 Examples

```
# known planet as a control (set P_min, P_max in Config to 0.5, 1.5 first)
python triceratops_validation.py --tic 100100827 --blind-search \
    --n-iter 5 --no-difference-imaging --ld-fit --err-floor-ppm 200 \
    --sensitivity --trilegal-fname trilegal_TIC100100827.dat \
    --outdir runs/WASP18b

# faint long-period candidate, ephemeris given for this target
python triceratops_validation.py --tic 165501544 --n-iter 20 \
    --ephemeris 132.42 1767.127 3228 4.8 \
    --vet-neighbours --ld-fit --sensitivity --n-samples 3000000 \
    --contrast-curve contrast_curve.csv --raw-aperture-flux

# blind search with several detrending windows
python triceratops_validation.py --tic 165501544 --blind-search \
    --detrend-windows 0.75 1.0 1.5 2.5
```

# Chapter 31

# Function-by-function additions

This chapter supplements Chapter 20. Names are as in the script.

- `_unit_shape_func(results, shape_model, time, P, T0, dur)`
  Returns the unit-depth profile $s(\phi)$ and a string naming its source (supplied model, TLS model, soft box).
- `per_transit_depths(time, flux, flux_err, results, cfg, P, T0, dur, shape_model=None, return_rejected=False, fit_tc=None)`
  Estimator of Chapter 25. Returns the structured array `dt` with fields `epoch, Tc, depth, e_depth, npts, coverage, dTc, e_dTc, snr, dchi2`, and, optionally, the list of rejected epochs with reasons. `results` may be `None`.
- `refine_ephemeris_from_transits(dt, P, T0, dur, cfg)`
  Weighted linear ephemeris from the measured mid-times; returns a dictionary with `updated`, `P`, `T0`, errors, $\chi^2_{\rm red}$, the O−C residuals and a reason string.
- `plot_per_transit_depths(dt, diag, cfg, tic)`
  Two panels: depths and measured − nominal mid-time.
- `_probs_from_lnZ`, `_pool_lnZ`, `_pooled_summary`
  Scenario probabilities, FPP and NFPP from log-evidences; log-mean-exp pooling; bootstrap and dominance diagnostic (Chapter 26).
- `_calc_probs_checked(target, lc, P_orb, sigma, cfg, max_retries=None)`
  As before, now passing `N=cfg.tri_N` when set.
- `run_triceratops(...)`
  Verifies star removal, runs the checked iterations, pools the evidences, logs both the pooled and the mean values, performs the pooled sensitivity scan, and returns `(target, FPPs, NFPPs, sigma_used)`; the arrays contain NaN for invalid runs.
- `nfpp_input_summary(target, depth, cfg, log, n_removed)`
  Log section of Chapter 27.
- `segment_edge_distance(time, gap=0.5)`
  Distance of every cadence to the nearest edge of its contiguous data segment.

- `candidate_edge_fraction(time, P, T0, dur, within=0.5, gap=0.5)`
  Fraction of the in-transit cadences within `within` days of a segment edge.
- `search_multi_window(sectors_data, star, cfg, log)`
  Multi-window blind search; returns the best TLS result, the chosen window, the stitched light curve and the table of results.
- `_lens_area`, `quad_ld_transit`, `limb_darkened_lc`
  The dependency-free limb-darkened transit model of Chapter 28.
- `plot_folded_lc(tp, fp, lc, cfg, tic, dur, model, model_label)`
  Folded light curve with residuals.
- `triceratops_flux_error(...)`
  Unchanged algorithm; called with the larger of the adopted and the measured duration.
- `extract_sector`
  Rejects cadences with relative errors above `err_clip_factor` times the median.
- `main(cfg)`
  Changed order of the central steps: (1) log of the ephemeris used; (2) per-transit analysis in all modes, with the trapezoid shape in known-ephemeris mode; (3) ephemeris refinement and second pass; (4) P/2 test; (5) adopted depth rule; (6) fold with $T_{14}^{\mathrm{fold}}$; (7) optional LD fit with guard; (8) $\sigma$; (9) TRICERATOPS target, difference imaging and vetting with verified removal; (10) `run_triceratops`.

# Chapter 32

# Worked example: TIC 165501544

## 32.1 Situation

TIC 165501544 is a faint long-period candidate ($P \approx 132$ d) with about ten sectors that contain a transit. The first complete run in known-ephemeris mode produced the following (October 2026 log).

- Limb-darkened fit of the fold: $R_p/R_\star = 0.0462$ ($\approx 6.6\,R_\oplus$), $b = 0.877 \pm 0.056$, $a/R_\star = 66 \pm 14$, T0 shift $0.0287 \pm 0.0048$ d, $\rho_\star = 0.314$ against 0.641 g cm$^{-3}$ from the TIC. Since $(66/84)^3 \approx 0.49$ for $a/R_\star = 84$ expected on a circular orbit, the density is consistent within $1.3\sigma$.
- The LD fit replaced the adopted depth (3228 → 1903 ppm) and $T_{14}$ (4.80 → 8.71 h). With the guard of Chapter 28 that update would not have been made, because of the 41-min shift.
- TRICERATOPS: FPP $= 0.45 \pm 0.30$, NFPP $= 0.089 \pm 0.118$ over 20 runs; leading evidences $\ln Z$ of EBx2P ($-73.4$), TP ($-76.6$), STP ($-77.6$), PTP and SEB ($-78.4$); the sensitivity scan (single runs) gave FPP 0.43, 0.95, 0.21, 0.19 for $\sigma \times 0.5, 1, 2, 4$ without a monotone trend. This is the signature of Monte Carlo noise, not of a $\sigma$ dependence, and motivated the pooling of Chapter 26.
- Difference imaging: only three sectors (17, 60, 85) had enough SNR (7.8, ∼20, ∼20; position errors 1.6–2.2″ in two of them); the other sectors had SNR $< 3$ and position errors of 30–60″, which constrain nothing.

## 32.2 The folded light-curve figure

The folded plot appeared to show nothing. The binned points and error bars were extracted from the vector PDF and quantified:

- out-of-transit scatter per 10-min bin: 625 ppm;
- a dip with a flat bottom of ≈ 1700–1900 ppm between $-0.10$ and $+0.15$ d, ingress starting near $-0.13$ d and egress ending near $+0.18$ d: a duration of about 7.5 h, centred near $+0.03$ d (about 40 min after the ephemeris), in agreement with the LD fit;

- 30 of the ∼2300 error bars had half-lengths above 0.05, against a typical 0.0019: cadences with blown-up errors that stretched the vertical axis (0.83–1.15) and hid the dip.

The dip is real in the data. The causes of the apparent failure were the plot (Chapter 29) and the unfiltered cadences; the shift from the nominal $T_0$ shows that the ephemeris needs refinement.

## 32.3 Suggested next run

```
python triceratops_validation.py --tic 165501544 --n-iter 20 \
    --ephemeris 132.42 1767.127 1900 7.5 \
    --ld-fit --sensitivity --n-samples 3000000 --vet-neighbours
```

and inspect in this order: the per-transit table and O−C panel, the refined ephemeris, the phase-0.5 depth, the LD-fit T0 shift (should now be small), the NFPP summary, the dominance line of the pooled evidences, and `plot_fits`.

# Chapter 33

# Tests and verification status

## 33.1 What was tested

The changes were exercised with synthetic data and mock objects. No full run with TESScut downloads, TRILEGAL and the real `calc_probs` was possible in the development environment.

- **Error model and checked runs.** Mock targets with all-$-\infty$ evidences below a $\sigma$ threshold, with and without the attributes `lnZ` and `FPP_degenerate` (older package versions), retries, indeterminate verdict when too few runs are valid.
- **Pooling.** Equality of the pooled evidence with the direct mean of $e^{\ln Z}$, handling of degenerate inputs, the dominance warning with strongly noisy evidences, `--no-pool`, pooled sensitivity scan, passing of `N`.
- **Star removal.** A mock target whose identifier column is integer (so that string matching fails): the manual removal is triggered and logged; with string identifiers it is not.
- **Per-transit estimator.** Synthetic light curve with a wrong ephemeris (Chapter 25); refinement; refusal to refine when the times are scrambled; phase-0.5 test; plot.
- **NFPP summary.** Run on the star table of the WASP-18 log.
- **Limb-darkened model.** Comparison with `batman`; recovery of $R_p/R_\star$, $b$, $a/R_\star$ and $T_{14}$ on batman-generated data in the three limb-darkening modes; effect on $\sigma$: with a trapezoid reference the inflation gave $\sigma = 343$ ppm, with the LD reference 34 ppm (for 27 ppm formal error).
- **Detection.** Simulation study of Chapter 29 with real `wotan` and TLS; the multi-window search on the two noise realizations on which the 1.5 d window failed.
- **Command line.** Mutual exclusion of `--ld-free` and `--ld-coeffs`, and of `--ephemeris` and `--blind-search`; help text.
- **Plot.** The folded-light-curve plot with data recovered from the uploaded figure.

## 33.2 What was not verified

- A complete pipeline run on real data after these changes.
- The cause of the inconsistency in the October 2026 log, where TIC 165501538 and 165501539 appeared in the list of stars excluded by difference imaging yet were still present in the TRICERATOPS field. The new verification step logs whether `remove_star` worked, but the original cause was not reproduced.
- The behaviour of the multi-window search on the real light curve; the simulation uses an assumed noise model.
- The scenario indices used for FPP and NFPP when pooling were taken from triceratops 1.1.0; they must be checked if another version is installed.

# Additional references


1. Kreidberg, L. 2015, "batman: BAsic Transit Model cAlculatioN in Python," Publications of the Astronomical Society of the Pacific, 127, 1161.
2. Mandel, K., & Agol, E. 2002, "Analytic Light Curves for Planetary Transit Searches," The Astrophysical Journal, 580, L171.
3. RAVEN: RAnking and Validation of ExoplaNets, arXiv:2509.17645; and "Automatic search for transiting planets in TESS-SPOC FFIs with RAVEN," arXiv:2603.22597 (MNRAS 2026). An independent, machine-learning validation method for TESS candidates (periods 0.5–16 d, depth $> 300$ ppm, SPOC FFI light curves up to sector 55, validation limited to $R < 8\,R_\oplus$).

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# Chapter 34

# Addendum 2: summary of the version 3.2 changes

This second addendum documents the changes made to `triceratops_validation.py` in late October 2026, after the first real-data run on TIC 229788204 (Chapter 42). That run exposed two independent problems: a known ephemeris that was slightly wrong, and TESScut files that the installed `triceratops` package could no longer read. Both are solved; in addition a folded grid search for the ephemeris is available as an option.

## 34.1 What changed

1. **A known ephemeris is checked against the data** before the transit-masked detrend (Chapter 35). Every expected transit is searched with a box of width $T_{14}$; a robust linear fit of the found mid-times gives $P$ and $T_0$. A period error of only 0.012 d moved the predicted transits of TIC 229788204 by up to 5 h over the TESS baseline; every later stage then measured noise instead of the transit.

2. **Folded $(P, T_0)$ grid search** (Chapter 36), for signals that are too shallow to be seen one transit at a time. It stacks all transits for each grid point and costs seconds, independent of the number of sectors. Selected with `--ephem-search grid`; the default `auto` uses it as a fall-back when the per-transit scan is inconclusive.

3. **TESScut / astrocut $\geq$1.4 compatibility** (Chapter 37). New cutouts lack the physical-pixel keywords `1CRV4P`/`2CRV4P`; `triceratops` then dropped every sector and `calc_depths` failed with `IndexError`. The cutout used by `triceratops` is now placed in the pixel frame of the extraction apertures through the WCS, and sectors that `triceratops` still drops are removed consistently.

4. **Dilution.** TESScut aperture photometry is passed to `triceratops` as *not* crowding-corrected (`dilution_corrected=False`) unless `--dilution-corrected` is given; a warning is printed when the depth handed to `calc_depths` does not match the depth seen in the folded light curve.

5. **Outlier clipping kept the bottom of deep eclipses out** (Chapter 38). Found in the

second run on TIC 229788204: every point deeper than continuum $-15\sigma$ inside the transit window, and deeper than $-5\sigma$ in the first detrending pass, was deleted. Fixed.

6. **Secondary-eclipse search at all orbital phases** (Chapter 39), for eccentric orbits, with a single-event check, $3\sigma$ depth limits and a companion brightness-temperature limit.

7. **Per-transit depths corrected for the dilution in each sector's aperture**, with odd-/even tests overall and per cadence; a warning when the limb-darkened fit is grazing (Chapter 40).

8. **Robustness.** TESScut downloads are retried (MAST answers "502 Bad Gateway" or "Request failed" now and then); per-transit mid-times that end on the edge of their search range are flagged; the log no longer prints the auto-excluded sectors twice.

## 34.2 Corrections to the earlier chapters

| Where | Earlier text | Current behaviour |
|---|---|---|
| §4.4, §20.12.2 | `dilution_corrected` default None; `False` only with `--raw-aperture-flux`. | With `--lc-source tesscut` (default) the script sets `False` itself; `--dilution-corrected` forces `True`. SPOC SAP/PDCSAP runs are unchanged. |
| §4.6, §7.4, §22.4 | The known ephemeris is used as given. | It is checked and, if needed, refined before the second detrend (Chapters 35, 36); `--ephem-search off` restores the old behaviour. |
| §5.3 item 3, §20.3.1 | Aperture pixels are in the full-FFI frame. | Only if the cutout carries the physical WCS keywords. For astrocut $\geq$1.4 files `lightkurve` returns `tpf.column = tpf.row = 0`, so the frame is the cutout frame. The script no longer depends on this (Chapter 37). |
| §12.2 | Garbled quotation marks around "signal on neighbour" and "signal on target". | Read: the neighbour is excluded if the measurement is inconsistent with "signal on neighbour" at more than `neighbour_sigma` standard deviations. |
| §15.1, §20.12.7 | `make_target` maps the apertures to `target.sectors`. | Also fixes the pixel frame of every `triceratops` cutout and removes sectors that `triceratops` skipped ("Ignoring for validation") from `target.sectors`. |
| §20.3 | A raw `\subsection` command appears in the text before `detrend_sector`. | Typesetting error; the paragraph belongs to a subsection "`detrend_sector(sd, window_length, cfg, mask_transits=None)`". |
| §22.1.2 | The last row of the table of automatic cuts is cut off. | `sector_max_outlier_frac` = 0.05: fraction of cadences with $\lvert f-1\rvert >$ `sector_outlier_level` (0.05). |

| Where | Earlier text | Current behaviour |
|---|---|---|
| §6.4, §22.3, `clip_outliers`, `fold_and_bin` | Positive-only clipping without an ephemeris; in transit only "absurd" depths below continuum $-15\sigma$ removed. | The code clipped downward at $5\sigma$ when no ephemeris was given, and the $-15\sigma$ floor removes the bottom of any eclipse deeper than $15\times$ RMS. Now: really positive-only without an ephemeris; in transit only fluxes below 10% of the continuum are removed (Chapter 38). |
| Ch. 25 (Addendum 1) | Mid-time free within $\pm 0.5\,T_{14}$. | Unchanged, but transits whose fitted mid-time ends on that limit are now flagged in the log: a sign that the ephemeris is wrong and the depth is not reliable. |

# Chapter 35

# Checking a known ephemeris: per-transit scan

## 35.1 Why a check is needed

An ephemeris from a catalogue, from ExoFOP or from a quick look in LcTools is rarely exact. With an error $\delta P$ in the period and $\delta T_0$ in the epoch, the predicted mid-time of epoch $n$ is off by

$$\Delta t(n) = \delta T_0 + n\,\delta P, \qquad \sigma_{\Delta t}(n) = \sqrt{\sigma_{T_0}^2 + n^2\sigma_P^2}.$$

TESS baselines now exceed 2000 d. For a 94 d orbit $n$ reaches 20, so a period error of 0.01 d already moves the last transits by 5 h. In known-ephemeris mode every later stage trusts the ephemeris: the transit mask of the second detrend, the per-transit fits (mid-time free within only $\pm 0.5\,T_{14}$), the fold, the limb-darkened fit and the depth handed to `calc_depths`. A wrong ephemeris then produces a smeared, too shallow fold and negative "depths" for individual transits, without any error message.

## 35.2 Algorithm

The check runs on the first-pass light curve (biweight window `window_length_search`, no transit mask), in which the transits are still intact, and before the second detrend.

1. **Box scan per epoch.** For every epoch $e$ with data, trial mid-times $c$ are placed on a grid with step $\max(T_{14}/40,\ 2\text{ min})$ within $T_c \pm R$, where $T_c = T_0 + eP$ and by default $R = \max(2\,T_{14},\ 0.5\text{ d})$ (`--ephem-scan-hours`). For each $c$

   $$d(c) = \text{median}\{f_i : 0.75\,T_{14} < |t_i - c| < 2.5\,T_{14}\} - \langle f_i\rangle_{|t_i-c|<T_{14}/2}, \qquad \text{S/N}(c) = \frac{d(c)}{\sigma_{\text{MAD}}/\sqrt{n_{\text{in}}}},$$

   with $\sigma_{\text{MAD}}$ the robust scatter of the baseline points. At least three in-transit points and two baseline points on each side are required. The $c$ with the largest S/N is refined by a parabola through its neighbours. A maximum on the edge of the range is flagged and not used.

2. **Robust linear fit.** Epochs with S/N $\geq$ `ephem_scan_min_snr` (5) enter a pair-consensus fit: every pair of epochs defines a line $T_c(e)$; the line that puts most epochs within $0.15\,T_{14}$ wins (ties: largest summed S/N). A least-squares fit of $T_c = T_0 + eP$ to those epochs gives $P$ and $T_0$; their errors follow from the residual scatter. A single spurious detection (an edge of a data segment, a bad sector) cannot pull this fit, unlike an ordinary least-squares fit.

3. **Second pass.** The scan is repeated around the new ephemeris with $R = \max(0.5\,T_{14},\, 4 \times \mathrm{rms}_{O-C})$.

4. **Adoption.** At least three consistent transits and $\mathrm{rms}_{O-C} < 0.25\,T_{14}$ are required. The refined ephemeris is adopted when the old one is off by more than $0.1\,T_{14}$ at any of the used epochs; otherwise the input is kept and reported as confirmed. $T_0$ keeps the epoch numbering of the input ephemeris; the log also gives the mid-time at the middle epoch, which is the best-constrained value.

## 35.3 Log output

```
--- Ephemeris check: box search of every expected transit within +-12.0 h ---
  epoch 0 predicted 1733.8557 found 1733.8549 ( -0.02 h) S/N = 140.6 ...
  epoch 12 predicted 2860.0185 found 2859.8594 ( -3.82 h) S/N = 25.1 ...
  epoch 19 predicted 3516.9469 found 3516.7099 ( -5.69 h) S/N = 9.2 ...
  linear fit of 8 transits (epochs [0, 1, 2, 3, 11, 12, 17, 19]):
    P = 93.834437 +/- 0.000428 d, T0 = 1733.85710 BTJD (epoch 0) ...
  ! the known ephemeris misses the transits by up to 5.65 h (= 1.27 T14);
    ADOPTING the refined ephemeris (P 93.846904 -> 93.834437 d ...)
```

## 35.4 Cost and limits

The scan only looks at a window of about one day around each expected transit and a few hundred trial mid-times per epoch; it takes well under a second for a quarter of a million cadences. It needs transits that are individually detectable (S/N $\gtrsim 5$ per transit). For shallower signals the grid search of Chapter 36 is the right tool. If the true offset exceeds the search range $R$, the transits are not found; increase `--ephem-scan-hours` using the published uncertainties and the $\sigma_{\Delta t}(n)$ formula above.

# Chapter 36

# Folded $(P, T_0)$ grid search

## 36.1 Idea

Instead of locating each transit, every trial ephemeris on a small grid around the input is evaluated with *all* transits stacked. The S/N of $N$ transits adds up as $\sqrt{N}$, so transits that are individually below the noise are still found. This is a box least-squares search restricted to a narrow region of $(P, T_0)$; it replaces a full TLS run when the period is approximately known.

## 36.2 Grid

- **Reference epoch.** $T_0$ is moved to the epoch $e_{\mathrm{ref}}$ closest to the median epoch of the data, $T_{\mathrm{ref}} = T_0 + e_{\mathrm{ref}} P$. At the middle of the baseline a period change hardly moves the transits, so $P$ and $T_{\mathrm{ref}}$ are nearly uncorrelated and the grid can be rectangular.
- **Ranges.** $T_{\mathrm{ref}} \pm \Delta T$ with $\Delta T = R$ (same range as the scan, `--ephem-scan-hours`); $P \pm \Delta P$ with by default $\Delta P = \Delta T / n_{\max}$, where $n_{\max}$ is the largest number of epochs between a data point and $e_{\mathrm{ref}}$: the same timing budget at the ends of the baseline as at its middle (`--ephem-grid-dp` to override).
- **Steps.** $T_{14}/20$ in $T_{\mathrm{ref}}$ and $T_{14}/(20\, n_{\max})$ in $P$, so that neighbouring grid points never differ by more than $T_{14}/20$ at any epoch. With the defaults the grid has about $110 \times 110$ points.

## 36.3 Evaluation

Only cadences that can lie within $\Delta T + 2.6\, T_{14}$ of a transit for some grid point are kept. For each trial period the phase $\phi = (t - T_{\mathrm{ref}} + P/2) \bmod P - P/2$ is computed once and the fluxes are histogrammed in fine phase bins ($T_{14}/80$). Cumulative sums then give, for all trial mid-times $c$ at once, the number of points and the flux sum in the box $|\phi - c| < T_{14}/2$ and in the two baseline windows $0.75\, T_{14} < |\phi - c| < 2.5\, T_{14}$:

$$d = \bar{f}_{\mathrm{out}} - \bar{f}_{\mathrm{in}}, \qquad \mathrm{S/N} = \frac{d}{\sigma_{\mathrm{MAD}} \sqrt{1/n_{\mathrm{in}} + 1/n_{\mathrm{out}}}}.$$

The cost is $O(N_P \times N_{\rm pts})$ for a few thousand to tens of thousands of retained cadences: 0.3–0.5 s for TIC 229788204 (42 sectors, $2.4 \times 10^5$ cadences). A full TLS search over the same baseline scales with the number of trial periods, which grows with the baseline, and takes minutes to hours.

A box of the full transit width is used deliberately. A shorter box core ($0.35\,T_{14}$ was tried first) gives a flat-topped S/N plateau: every mid-time for which the core still lies inside the transit scores the same, the maximum becomes arbitrary within $\pm 0.15\,T_{14}$, and with many epochs this plateau extends along a ridge in $(P, T_0)$. In injection tests (below) this produced mid-time errors of 30–45 min; the full-width box reduced them to 5–10 min. The per-transit scan uses the same full-width box. Remaining structure in the plateau is handled by taking the middle of the contiguous region with S/N within 1 of the maximum at the best period.

## 36.4 Result and adoption

The log gives the S/N at the input ephemeris and at the best grid point, the depth, the number of distinct transits inside the box, and $P$, $T_{\rm ref}$ and $T_0$ (epoch 0) with an error estimated from the extent of the region with S/N $\geq$ peak$-1$. A map of the S/N is written to `ephem_grid_<TIC>.pdf` (input ephemeris marked "+", best point "x"; Figure 36.1).

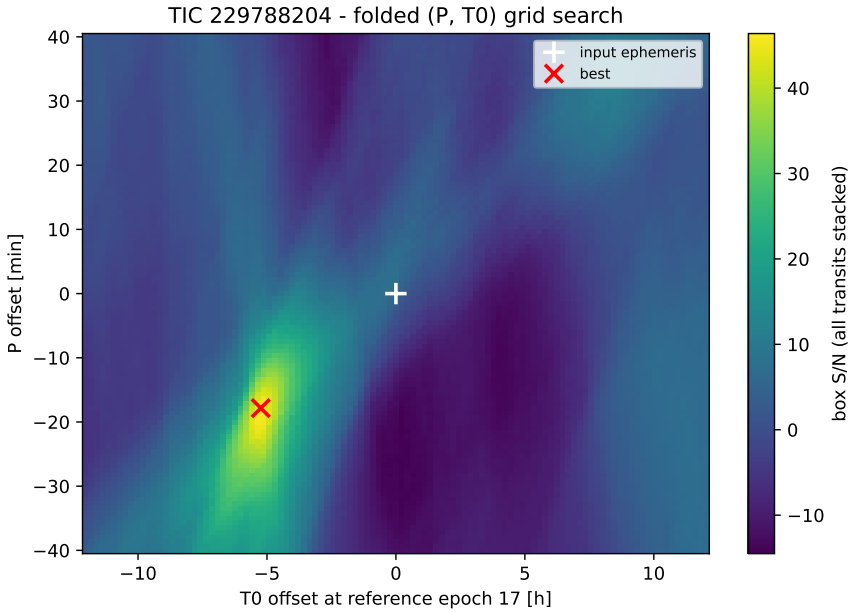



Figure 36.1: S/N map of the grid search for TIC 229788204, starting from the wrong period 93.846904 d. The peak lies 18 min per orbit below the input period and 5 h earlier at the reference epoch 17. The diagonal ridges are ephemerides that still catch part of the transits.

The grid result is adopted when

- the best point is not on the edge of the grid (otherwise the real ephemeris may lie outside it; widen the ranges);
- S/N $\geq$ `ephem_grid_min_snr` (7) and at least three transits contribute;

- the S/N exceeds the S/N at the input ephemeris by more than 2, and the predicted mid-times differ by more than $0.1\,T_{14}$ somewhere in the data.

## 36.5 Choosing the method

| `--ephem-search` | **Behaviour** |
|---|---|
| `auto` (default) | Per-transit scan; if it is inconclusive (fewer than three consistent transits), the grid search. |
| `scan` | Per-transit scan only. Gives individual mid-times (useful for transit-timing variations). |
| `grid` | Grid search only. Best for shallow transits and for many sectors with few points per transit. |
| `off` | Use the input ephemeris unchanged (old behaviour; also `--no-ephem-scan`). |

A very wide search (many hours, many epochs) increases the chance of a false peak from systematics for weak signals. Use the published uncertainties to set the range, and inspect the S/N map: a single compact peak is convincing; a ridge or several comparable peaks are not.

## 36.6 Tests

- **TIC 229788204** (deep transit, wrong input period 93.846904 d): best $P = 93.83452$ d, S/N 46 against 5 at the input ephemeris, ten transits in the box; adopted. The per-transit scan gives $P = 93.83444$ d; both agree with a manual measurement, $93.83463 \pm 0.00013$ d.
- **Injection** into the same light curve with the eclipses masked: a box transit of $T_{14} = 4.46$ h, $P = 50.123$ d, searched from an input that is off by $+0.004$ d in $P$ and $+86$ min in $T_0$.

| depth | `auto`: per-transit scan | grid search |
|---|---|---|
| 800 ppm | 4 consistent transits; adopted, $\delta P = 6 \times 10^{-4}$ d, $\delta T_{\rm ref} = -8$ min | (not needed; grid alone: S/N 10.5, $\delta T_{\rm ref} \approx 9$ min) |
| 500 ppm | inconclusive | S/N 6.6 at the right period ($\delta P = 6 \times 10^{-4}$ d), below the threshold of 7; input kept |
| 300 ppm | inconclusive | S/N 4.1; input kept |

For 500 and 300 ppm the scan found no consistent set of transits and `auto` fell back to the grid, as intended. At 500 ppm the grid peak lies at the right ephemeris but is not adopted with the default threshold; lower `--ephem-grid-min-snr` only after inspecting the S/N map.

# Chapter 37

# TESScut files, astrocut 1.4 and the TRICERATOPS pixel frame

## 37.1 Symptom

```
Getting TessCut for sector 14
Sector 14 raised exception. Ignoring for validation.
KeyError: "Keyword '1CRV4P' not found."
...
  File ".../triceratops.py", line 777, in calc_depths
    mu_x = self.pix_coords[k][i, 0]
IndexError: list index out of range
```

## 37.2 Cause

1. TESScut cutouts are produced by `astrocut`. Files made with version 1.4.0 (all downloads of October 2026) no longer contain the physical-WCS keywords `1CRV4P`, `2CRV4P` (or `1CRV5P`, `2CRV5P`) that give the position of the cutout on the full FFI.

2. `triceratops.target` reads `1CRV4P`/`2CRV4P` for every sector; the `KeyError` is caught, the sector is skipped with "Ignoring for validation", *but it stays in* `target.sectors`. With all sectors skipped, `pix_coords` is empty while `all_ap_pixels` has one entry per sector: `IndexError` in `calc_depths`.

3. `lightkurve` returns `tpf.column = tpf.row = 0` for such files. The pipeline's aperture pixels are therefore in the frame of the $10 \times 10$ extraction cutout, whereas `triceratops` uses its own, larger cutout ($22 \times 22$ here) whose origin lies elsewhere. Even with the keyword problem patched by setting zero, the apertures and the star positions would be shifted by several pixels: wrong flux ratios, wrong `tdepth`, wrong NFPP — without an error message.

## 37.3 Solution

The adapter that routes `triceratops`' cutout requests through the pipeline's cache (`_triceratops_tesscut_cache`) now calls `_fix_pixel_frame` for every cutout it hands to `triceratops`:

$$\Delta_{\mathrm{col}} = c_0 + x_{\mathrm{s}} - x_{\mathrm{t}}, \qquad \Delta_{\mathrm{row}} = r_0 + y_{\mathrm{s}} - y_{\mathrm{t}},$$

where $(x_{\mathrm{s}}, y_{\mathrm{s}})$ is the target position in the extraction cutout (its WCS), $(c_0, r_0) =$ (`tpf.column`, `tpf.row`) of that cutout, and $(x_{\mathrm{t}}, y_{\mathrm{t}})$ the target position in the `triceratops` cutout. The rounded offsets are written to `1CRV4P`, `2CRV4P` of the `triceratops` cutout in memory. `triceratops` then computes `pix_coords` in exactly the frame of `all_ap_pixels`, for any combination of old and new files. Both cutouts are centred on the same pixel grid, so the offset is an integer; a warning is printed if it differs from an integer by more than 0.25 px. Check on sector 57 of TIC 229788204: offset $(-6, -6)$; target position $(4.712, 4.796)$ in both frames (difference 0.001 px).

In addition, `make_target` compares the number of `pix_coords` with `target.sectors`. Sectors that `triceratops` skipped for any other reason are removed from `target.sectors` (and from the aperture list) and logged; if the skipped sectors cannot be identified, the script stops with a clear message instead of an `IndexError`.

## 37.4 Download retries

`_load_tesscut_tpf` now tries a download up to `tesscut_retries` (3) times with waits of 15 and 30 s. In the TIC 229788204 run, sector 15 (which contains a transit) was lost to a single "502 Bad Gateway".

## 37.5 Dilution of aperture photometry

The FFI aperture light curve contains the light of every star that falls in the aperture. SPOC PDCSAP light curves are corrected for this (CROWDSAP), so their transit depth is larger. For TIC 229788204 the PDCSAP depth (LcTools) is 67 000 ppm, the depth in the pipeline's aperture about 45 000 ppm: about a third of the aperture flux comes from other stars.

`triceratops` must be told which kind of depth it gets (`dilution_corrected`), and the depth passed to `calc_depths` must be the depth of the light curve passed to `calc_probs`. Therefore:

- with `--lc-source tesscut` the script sets `dilution_corrected=False` (override: `--dilution-corrected`);
- when a depth given with `--ephemeris` differs by more than 30% from the bottom of the binned fold, a warning is logged. Give the depth measured in the aperture light curve, or let the limb-darkened fit update it (`--ld-fit`; this happens only when the fold is not smeared).

# Chapter 38

# Outlier clipping and deep eclipses

## 38.1 Symptom

In the second run on TIC 229788204 (correct ephemeris) only two of ten transits passed the per-transit fit ("bottom not sampled at the best shift"), the per-transit scan found box depths of 3400–4900 ppm instead of about 19 000 ppm, the binned fold reached only 2.6% while single 200-s cadences in sector 57 reach 4–5%, and the limb-darkened fit returned a depth of 25 400 ppm. In sector 57, 34 cadences lie within ±0.04 d of mid-transit in the background-corrected light curve, but only one of them survived in the detrended, stitched light curve.

## 38.2 Cause

`clip_outliers` (after every detrending pass) and the clip in `fold_and_bin`:

- without an ephemeris (first pass, blind search) the downward cut at $5\sigma$ applied to *all* points, although the log called it "positive-only" and §6.4 describes it so. With RMS = 1900 ppm every point deeper than 0.95% was deleted;
- inside the transit window points deeper than continuum $-15\sigma$ were removed as "pathological". With RMS = 1800–2050 ppm that is 2.7–3.1%.

Any eclipse deeper than about 15× RMS lost its bottom. The light curve handed to TLS, to the per-transit fits, to the limb-darkened fit and to `calc_probs` then showed a truncated, flat-bottomed dip. For a V-shaped, possibly stellar eclipse this is the worst possible distortion: it makes the signal look like a box-shaped planet transit.

## 38.3 Fix

- No ephemeris: only upward spikes are clipped.
- With an ephemeris: outside the transit window two-sided as before; inside it only upward spikes and fluxes below 10% of the continuum.

- The same rule in the fold clip.

Check on sector 57: before the fix 0 (first pass) and 2 (transit-aware) of the 34 core cadences survived; after the fix all 34. Runs on deep transits made with earlier versions (including blind TLS searches) should be repeated.

# Chapter 39

# Secondary eclipse at any phase

## 39.1 Why

The phase-0.5 test (§22.5) only looks where a circular orbit puts the secondary eclipse. For an eccentric orbit the secondary is displaced and its duration differs from that of the primary:

$$\phi_{\rm sec} - 0.5 \simeq \frac{2}{\pi}\, e\cos\omega, \qquad \frac{T_{\rm sec}}{T_{\rm pri}} \simeq \frac{1+e\sin\omega}{1-e\sin\omega}.$$

For a candidate that may be a low-mass star, a secondary eclipse anywhere in phase is the most direct photometric evidence for a self-luminous companion.

## 39.2 Algorithm (`secondary_eclipse_search`)

1. The *first-pass* light curve (biweight window `window_length_search`, 1.5 d) is used: the second-pass window ($3\,T_{14}$) is tuned to the primary and would partly absorb a longer secondary.

2. The light curve is folded on $P$ and histogrammed in phase bins of $T_{14}/80$; cumulative sums over three wrapped copies give box sums at every trial phase without loops over the data.

3. Trial phases with step $T_{14}/20$; trial widths $w = \{0.5, 0.75, 1, 1.5, 2\} \times T_{14}$ (`secondary_dur_factors`). For each, depth = mean of the baselines $0.75w < |\Delta\phi| < 2w$ on both sides minus the mean in the box, with S/N from the robust point scatter. The region of the primary eclipse (plus $2w$) is excluded.

4. The forty strongest positive local maxima are re-measured *event by event*, each event with its own quadratic baseline fitted within $0.75w$–$3w$ on both sides and its own noise from the residuals (v3.2.1; a flat baseline let residual trends on the scale of a wide box mimic an eclipse). A candidate is *coherent* when (i) the weighted mean of the events has S/N $\geq$ `secondary_min_snr` (7), (ii) it rests on at least two events, (iii) it keeps S/N $\geq 3.5$ after removing the event with the largest contribution, (iv) S/N $\geq 4.9$ with errors inflated by $\sqrt{\max(\chi^2_{\rm red}, 1)}$ of the event depths, and (v) at least 60% of the events show a dip. Each rejected candidate is logged with its reasons. This rejects single dips (scattered light, momentum dumps, segment edges) that otherwise dominate a fold over many sectors.

5. Output: phase coverage, the candidates with their event counts, the $3\sigma$ depth limit for $w = T_{14}$ versus phase (median, at 0.5, worst case), and for a coherent candidate $e\cos\omega$ and a coarse $e\sin\omega$.

6. **Brightness temperature.** With equal occulted areas in both eclipses, the ratio of secondary to primary depth equals the ratio of the surface brightnesses at $\approx$ 800 nm. Inverting the Planck function gives the companion's brightness temperature (detection) or an upper limit (no detection). Dilution cancels in the ratio. A self-luminous M dwarf has 2800–3500 K.

Products: `secondary_search_<TIC>.pdf` (S/N and limit versus phase, fold at the best phase) and `secondary_search_<TIC>.txt`. The search takes well under a second. Disable with `--no-secondary-search`; threshold `--secondary-min-snr`.

## 39.3 Tests

On the stitched light curve of TIC 229788204 (second run):

- no coherent secondary. The strongest peak (phase 0.680, S/N 8.3) comes from a single event in sector 59 (4460 ppm, S/N 16.7); without it S/N is 2.1. Rejected as single event. $3\sigma$ limit 320 ppm (median over phases, coverage 99%), 280 ppm at phase 0.5;
- injected box eclipses: 1500 ppm at phase 0.37 ($0.8\,T_{14}$), 800 ppm at 0.50 and 600 ppm at 0.81 ($1.5\,T_{14}$) are recovered at the right phase, the 600 ppm case with the right width; 400 ppm (S/N $\approx$ 4) is not.

## 39.4 Fourth run and the v3.2.1 tightening

The fourth run on TIC 229788204 (v3.2) reported a "coherent" secondary at phase 0.895 (width $2\,T_{14}$, 1557 ppm, S/N 19.4, brightness temperature $\approx$ 2300 K). Re-measuring each event with a local quadratic baseline showed one event (sector 74, 2780 ppm) carrying the signal: event depths $\chi^2_{\rm red} = 40$, a dip in only 55% of the events, weighted mean $230 \pm 80$ ppm. Criteria (iv) and (v) above were added. With them, the first-pass light curve gives no coherent secondary ($3\sigma$ limit 335 ppm), while injected eclipses of 1500, 1000, 800, 600 and 2000 ppm at phases 0.37, 0.62, 0.50, 0.81 and 0.20 are recovered; a 1000 ppm, $2\,T_{14}$-wide eclipse placed on the systematics at phase 0.30 is not.

## 39.5 Limits of the interpretation

- An eccentric, inclined orbit can avoid a secondary eclipse altogether; a non-detection excludes a self-luminous companion only for orbits that produce one.
- Phases inside data gaps (here 1%) are not tested.
- The width grid assumes $0.5 \le T_{\rm sec}/T_{\rm pri} \le 2$.

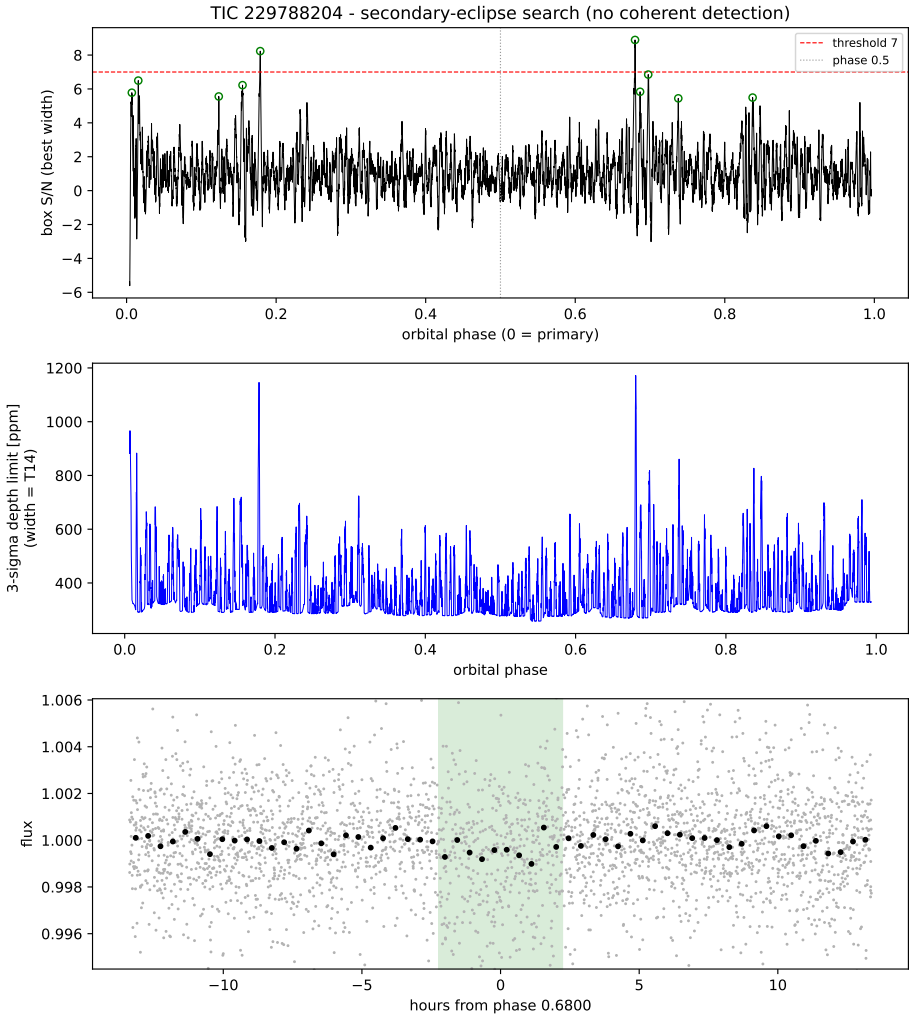



Figure 39.1: Secondary-eclipse search for TIC 229788204. Top: S/N versus phase (open circles: rejected candidates). Middle: $3\sigma$ depth limit for a width of $T_{14}$. Bottom: fold at the strongest (rejected) peak.

# Chapter 40

# Per-sector dilution and grazing eclipses

## 40.1 Per-transit depths in different apertures

The aperture is the set of pixels above the 85th percentile of the median image of each sector's cutout. Its shape changes from sector to sector, and so does the light of neighbours inside it. A raw aperture depth $\delta_s$ therefore depends on the sector:

$$\delta_s = g_s\,\delta_\star,$$

with $g_s$ the flux fraction of the target in the aperture of sector $s$ and $\delta_\star$ the on-target depth. In the third run on TIC 229788204 the raw depths of the same eclipse ranged from 27 500 to 49 700 ppm ($\chi^2_{\rm red} = 107$), and the odd/even test gave $4.9\sigma$ — both uninterpretable without $g_s$.

After `make_target`, `sector_target_fractions` computes $g_s$ for every sector with the PSF model of `triceratops` (circular Gaussian, $\sigma =$ `psf_sigma`), the star positions `target.pix_coords` (same pixel frame as the apertures, Chapter 37) and all stars of the field. `dilution_corrected_depths` then logs, per transit, the sector, cadence, $g_s$, the raw and the on-target depth, the weighted mean on-target depth, the constant-depth $\chi^2_{\rm red}$ before and after the correction, and the odd/even difference overall and *per cadence* (30-min, 10-min and 200-s data smear a V-shaped minimum differently, so odd and even should be compared within one cadence). If $\chi^2_{\rm red} > 5$ remains, the log warns that the depths are still inconsistent (systematics, a varying eclipse, or an imperfect flux model of the neighbours).

## 40.2 Grazing limb-darkened fits

When the fit gives $b > 1 - R_p/R_\star$ the eclipse is grazing: there is no flat bottom, $T_{23}$ is undefined, and $R_p/R_\star$ and $b$ are degenerate (third run: $R_p/R_\star = 0.45 \pm 10.9$, $b = 1.17 \pm 12.7$). The log now says so. Depth and $T_{14}$ of the fit remain usable; a radius derived from $R_p/R_\star$ is not. V-shaped, grazing eclipses are typical of eclipsing binaries.

## 40.3 Secondary-eclipse widths

The search of Chapter 39 uses $\max(T_{14,\mathrm{adopted}}, T_{14,\mathrm{trapezoid}})$ as the unit of width. For a V-shaped eclipse the trapezoid fit returns a short $T_{14}$ (2.3 h instead of 4.5 h), which would have limited the trial widths to 1.1–4.5 h.

# Chapter 41

# Configuration, command line and functions

## 41.1 New configuration fields

| Field | Default | Meaning |
|---|---|---|
| `ephem_search` | `"auto"` | `auto`, `scan`, `grid` or `off` (Chapter 36). |
| `ephem_scan` | True | False is the same as `ephem_search = "off"` (backwards compatibility). |
| `ephem_scan_range_d` | None | Half-range in mid-time [d] for scan and grid; None: $\max(2\,T_{14}, 0.5$ d). |
| `ephem_scan_min_snr` | 5.0 | Minimum S/N of a single transit for the linear fit. |
| `ephem_grid_dP` | None | Grid half-range in $P$ [d]; None: $\Delta T/n_{\max}$. |
| `ephem_grid_min_snr` | 7.0 | Minimum stacked S/N to adopt the grid result. |
| `tesscut_retries` | 3 | Download attempts per TESScut cutout. |
| `secondary_search` | True | Secondary-eclipse search at all phases. |
| `secondary_dur_factors` | 0.5–2.0 | Trial widths in units of $T_{14}$. |
| `secondary_min_snr` | 7.0 | Stacked S/N for a coherent secondary. |
| `dilution_corrected` | None→False | Set to False for TESScut photometry at run time. |

## 41.2 New command-line options

| Option | Type, default | Effect |
|---|---|---|
| `--ephem-search` | choice, `auto` | `auto \| scan \| grid \| off`: check/refine a known ephemeris before detrending. |

| Option | Type, default | Effect |
|---|---|---|
| `--no-ephem-scan` | flag | Same as `--ephem-search off`. |
| `--ephem-scan-hours H` | float | Half-range in mid-time [h] (scan and grid). |
| `--ephem-grid-dp D` | float | Grid half-range in $P$ [d]. |
| `--ephem-grid-min-snr X` | float, 7 | Minimum stacked S/N to adopt the grid result. |
| `--no-secondary-search` | flag | Skip the secondary-eclipse search. |
| `--secondary-min-snr X` | float, 7 | Detection threshold of that search. |
| `--dilution-corrected` | flag | Treat the light curve as crowding-corrected; excludes `--raw-aperture-flux`. |

All options listed in Chapter 30 (Addendum 1) remain valid.

## 41.3 Examples

```
# known ephemeris, default check (scan, grid as fall-back)
python triceratops_validation.py --tic 229788204 \
    --ephemeris 93.83463 1733.8576 45000 4.46 \
    --ld-fit --refine-chi2-max 20 --sensitivity --n-iter 20

# shallow candidate, ephemeris uncertain by ~0.5 d at the end of the baseline
python triceratops_validation.py --tic 165501544 \
    --ephemeris 132.42 1767.127 3228 4.8 \
    --ephem-search grid --ephem-scan-hours 18

# keep the input ephemeris exactly
python triceratops_validation.py --tic 229788204 \
    --ephemeris 93.83463 1733.8576 45000 4.46 --ephem-search off
```

## 41.4 New and changed functions

- `scan_transit_times(time, flux, flux_err, P, T0, dur, search_range, min_snr=5.0, step=None)`
  Box scan of every expected transit; returns a list of dictionaries (`epoch`, `Tc_pred`, `Tc`, `snr`, `depth`, `n_in`, `at_edge`).
- `fit_linear_ephemeris(rows, P, T0, dur, min_snr=5.0, tol_frac=0.15, max_rms_frac=0.25)`
  Pair-consensus linear fit of the scanned mid-times; returns $P$, $T_0$, errors, used and rejected epochs, O–C residuals and an `ok` flag.
- `check_known_ephemeris(time, flux, flux_err, P, T0, dur, cfg, log=None)`
  Two-pass scan and adoption rules of Chapter 35; returns `(P, T0, fit)`.
- `fold_grid_search(time, flux, P, T0, dur, dT0, dP=None, step_frac=20.0, core_frac=0.5)`
  Folded box search on the $(P, T_{\rm ref})$ grid with cumulative-sum box sums; returns the best

ephemeris, S/N, depth, number of transits, S/N at the input, errors, the S/N map and the grid.

- `plot_ephem_grid(res, P_in, T0_in, cfg, tic)`
  Writes `ephem_grid_<TIC>.pdf`.
- `grid_known_ephemeris(time, flux, P, T0, dur, cfg, log=None)`
  Runs the grid search, logs it and applies the adoption rules of Chapter 36.
- `refine_known_ephemeris(time, flux, flux_err, P, T0, dur, cfg, log=None)`
  Dispatcher on `cfg.ephem_search`; called in `main` right after the first detrend in known-ephemeris mode.
- `secondary_eclipse_search(time, flux, P, T0, dur, dur_factors, step_frac=20.0, gap_frac=0.25, min_snr=7.0, n_peaks=10)`, `_secondary_epoch_depths`, `_stack_stats`, `_planck_ratio_temperature`, `plot_secondary_search` — Chapter 39.
- `sector_target_fractions(target, all_ap, sigma=0.75)`, `_sector_of_time`, `dilution_corrected_depths(dt, fractions, sectors_data, log=None)` — Chapter 40; called in `main` after `make_target`.
- `fit_limb_darkened_lc` — grazing warning, `fit["grazing"]`.
- `_load_tesscut_tpf` — download retries.
- `_triceratops_tesscut_cache` — new inner function `_fix_pixel_frame`; records the sectors handed to `triceratops` in `cfg._tri_loaded_sectors`.
- `make_target` — removes sectors skipped by `triceratops`.
- `main` — stores `cfg._sector_frames` (WCS and pixel origin of every extraction cutout) before `make_target`; dilution default and depth warning; flag for per-transit mid-times at the search limit.
- `filter_bad_sectors` — prints each line once.

# Chapter 42

# Worked example: TIC 229788204

## 42.1 Candidate

K/M dwarf, $T_{\rm eff} = 4026$ K, $R_\star = 0.654\,R_\odot$, $M_\star = 0.629\,M_\odot$, $T_{\rm mag} = 12.10$, near the northern continuous viewing zone (42 sectors with FFI coverage between sectors 14 and 86). Input from the candidate sheet: $P = 93.846904$ d, $T_0 = 1733.798618$ BTJD, depth 66 678 ppm (PDCSAP), $T_{14} = 4.125$ h. The transit is V-shaped; the ExoPlanetPie fit of cycle 1 gives $R_p/R_\star = 0.31$, $b = 0.83$, so $b + R_p/R_\star \approx 1.14 > 1$: a grazing geometry. The implied radius ($\approx 1.8\,R_{\rm J}$ at $T_{\rm eq} \approx 265$ K) is implausible for a planet; an eclipsing binary with a low-mass companion is the likely explanation.

## 42.2 First run and diagnosis

The candidate sheet's $T_0$ marks the start of the transit, not its middle: the minimum lies at 1733.856. With $T_0 = 1733.8557$ and $P = 93.846904$ the first run showed:

- a trapezoid on the fold with $T_{14} = 1.34$ h and 6700 ppm, and an LD fit with $R_p/R_\star = 0.04$ and 1078 ppm — a smeared fold;
- per-transit mid-times on the ±40 min limit and negative depths of −10 000 to −12 000 ppm for epochs 7, 11 and 12;
- `KeyError: '1CRV4P'` for every sector in `triceratops` and an `IndexError` in `calc_depths` (Chapter 37).

Measuring the mid-times directly in the background-corrected light curves gave a linear ephemeris with an O–C scatter of about 4 min:

$$P = 93.83463 \pm 0.00013 \text{ d}, \qquad T_0 = 1733.8576 \pm 0.0019 \text{ BTJD}.$$

The input period was too long by 0.0123 d (18 min per orbit), i.e. 5.7 h at epoch 19, more than $T_{14}$. Transits were also found at epochs 1 (sector 19, 1827.690) and 3 (sector 26, 2015.362), which the candidate sheet lists as not observed, and at epochs 19 (3516.716) and 20 (3610.58, partial).

## 42.3 Other findings

- The aperture depth at mid-transit is about 42 000–45 000 ppm in 200-s data (less in 30-min data, which smear the V-shaped minimum), against 67 000 ppm in PDCSAP: roughly a third of the aperture flux comes from neighbours.
- In-transit centroid shifts are significant in several sectors (up to $24\sigma$); with a 4.5% deep eclipse and contaminating flux in the aperture this is expected even for an on-target signal, so difference imaging has to decide.
- The phase-0.5 test of the first run (806 ppm, $3.8\sigma$) and the odd/even test were made with the wrong ephemeris and are not meaningful; they must be repeated. A significant secondary eclipse or an odd/even difference would confirm the eclipsing-binary interpretation.

## 42.4 Second run (correct ephemeris, before the clipping fix)

- Ephemeris check: input confirmed (scan $P = 93.83480 \pm 0.00072$ d).
- TESScut/TRICERATOPS: all 41 sectors loaded, no `KeyError`.
- Difference imaging excluded the three neighbours with `tdepth` $> 0$ (89–179″ away, 50–1200$\sigma$); all other stars are too faint to cause the dip, so NFPP $= 0$ follows from the flux budget.
- Per-transit fits, odd/even ($4.9\sigma$ from two transits of different cadence), LD fit (depth 25 400 ppm; $\rho_\star = 6.4$ against 3.2 g cm$^{-3}$ from the TIC) and the light curve passed to `calc_probs` are invalid because of the clipping bug of Chapter 38.
- TRICERATOPS: FPP $\approx 0$, "PASSED (provisional)", but the evidence was dominated by a single run and FPP rose to 0.65 at $4\sigma$.

## 42.5 Interpretation of a "PASSED" verdict for this candidate

Even with a correct light curve a low FPP would not validate a planet here. The best TP solutions have $R_p \approx 15$–$18\,R_\oplus$ (1.4–1.6 $R_\mathrm{J}$). Brown dwarfs and stars below $\sim 0.15\,M_\odot$ have the same radii; their transits are photometrically planet-like and TRICERATOPS does not separate them from planets. What the photometry can test:

- **Secondary eclipse.** A self-luminous M dwarf ($T \approx 3000$ K) with $k^2 \approx 0.067$ (PDCSAP depth) has a TESS-band surface-brightness ratio of about 0.2 relative to the 4026 K primary, i.e. a secondary eclipse of about 1.4% on target, $\approx 0.8$% in the aperture. The phase-0.5 test gives $31 \pm 156$ ppm: a circular-orbit M-dwarf companion is excluded. A brown dwarf or planet (negligible thermal emission at this separation) and an eccentric orbit with the secondary away from phase 0.5 are not. The search at all phases (Chapter 39) found no coherent secondary anywhere ($3\sigma$ limit $\approx 320$ ppm, 99% phase coverage); with a primary depth of about 4% this limits the brightness temperature of an eclipsed companion to $\lesssim 1900$ K, below that of any M dwarf. Only an orbit that avoids a secondary eclipse leaves a stellar companion open.

- **Eccentricity.** The short duration relative to a circular orbit (stellar density from the transit about twice the TIC value) and the TRICERATOPS TP draws ($e \approx 0.4$–$0.5$) point to an eccentric orbit or a high impact parameter; then a secondary eclipse can lie elsewhere in phase.

The mass needs radial velocities. For $M_\star = 0.63\,M_\odot$ and $P = 93.8$ d, $K \approx 61$ m s$^{-1}$ per Jupiter mass (circular), about 0.8–5 km s$^{-1}$ for a brown dwarf and $\approx 8$ km s$^{-1}$ for a $0.15\,M_\odot$ star; even low-precision RVs ($V = 13.4$) separate the cases.

## 42.6 Third run (clipping fixed)

- Ephemeris refined from ten transits: $P = 93.83471 \pm 0.00005$ d, $T_0 = 1733.85618 \pm 0.00055$ BTJD, O–C rms 1.4 min, $\chi^2_{\rm red} = 1.0$.
- Raw per-transit depths 27 500–49 700 ppm; odd/even $4.9\sigma$, not interpretable without the per-sector dilution (Chapter 40). Phase 0.5: $-8 \pm 158$ ppm.
- Limb-darkened fit grazing and degenerate ($b = 1.17$, $R_p/R_\star = 0.45$, both with errors $> 10$).
- TRICERATOPS: FPP = 1.000 (0.98–1.00 for $\sigma \times 0.5$–4), NFPP = 0. Pooled probabilities: SEB 0.73, SEBx2P 0.13, EB 0.11, PEB 0.01; TP $\sim 10^{-8}$ (it would need a grazing $19.6\,R_\oplus$ planet). The Monte Carlo had not converged (one run held 69% of the evidence), so the ranking *among* the eclipsing-binary scenarios is uncertain; the classification as a false positive is not.

The "PASSED" of the second run was an artefact of the truncated eclipse. Once the V shape is preserved, the signal is classified as an eclipsing binary on the target or on an unresolved bound companion. Without a secondary eclipse at phase 0.5, either the orbit is eccentric (secondary elsewhere, tested by Chapter 39) or the true period is $2P = 187.67$ d with two similar eclipses (EBx2P; tested by the dilution-corrected odd/even comparison). Independent checks: Gaia DR3 RUWE / non-single-star solution and the position in the colour–magnitude diagram (an equal-mass binary is $\approx 0.75$ mag above the main sequence), and one spectrum (double-lined or not).

## 42.7 Fourth run (dilution correction and secondary search)

- Dilution-corrected depths: target fraction 0.36–0.66 per sector; on-target depth 72 600 ± 440 ppm (PDCSAP 66 700); $\chi^2_{\rm red}$ from 107 to 15; odd/even $0.9\sigma$ overall, $+1.1\sigma$ (200 s) and $-1.5\sigma$ (30 min) per cadence: no odd/even difference.
- No secondary eclipse at any phase after the v3.2.1 tightening ($3\sigma \approx 335$ ppm): an eclipsed companion would be cooler than $\approx 1950$ K.
- LD fit flagged as grazing.
- TRICERATOPS: FPP = 1.00 (0.96–1.00 for $\sigma \times 0.5$–4), NFPP = 0; EB 0.55, EBx2P 0.31, PEBx2P 0.14 (Monte Carlo not converged).

Most consistent picture: two similar low-mass stars in a grazing orbit of 187.67 d (EBx2P), which explains the equal odd/even depths and the missing secondary; an eccentric orbit without a secondary eclipse is not excluded.

## 42.8 Suggested run

```
python triceratops_validation.py --tic 229788204 \
    --ephemeris 93.83463 1733.8576 45000 4.46 \
    --ld-fit --refine-chi2-max 20 --sensitivity --n-iter 20
```

## 42.9 Verification status

Tested in the development environment (no MAST, Gaia or TRILEGAL access):

- the scan, the robust fit, the grid search and the adoption rules, on the stitched light curve of the first run and with injected shallow transits;
- `_fix_pixel_frame` with the cached $10 \times 10$ and $22 \times 22$ cutouts of sector 57 and the patched `lightkurve.search_tesscut` call that `triceratops` makes;
- syntax and static checks of the whole script.

After the second run: the clipping fix was checked on sector 57 (all 34 core cadences retained). Not verified: a complete run with downloads, the `triceratops` target construction on all sectors, and `calc_probs`.

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# Chapter 43

# Addendum 3: version 4.0

Version 4.0 is meant as the stable form of `triceratops_validation.py`. It addresses the weak points left after the analysis of TIC 229788204: two that could change the false-positive probability itself (exposure smoothing and mixed dilution in the folded light curve), the convergence of the TRICERATOPS Monte Carlo, the aperture, the difference imaging, the per-transit depths, and a timing analysis. A regression test suite protects the whole.

## 43.1 What changed

| **Topic** | **Change (chapter)** |
|---|---|
| Aperture | S/N-optimal aperture from a catalogue PSF model of the field; the 85th-percentile mask remains available (Chapter 44). |
| Dilution | Every sector is put on the target's flux scale with its own target fraction $g_s$ before folding; TRICERATOPS receives `dilution_corrected=True` (Chapter 44). |
| TRICERATOPS models | Smoothed with the effective exposure of the binned fold instead of the package default of 2 min (Chapter 45). |
| Convergence | Runs are added until no single run dominates the pooled evidence; class probabilities (Chapter 45). |
| Difference imaging | Systematic floor on the centroid error, direct-image fit held on the target, differential offset as the main metric (Chapter 46). |
| Per-transit depths | Refitted with the limb-darkened shape smoothed with each sector's cadence; per-sector table and odd/even per cadence (Chapter 47). |
| Timing | Eclipse timing and depth variations, ground-based times (Chapter 47). |
| Output | `SUMMARY` block and `summary_<TIC>.json`; field catalogue cached as `field_TIC<tic>.csv`; `plot_fits_<TIC>.pdf` (no double extension); figures closed after use (Chapter 48). |
| Secondary search | The primary eclipse ($\pm T_{14}$) is removed from the data first: at the edge of an event window it faked a "secondary" 1.1 d before the primary in testing. |

| Topic | Change (chapter) |
|---|---|
| Tests | `tests/test_triceratops_validation.py`, 15 tests on synthetic data (Chapter 48). |

## 43.2 Corrections to the earlier chapters

| Where | Earlier text | Current behaviour |
|---|---|---|
| §5.3, §20.3.1 | Aperture = pixels above the 85th percentile. | Default `--aperture optimal`; `percentile` on request or when the catalogue is unavailable. |
| §16, Ch. 26 | `n_iter` runs, then a warning if one run dominates. | Runs are added (up to `tri_max_iter`, `tri_time_budget`) until the largest single-run share is $\leq$ `tri_conv_max_share`. |
| Ch. 37, Ch. 40 | TESScut photometry passed as raw aperture flux; depths divided by $g_s$ only in a diagnostic table. | The light curve itself is corrected per sector (`--dilution-mode sector`); `none` restores the raw flux. |
| Ch. 40 | Per-sector odd/even with the trapezoid depths. | With the cadence-smoothed limb-darkened depths. |
| §12 | Neighbour vetting compared with the aperture depth. | The on-target depth is passed, as the test assumes. |

# Chapter 44

# Optimal aperture and per-sector dilution

## 44.1 Field model

`field_catalog` queries the TIC within `field_radius_arcsec` ($220''$) of the target, drops artefacts and duplicates, moves the stars to epoch 2022 with their proper motions, puts the target in row 0 and caches the table. In every sector the stars are projected into the aperture frame with the cutout's WCS (`star_pixels`) and represented by the same circular Gaussian PSF ($\sigma = 0.75$ px) that TRICERATOPS uses.

## 44.2 Optimal aperture

For each pixel $j$ the target flux is $T_j = (I_j - B)\, f_{\star,j}$, with $I_j$ the median image, $B$ its 10th percentile and $f_{\star,j}$ the model fraction of the target in that pixel; its variance is taken as $I_j$ (photon noise including sky). Pixels are sorted by $T_j/\sqrt{I_j}$ and added while

$$\mathrm{S/N}(n) = \frac{\sum_{j \le n} T_j}{\sqrt{\sum_{j \le n} I_j}}$$

increases; at least `aperture_min_pix` (4) pixels are kept because of pointing jitter (`optimal_aperture`).

## 44.3 Per-sector dilution correction

$g_s$ = the target's share of the model flux in the aperture of sector $s$ (`target_fraction`). After every detrending pass each sector is transformed to the target's scale,

$$f \;\to\; 1 + \frac{f - 1}{g_s}, \qquad \sigma_f \;\to\; \sigma_f / g_s,$$

exactly once per detrend (`apply_dilution_correction`). Sectors without a target fraction are dropped when others have one. All depths in the log are then on-target depths, the fold is no longer a mixture of differently diluted sectors, and TRICERATOPS gets `dilution_corrected=True`. The log cross-checks $g_s$ against the TRICERATOPS field model.

## 44.4 Result on TIC 229788204

Integration test on the real TESScut cutouts of twelve sectors (offline, with the star table of the previous run as field; Chapter 48):

| | 85th-percentile aperture (v3.2) | optimal aperture (v4.0) |
|---|---|---|
| pixels | $\approx 15$ | 6–9 |
| target fraction $g_s$ | 0.36–0.66 | 0.978–0.988 |
| on-target depth | $72\,600 \pm 440$ ppm | $72\,300 \pm 195$ ppm |
| constant-depth $\chi^2_{\mathrm{red}}$ | 14.6 | 4.8 |
| odd/even (all / 200 s / 30 min) | $0.9/+1.1/-1.5\,\sigma$ | $1.3/+0.5/-0.5\,\sigma$ |

The three neighbours that the earlier run removed from the star table (89–179″) were missing from this offline field; at 4–8 px from the target they hardly enter a 6–9-pixel aperture.

# Chapter 45

# TRICERATOPS: exposure, convergence and classes

## 45.1 Exposure smoothing

`calc_probs` averages its models over `exptime` (default 0.00139 d = 2 min) with `nsamples` subsamples. The folded light curve of the pipeline mixes 30-min, 10-min and 200-s data and is binned to 10 min, so a V-shaped eclipse is smoothed over $\approx$ 14 min. With 2-min models the shape comparison between planet and binary scenarios was made against models sharper than the data. v4.0 passes

$$t_{\rm exp} = \sqrt{t_{\rm bin}^2 + \langle t_{\rm cad}^2 \rangle},$$

the point-weighted mean square of the native cadences (`effective_exposure_d`, the same value as the limb-darkened fit), and `nsamples` = 20. Override with `--tri-exptime-min`; 0 restores the package default.

## 45.2 Convergence

After `n_iter` runs the largest share of a single run in the pooled evidence of the leading scenario is computed. While it exceeds `tri_conv_max_share` (0.5), runs are added in blocks of five, up to `tri_max_iter` (default 3×`n_iter`) or `tri_time_budget` (3600 s). The log states whether and when it converged.

## 45.3 Probability per class of explanation

Because the ranking among similar scenarios can stay unstable while the classification is robust, the pooled probabilities are also summed per class: planet on the target or an unresolved companion (TP, PTP, DTP); eclipsing binary on the target or a bound companion (EB, EBx2P, PEB, PEBx2P, SEB, SEBx2P, STP); unresolved background or foreground star (DEB, DEBx2P, BTP, BEB, BEBx2P); resolved nearby star (NTP, NEB, NEBx2P).

# Chapter 46

# Difference imaging in version 4.0

- **Systematic floor.** The Monte Carlo centroid error describes photon noise only (0.007 px in the deepest sectors). The Gaussian PRF model and pointing add $\approx 0.1$ px; `di_sys_floor_px` $= 0.1$ is added in quadrature.
- **Direct image held on the target.** The fit to the out-of-eclipse image starts at the target's catalogue position and may move by at most 1 px (`fit_source_position(.., guess, box)`). Before, it jumped to a brighter star 6–7 px away in sectors 15, 22 and 81.
- **Differential offset** (difference-image centroid minus the target's own PRF centre in the direct image) is the main metric: it cancels most of the PRF-model error. The log reports its median in units of the error over sectors with S/N $\geq 3$.
- Sectors without an eclipse are listed once; the list of excluded stars is summarised.

Integration test on TIC 229788204 (ten sectors with an eclipse): differential offsets 0.03–0.09 px, median $0.5\sigma$ over nine sectors with S/N $\geq 3$: the light is lost at the target's position. (v3.2 reported a mean of $5.3\sigma$ from catalogue offsets with photon-noise errors only.)

# Chapter 47

# Per-transit depths and eclipse timing

## 47.1 Depths with the limb-darkened shape

`per_transit_depths_ld` refits every eclipse after the limb-darkened fit, with the intrinsic model averaged over the cadence of its own sector and normalised to the *intrinsic* depth (so that a 30-min and a 200-s sector return the same depth). The coverage criterion is judged against the highest value the smoothed shape can reach. `report_depths_per_sector` then lists sector, cadence, aperture, $g_s$ and on-target depth, the constant-depth test and the odd/even comparison overall and per cadence.

## 47.2 Timing analysis

`etv_analysis` uses the mid-times of these fits (and ground-based times from `--extra-times`: one line per eclipse, `BJD_TDB err_min [label]`; full BJD is converted to BTJD):

- linear ephemeris, O–C rms and $\chi^2_{\rm red}$;
- odd minus even mid-times; for a binary at $2P$ this gives $e\cos\omega \simeq \pi\,\Delta t/(4P)$;
- quadratic ephemeris ($\mathrm{d}P/\mathrm{d}E$, $\Delta\chi^2$);
- light-travel-time effect: a sinusoid plus a linear ephemeris on 400 outer periods between $3P$ and three times the baseline; best period, amplitude and $\Delta\chi^2$ (four extra parameters), and the $3\sigma$ amplitude limit for outer periods within the baseline, converted to $a_{12}\sin i$ (1 d of light time = 173.1 AU);
- eclipse-depth trend (a changing inclination makes grazing eclipses deeper or shallower).

Outputs: `etv_<TIC>.pdf` and `etv_<TIC>.txt`.

## 47.3 Result on TIC 229788204 (integration test)

Ten eclipses: $P = 93.834742 \pm 0.000029$ d, O–C rms 1.0 min; odd minus even $+0.25 \pm 0.40$ min, i.e. $e\cos\omega = (1.4 \pm 2.3) \times 10^{-6}$ if the true period is 187.67 d; no period change ($-0.01 \pm 1.03$ s

per cycle); no significant LTTE (best $\Delta\chi^2 = 8.3$), $3\sigma$ limit 2.1 min or $a_{12}\sin i < 0.25$ AU for outer periods up to 1877 d; depth trend $-146 \pm 251$ ppm yr$^{-1}$.

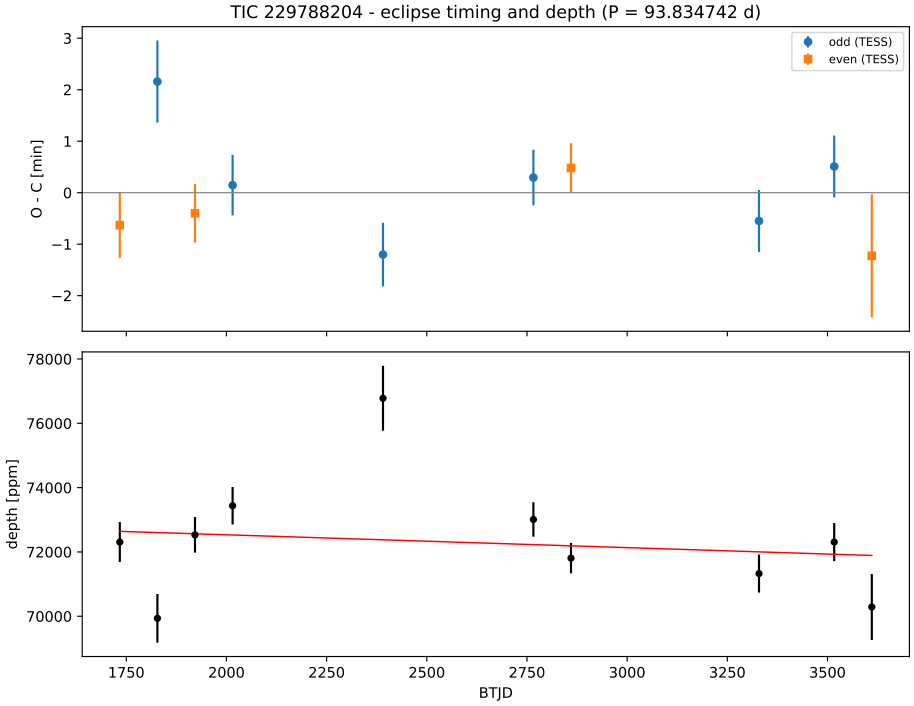



Figure 47.1: O–C of the ten eclipses (odd and even marked) and the on-target depths with their linear trend; integration test of v4.0 on TIC 229788204.

# Chapter 48

# Output, command line, tests and verification

## 48.1 Summary output

At the end of a run the log has a `SUMMARY` block with the key results (stellar parameters, ephemeris, depths, $g_s$ range, limb-darkened fit, secondary-eclipse result and limit, odd/even, timing, difference-image offset, FPP, NFPP, class probabilities, verdict, number of runs, convergence); the same is written to `summary_<TIC>.json` for tables over many candidates.

## 48.2 New command-line options

| Option | Default | Effect |
|---|---|---|
| `--aperture` | `optimal` | `optimal \| percentile`. |
| `--dilution-mode` | `sector` | `sector \| none`; `--raw-aperture-flux` implies `none`. |
| `--tri-exptime-min M` | bin + cadences | Model exposure for TRICERATOPS [min]; 0 = package default. |
| `--tri-max-iter N` | 3×`n-iter` | Maximum runs while not converged. |
| `--tri-time-budget S` | 3600 | Time budget for extra runs [s]. |
| `--no-etv` | – | Skip the timing / depth-variation analysis. |
| `--extra-times FILE` | – | Ground-based mid-times for the timing analysis. |

New `Config` fields: `aperture_mode`, `aperture_min_pix`, `dilution_mode`, `field_radius_arcsec`, `tri_conv_max_share`, `tri_max_iter`, `tri_time_budget`, `tri_exptime_d`, `tri_nsamples`, `di_sys_floor_px`, `etv_analysis`, `extra_times_file`.

## 48.3 Regression tests

`python -m pytest -q tests` in the script folder runs 15 tests on synthetic data (about 10 s, no internet): clipping of deep eclipses; ephemeris scan and grid search; optimal aperture and dilution (applied once); held direct-image fit; secondary search (detection, single event, primary not confused with a secondary); depths of 30-min and 200-s eclipses with the cadence-smoothed shape; timing (LTTE recovery, odd/even offset); exposure arguments; class probabilities; the TRICERATOPS convergence loop with a mock target whose evidences are heavy-tailed. Run them after every change.

## 48.4 Verification status

- All 15 regression tests pass (Python 3.13, numpy 2); the script needs Python $\geq$ 3.8.
- Integration test of `main` on the real TESScut cutouts of TIC 229788204 (12 sectors) with the network calls replaced (stellar parameters, sector list, field table from the previous log): extraction, optimal apertures, dilution, ephemeris check, per-transit fits, secondary search, limb-darkened fit, per-sector depths, timing, difference imaging and the summary ran without error; results in Chapters 44, 46 and 47.
- Not verified here: `calc_probs` of the real package with `exptime`, the TRICERATOPS target construction with the new apertures, and a run with network access. The first full run of v4.0 should be compared with the v3.2 results.

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# Chapter 49

# Addendum 4: version 4.1, ground-based photometry

A TESS pixel is 21″ wide. Stars a few arcseconds apart, such as TIC 65344135 and its neighbour TIC 65344134 at 5.9″ (AAVSO campaign 942), end up in the same aperture. TESS alone then cannot tell which star dims. Ground-based photometry under 2–3″ seeing separates such stars. Version 4.1 reads these light curves and uses them in four ways:

1. **Star clearance.** For every star near the target the pipeline computes the depth the ground aperture would show if that star hosted the TESS signal, and compares it with the measured depth. Stars that are inconsistent are removed before `calc_probs`, as an SG1 "NEB check" would do.
2. **Chromaticity.** The intrinsic depth is compared across bands and with TESS. A planet transit is nearly achromatic; a blend or an eclipsing binary often is not.
3. **Timing.** Ground mid-times enter the ETV analysis and update the ephemeris.
4. **TRICERATOPS+.** When the installed package is TRICERATOPS+ [1], up to seven light curves (one per band) also enter the likelihood of every scenario.

## 49.1 What changed

| Topic | Change (chapter) |
|---|---|
| Input | `--ground-lc SPEC` (repeatable) and `--ground-list FILE`; AstroImageJ, AAVSO/EXOTIC and plain tables; fluxes or magnitudes; BJD_TDB, JD_UTC, HJD_UTC (Chapter 50). |
| Fit | TESS transit shape with free depth and mid-time on a grid, polynomial and airmass baseline, red-noise factor, coverage checks (Chapter 51). |

| Topic | Change (chapter) |
|---|---|
| Field | Flux fraction of every TRICERATOPS star in the ground aperture (Gaussian seeing, circular aperture, band magnitudes) (Chapter 51). |
| Clearance | Stars excluded before `calc_probs`; `--ground-no-clear` only reports; `--exclude-stars` for stars cleared elsewhere (Chapter 51). |
| Chromaticity | $\chi^2$ test of a band-independent intrinsic depth, TESS included (Chapter 51). |
| Timing | Ground mid-times are added to `etv_analysis` and written to `ground_times_<TIC>.txt` (Chapter 51). |
| TRICERATOPS+ | Detected automatically; input files and `fluxratio_<band>` columns are prepared (Chapter 52). |
| Dilution | If the installed triceratops has no `dilution_corrected` argument (TRICERATOPS+, or versions before 1.1), `--dilution-mode none` is used automatically, so the dilution is not corrected twice (Chapter 52). |
| Output | `ground_lightcurves_<TIC>.pdf`, `ground_clearance_<TIC>.pdf/.csv`, `ground_chromaticity_<TIC>.pdf`, `ground_times_<TIC>.txt`, and a `ground` block in `summary_<TIC>.json`. |
| Tests | `tests/test_ground_photometry.py`, 13 tests; 28 in total (Chapter 53). |

## 49.2 Where it sits in the pipeline

The ground fits are step 7d. They run after the limb-darkened fit and the per-transit depths, because they use the TESS transit shape and the covariance of the TESS ephemeris, and before the timing analysis, which receives their mid-times.

The clearance is step 8c. It runs after `make_target`, because it needs the TRICERATOPS star table with `tdepth`, and after difference imaging (8a) and neighbour vetting (8b). All three add the stars they exclude to the same set, `drop_stars`. The target itself is never removed.

# Chapter 50

# Preparing ground light curves

## 50.1 What the observer should deliver

- **The whole transit.** Include at least 45–60 min of baseline before ingress and after egress. A light curve with no baseline on either side is fitted but not used for clearance.
- **Separate apertures.** Measure the target and every neighbour that could host the signal, each in its own aperture. In AstroImageJ, mark them `T1`, `T2`, ...
- **Aperture radius and seeing.** Report the aperture *radius* and the seeing FWHM in arcseconds. These set the dilution model. Without them, FWHM = 3″ (`--ground-fwhm`) and a radius of 1.7×FWHM are assumed and flagged in the log.
- **Bands.** A red band (Rc or Sloan $r'$) for the depth, and if possible a blue one (B, V, $g'$) for the chromaticity test.
- **Airmass.** Include an airmass column. It is used in the baseline automatically.
- **Times.** Give BJD_TDB. JD_UTC is accepted and converted.

## 50.2 File formats

**AstroImageJ measurement table.** Tab separated, header with an empty first cell. Columns `BJD_TDB`, `rel_flux_T1`, `rel_flux_err_T1`, `AIRMASS`. Another star of the same table is chosen with `fcol=rel_flux_T2`; its error column is found automatically. With `pixscale=`, `Source_Radius` and `FWHM_Mean` are converted to arcseconds.

**AAVSO / EXOTIC exoplanet report.** Header lines `#KEY=VALUE` (`DATE_TYPE`, `FILTER`, `OBSCODE`, `MEASUREMENT_TYPE`, `DELIM`), then a column line `#DATE,DIFF,ERR,DETREND_1,...`. The band is taken from `FILTER` when none is given. Magnitudes are recognised from `MEASUREMENT_TYPE`.

**Plain table.** Time, flux and error, comma or blank separated, with or without a header. Columns named `mag` or flagged with `mag=1` are magnitudes.

Times above $2.4 \times 10^6$ are full JD. Values between $5 \times 10^4$ and $10^5$ are MJD. Values near $10^4$ are taken as JD − 2450000 or JD − 2400000. Anything smaller is BTJD. `time=JD_UTC` applies the barycentric correction for the target direction (geocentre, astropy). `time=HJD_UTC` adds only TDB − UTC; heliocentric and barycentric times differ by less than about 4 s.

Points are clipped at $5\sigma$ from an 11-point running median, which leaves an hours-long transit intact. Missing errors are replaced by the point-to-point scatter.

## 50.3 The specification

Each light curve is one specification: the path, then `key=value` options separated by commas or blanks. Quote a path that contains blanks.

| Key | Meaning |
|---|---|
| `band`, `filter` | B V R I (Johnson–Cousins; Rc = R), g r i z (Sloan; `r'`, `SR`, `rp` also accepted), J H K, CBB, CV. |
| `star` | TIC ID at the aperture centre (default: the target); `star=TIC 65344134` is accepted. |
| `ra`, `dec` | Explicit aperture centre [deg], instead of `star`. |
| `ap` | Aperture *radius* [arcsec]. |
| `fwhm` | Seeing FWHM [arcsec]. |
| `obs` | Observer or code, for the log. |
| `tcol`, `fcol`, `ecol` | Column names (or 0-based numbers) for time, flux and error. |
| `time` | `BJD_TDB` (default), `JD_UTC`, `HJD_UTC`. |
| `mag` | `1` if the flux column holds magnitudes. |
| `detrend` | `auto` (linear + airmass if present), `poly0`, `poly1`, `poly2`, `none`, plus covariate column names: `detrend=poly1,AIRMASS,FWHM_Mean`. |
| `plus` | Band passed to TRICERATOPS+, overriding the automatic mapping. |
| `label` | Name in the log and the figures (default: the file name). |

A list file (`--ground-list`) holds one specification per line, and `#` starts a comment. Relative paths are relative to the list file. A template, `ground_lcs_example.txt`, is in the script folder:

```
# TIC 65344135, AAVSO campaign 942: one light curve per line
"AAVSO_TIC65344135_20261107_A.txt" band=R ap=3.5 fwhm=2.4 obs=A
"Measurements_20261107_B.tbl" band=r' ap=3.0 fwhm=2.2 obs=B
"Measurements_20261107_B.tbl" band=r' ap=3.0 fwhm=2.2 obs=B fcol=rel_flux_T2 star=65344134 label=
    B_neighbour
```

# Chapter 51

# Method

## 51.1 Transit fit

The model is the TESS transit shape $u(t)$, normalised to a unit depth. It is taken from the limb-darkened fit if one ran (`--ld-fit`), otherwise from the trapezoid of the per-transit fits, otherwise from a trapezoid with $T_{12} = 0.25\,T_{14}$. The baseline is a polynomial in time plus the covariates $c_k$:

$$f(t) = \sum_j a_j \left(\frac{t-\bar{t}}{\Delta t}\right)^j + \sum_k b_k\,\hat{c}_k(t) - A\,u(t-T_c). \tag{51.1}$$

For each $T_c$ on a 1-min grid the model is linear, and is solved by weighted least squares. The grid covers $T_{c,\mathrm{pred}} \pm W$, with

$$W = \max\!\big(3\,\sigma_{T_c,\mathrm{pred}},\ W_{\min}\big) \le T_{14}, \tag{51.2}$$

with $W_{\min} =$ `ground_tc_window_min` (`--ground-tc-window`), where $\sigma^2_{T_c,\mathrm{pred}} = c_{00} + 2E\,c_{01} + E^2 c_{11}$ comes from the covariance of a weighted linear fit of the TESS mid-times (scaled by $\chi^2_{\mathrm{red}}$ if above 1), and $E$ is the epoch of the ground transit. $W_{\min}$ defaults to 60 min.

The depth errors are multiplied by $\sqrt{\max(\chi^2_{\mathrm{red}}, 1)} \cdot \beta$. Here $\beta \ge 1$ is the time-averaging red-noise factor of the residuals, binned over 5, 10, 20 and 30 min.

A light curve counts as a **detection** if all of the following hold:

- $A/\sigma_A \ge 5$ (`--ground-min-snr`) and $\Delta\chi^2 \ge 25$ relative to the baseline-only fit;
- at least 30% of the transit window has data;
- there is baseline on at least one side.

The **mid-time** is reported only for a detection in which ingress or egress is at least half covered. Its error is the $\Delta\chi^2 = 1$ interval on the scaled $\chi^2$ curve. A light curve is **usable** for clearance when at least one $T_c$ in the window has 50% coverage and there is baseline on one side. Three depths are kept:

- the best $A \pm \sigma_A$;

- $A$ at the predicted time;
- the window upper limit $A_{\mathrm{up}} = \max_{T_c}(A + 3\sigma_A)$ over the grid points with 50% coverage.

## 51.2 Flux fractions in the ground aperture

The seeing PSF is taken to be Gaussian, with $s = \mathrm{FWHM}/2.3548$. A star at distance $d$ from the centre of a circular aperture of radius $r$ contributes the encircled fraction

$$\epsilon(d) = P\left[\chi'^2_2\left(\lambda = d^2/s^2\right) < r^2/s^2\right], \tag{51.3}$$

which is the non-central $\chi^2$ CDF with two degrees of freedom. Its flux fraction in the aperture is

$$\phi_j = \frac{10^{-0.4 m_j}\,\epsilon(d_j)}{\sum_i 10^{-0.4 m_i}\,\epsilon(d_i)}. \tag{51.4}$$

The test suite checks $\epsilon$ against a numerical integral. The stars are those of the TRICERATOPS table. Band magnitudes come from, in order of preference:

1. the catalogue column for that band (`Vmag`, `Jmag`, …; TRICERATOPS+ also provides $B$, $g$, $r$, $i$, $z$);
2. Gaia $G$ and $G_{BP} - G_{RP}$ with the relations of Riello et al. (2021) for $V$, $R_C$, $I_C$, $g$, $r$, $i$;
3. the nearest available magnitude, flagged as an approximation in the log.

Only the *relative* brightness of nearby stars enters $\phi_j$, so small colour-term errors are of little consequence.

## 51.3 Clearance

TRICERATOPS' `calc_depths` gives, for every star $j$, the intrinsic depth $\texttt{tdepth}_j$ needed to produce the TESS depth. If $j$ were the host, the ground aperture would show

$$\delta_{\mathrm{exp},j} = \mathrm{tdepth}_j \cdot \phi_j. \tag{51.5}$$

With a tolerance $\tau$ (`--ground-depth-tol`, default 0.25), which covers the dilution model, band and limb-darkening differences, star $j$ is classified per light curve:

| **Status** | **Condition** |
|---|---|
| excluded | $(1-\tau)\,\delta_{\mathrm{exp},j} > A_{\mathrm{up}}$: the dip it would cause is not there, at any $T_c$ in the window. Or a dip is detected with $A - 3\sigma_A > (1+\tau)\,\delta_{\mathrm{exp},j}$: it cannot produce the dip that is seen. |
| consistent | Detected, and $\lvert A - \delta_{\mathrm{exp},j}\rvert \le 3\sqrt{\sigma_A^2 + (\tau\,\delta_{\mathrm{exp},j})^2}$. |
| allowed | No conflict and no detection, or too little signal to test. |

| Status | Condition |
|---|---|
| unconstrained | The light curve does not cover the transit window. |

The combined verdict for a star is:

- *excluded* if any light curve excludes it;
- *conflict* if one light curve excludes it and another finds it consistent. A conflict is not removed and is reported.

Excluded stars, except the target, are removed before `calc_probs`. `--ground-no-clear` only reports them.

If the target itself is excluded, the log says so in capitals and the target stays in TRICERATOPS. If a neighbour is then consistent, the log reports a probable nearby eclipsing binary. Check the aperture, filter, comparison stars and the time system before believing such a result.

## 51.4 Chromaticity

For each candidate host $h$ (the target, and any neighbour found consistent), every detected light curve gives an intrinsic depth $A/\phi_h$. The light curve is used only if $\phi_h \geq 0.05$. The error of each value includes a 3% systematic for the flux fraction. The values are averaged per band, and the TESS intrinsic depth $\texttt{tdepth}_h$ is added as one more point. A $\chi^2$ test of a constant depth gives $p$, and the candidate is flagged *chromatic* if $p < 0.0027$ ($3\sigma$). The ratio of the reddest to the bluest ground band is also reported.

Limb darkening makes a planet's depth slightly band dependent. For large or grazing bodies this can reach a few per cent, so a $p$ just below the threshold for a V-shaped event is not conclusive by itself.

## 51.5 Timing

Ground mid-times with `timing_ok` are appended to the TESS mid-times in `etv_analysis`, which handles the linear ephemeris, O−C and the LTTE limits. They are also written to `ground_times_<TIC>.txt`, in the `--extra-times` format.

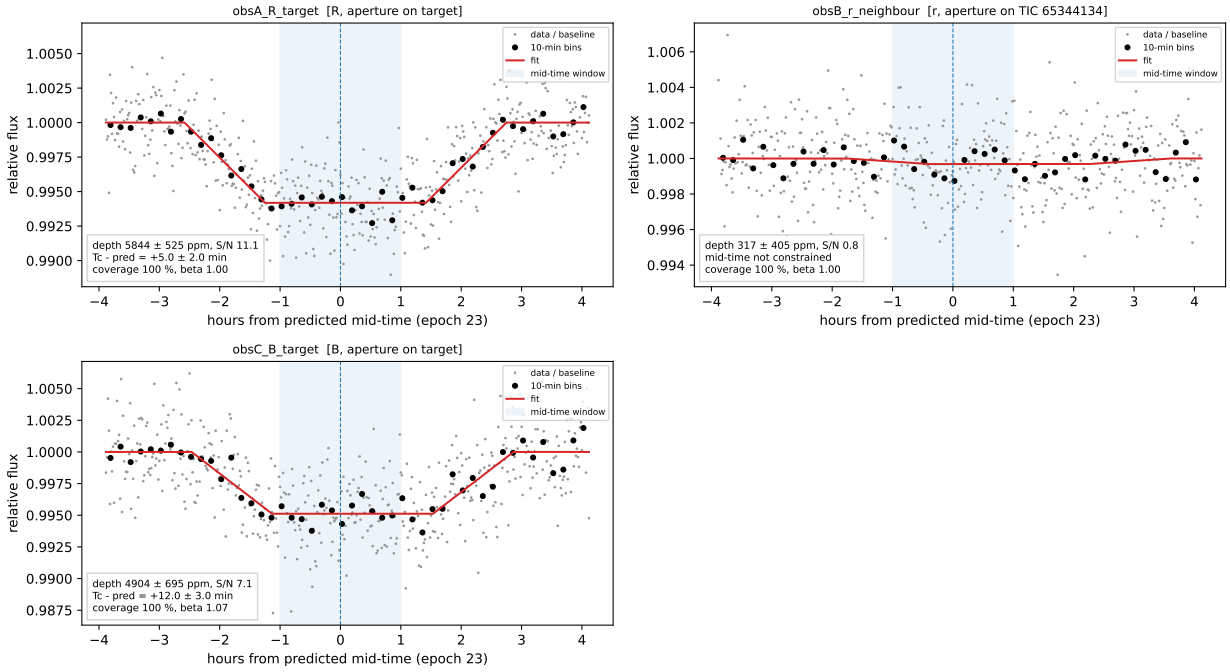



Figure 51.1: `ground_lightcurves_<TIC>.pdf` for the synthetic test of Chapter 53. Grey: data divided by the fitted baseline; black: 10-min bins; red: fit; blue band: mid-time search window around the predicted time.

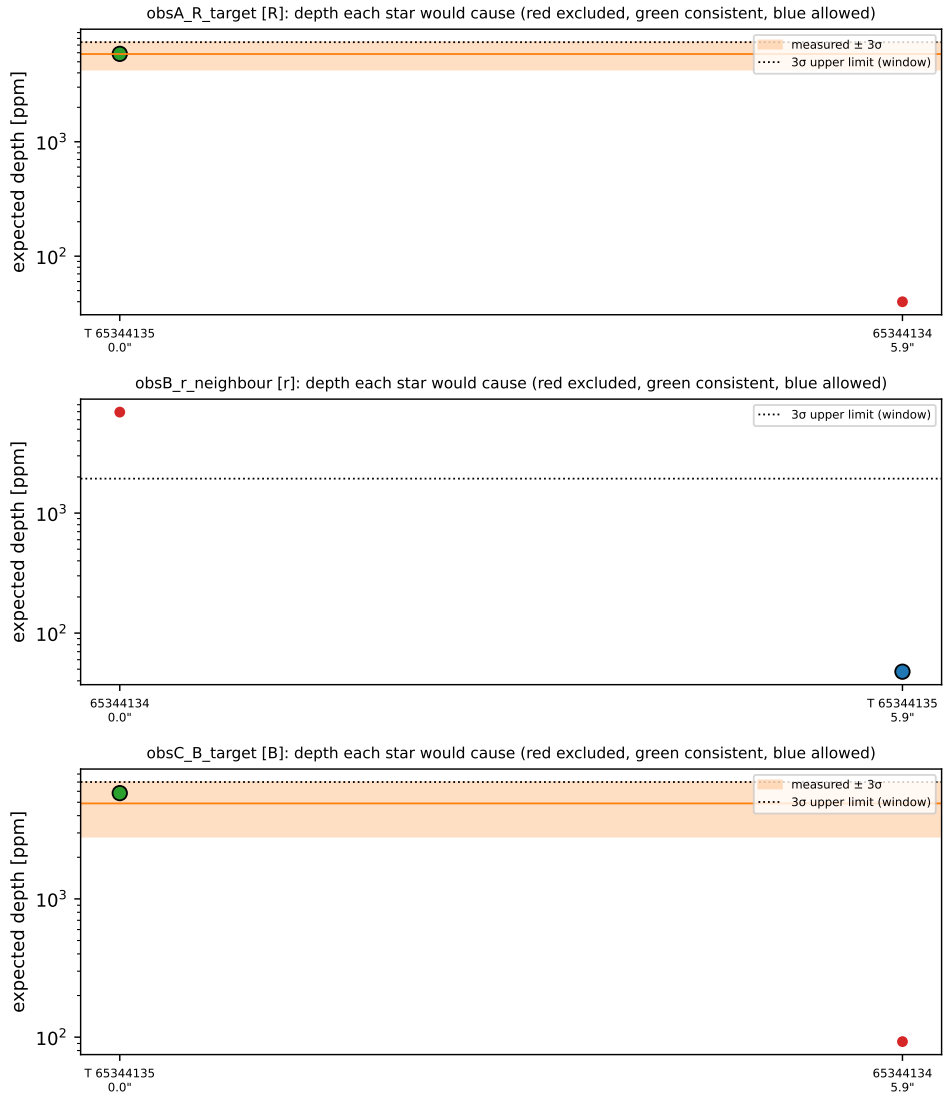



Figure 51.2: `ground_clearance_<TIC>.pdf`. For each light curve, the depth each star would produce in that aperture if it hosted the signal (red excluded, green consistent, blue allowed; the target has a black edge). The orange band is the measured depth $\pm 3\sigma$; the dotted line is the window upper limit. Top: the target aperture sees the expected dip. Middle: the neighbour's aperture shows no dip, so the neighbour is excluded.

# Chapter 52

# TRICERATOPS and TRICERATOPS+

## 52.1 With the standard package

With triceratops 1.1, the ground data reach the validation only through the star clearance, that is, through `remove_star`. Removing a neighbour eliminates its NTP, NEB and NEBx2P scenarios, and this is usually what brings NFPP down.

## 52.2 TRICERATOPS+

TRICERATOPS+ [1] adds the ground light curves to the likelihood of each scenario, with band-specific limb darkening and flux ratios for the companion and the eclipsing binary. Install it from GitHub:

```
git clone https://github.com/JGB276/TRICERATOPS-plus.git
cd TRICERATOPS-plus
pip install .
```

Install it in a separate environment. The package is also called `triceratops` (version 1.0.17), so it replaces triceratops 1.1.

The script detects TRICERATOPS+ from the `calc_probs` signature (`external_lc_files`). With `--tri-plus auto` (default) or `on`, it then does the following:

- It selects at most seven light curves (the limit in the TRICERATOPS+ code; its docstring still says four), one per band. TRICERATOPS+ knows $g\,r\,i\,z\,J\,H\,K$; Rc is passed as $r$ and Ic as $i$. TESS is not an external band: the TESS light curve is the main input of `calc_probs`, and TESS (or Vis) exists only as the filter of a contrast curve. B, V, CBB, clear and TESS-band ground light curves are not supported, and these light curves are used only for clearance and chromaticity. Per band it takes the most precise usable light curve, with target-centred apertures first.
- It writes `ground_plus_<TIC>_<band>_<n>.txt`: time from mid-time [d], flux divided by the

fitted baseline, and the scaled error. This is the three-column format that TRICERATOPS+ reads.

- It sets `target.stars["fluxratio_<band>"]` from the aperture model of that light curve.
- It passes `external_lc_files` and `filt_lcs` to every `calc_probs` call.

TRICERATOPS+ renormalises the external light curves only when more than one possible host remains. When the clearance leaves only the target, the script therefore divides out the residual dilution itself: $(f - (1 - \phi_t))/\phi_t$.

## 52.3 The dilution flag

TRICERATOPS+ is based on triceratops 1.0.17, which has no `dilution_corrected` argument. A light curve already put on the target's scale (v4.0 default, `--dilution-mode sector`) would then be corrected a second time. When the argument is missing, v4.1 switches to `--dilution-mode none` at the start of the run and says so. Give the depth on the aperture scale in that case, as the depth check in step 8 reminds you.

## 52.4 Other differences handled by the script

TRICERATOPS+ differs from triceratops 1.1 in two more places. The script (v4.1, 4 November 2026) handles both, so no changes are needed after installing the package:

- **TESScut cutouts.** TRICERATOPS+ requests its cutouts directly with astroquery's `Tesscut.get_cutouts` and reads the keywords `1CRV4P`/`2CRV4P`. Cutouts made by astrocut $\geq 1.4$ no longer contain these keywords (Chapter 37). During target construction the script serves these requests from its own TESScut cache, with the same pixel-frame correction as for triceratops 1.1.
- **Evidences.** TRICERATOPS+ keeps the evidences only in `target.probs["lnZ"]`, not in `target.lnZ`. It also normalises the probabilities with $e^{\ln Z}/\sum e^{\ln Z}$, which can underflow to NaN. The script copies the evidences to `target.lnZ`, which the convergence loop needs, and recomputes the probabilities with the stable softmax when the package result is not finite.

The TRICERATOPS+ example notebooks (`examples/` on GitHub) are in the script folder as PDF: `TRICERATOPS-plus_TOI4051_01_example.pdf`, `TRICERATOPS-plus_TOI4155_01_example.pdf`, and the general `example` and `kepler_example` notebooks.

# Chapter 53

# Example, options, tests and status

## 53.1 AAVSO campaign 942: TIC 65344135

The target has $V = 12.98$. The neighbour TIC 65344134, at $5.9''$, has $V = 13.17$. TESS sees a 3.2 ppt dip in the combined aperture. The neighbour is 0.84 times as bright as the target, so the on-star depth would be about 5.9 ppt if the target is the host, or about 7.0 ppt if the neighbour is.

The synthetic test reproduces this situation with three light curves:

- target, R band, $3.5''$ aperture, seeing $2.4''$: depth 5.8 ppt;
- neighbour, $r'$ band: no dip;
- target, B band.

The log then shows:

```
obsA_R_target R 65344135 0.0 0.994 5866 5844 525 7419 consistent
obsB_r_neighbour r 65344134 0.0 0.992 6943 317 405 1937 excluded
obsC_B_target B 65344135 0.0 0.987 5822 4904 695 7003 consistent
  2 star(s) cleared by the ground photometry (removed before calc_probs)
  intrinsic depth if TIC 65344135 is the host [ppm]: R: 5877 +/- 557,
      B: 4970 +/- 720; TESS 5900 ... p = 0.455 -> achromatic
```

In the opposite case, with no dip on the target and a 6.9 ppt dip on the neighbour, the target is reported as excluded and the neighbour as consistent, with the warning "probable nearby eclipsing binary" (Figure 53.1). Both cases are regression tests.

A run for the campaign, once the light curves have arrived:

```
python triceratops_validation.py --tic 65344135 --ephemeris 36.88 4352.3681 5900 5.33 --ephem-scan
    -hours 16 --ld-fit --ground-list ground_942.txt
```

The ephemeris gives the period [d], T0 [BTJD] (BJD 2461352.3681 is BTJD 4352.3681), the depth [ppm] and T14 [h].

- **Depth.** The value is on the target scale, which matches the default per-sector dilution

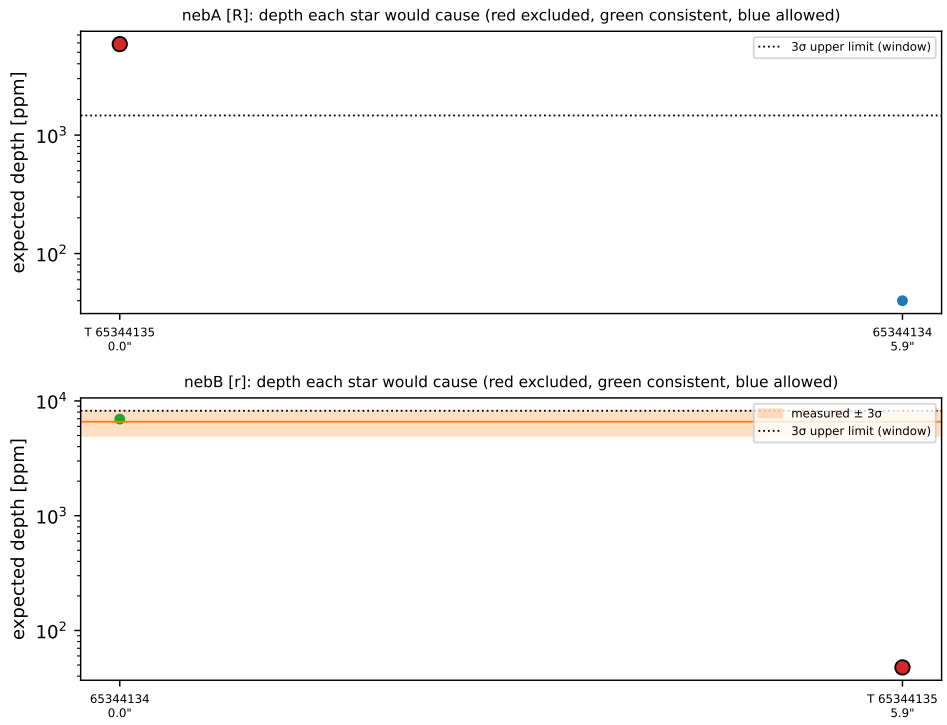



Figure 53.1: Clearance when the signal is on the neighbour: no dip in the target aperture (target excluded), the expected 6.9 ppt dip in the neighbour's aperture (consistent).

correction; `--ld-fit` re-measures it. With `--dilution-mode none` (or TRICERATOPS+), use the aperture depth of about 3200 ppm.

- **Scan window.** The period is known only to 0.01 d and is extrapolated over about 79 cycles back to 2018, so the ephemeris scan needs a window of 16 h.

## 53.2 New command-line options

| Option | Meaning |
|---|---|
| `--ground-lc SPEC` | One ground light curve (repeatable); syntax in Chapter 50. |
| `--ground-list FILE` | File with one specification per line. |
| `--ground-no-clear` | Report the clearance, do not remove stars. |
| `--ground-tc-window MIN` | Minimum half-width of the mid-time search (60). |
| `--ground-min-snr X` | Detection threshold (5). |
| `--ground-depth-tol X` | Fractional tolerance on the expected depth (0.25). |
| `--ground-fwhm ARCSEC` | Seeing if a light curve gives none (3.0). |
| `--tri-plus auto\|on\|off` | Use TRICERATOPS+ when installed (auto), require it (on), or never (off). |

| Option | Meaning |
|---|---|
| `--exclude-stars TIC ...` | Remove stars cleared by other follow-up, such as high-resolution imaging or spectroscopy. |

## 53.3 Tests

`tests/test_ground_photometry.py` adds 13 tests on synthetic data:

- specification and band parsing;
- the AstroImageJ, AAVSO/EXOTIC and magnitude readers;
- JD_UTC to BJD_TDB conversion;
- recovery of depth and mid-time with an airmass trend;
- aperture fractions against a numerical integral;
- clearance with the signal on the target, on the neighbour, and with no coverage;
- chromaticity with achromatic and chromatic depths;
- the TRICERATOPS+ inputs, with a mock target whose `calc_probs` has the TRICERATOPS+ signature;
- ground mid-times in the ETV analysis;
- a TRICERATOPS+-style result (evidences only in `probs`, NaN probabilities) and TESScut requests through `Tesscut.get_cutouts`.

`python -m pytest -q tests` runs all 28 tests in about 12 s.

## 53.4 Verification status

- All 28 regression tests pass (Python 3.13, numpy 2). The script still needs only Python $\geq$ 3.8.
- **File formats.** The readers follow the published structure of AstroImageJ tables and AAVSO/EXOTIC reports. They have been tested on files written in that structure, not yet on real campaign files. Please send the first real light curves along, so that the readers can be checked against them.
- **TRICERATOPS+.** The input files and arguments were checked against the TRICERATOPS+ source (`calc_probs`, `renorm_flux`, `fluxratio_<band>`) and with a mock target. A real TRICERATOPS+ `calc_probs` run, which needs network access, has not been done here.
- **Limitations.**

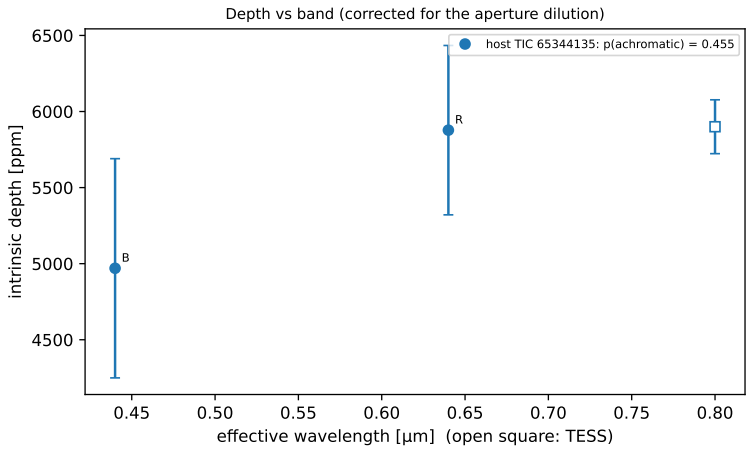



Figure 53.2: `ground_chromaticity_<TIC>.pdf`: intrinsic depth per band (filled) and TESS (open square) for each candidate host.

– The seeing PSF is a Gaussian. Wings and a donut-shaped PSF change the fractions of stars near the aperture edge, which is why the tolerance is 25%.

– Comparison-star variability and sky-annulus contamination are not modelled.

– The clearance uses the TESS shape; a strongly chromatic grazing event can change shape as well as depth.

# Bibliography

[1] J. Gómez Barrientos et al., "Validation of TESS Planet Candidates with Multicolor Transit Photometry and TRICERATOPS+", AJ 170, 148 (2025); arXiv:2508.02782; `https://github.com/JGB276/TRICERATOPS-plus`.

[2] M. Riello et al., "Gaia Early Data Release 3: Photometric content and validation", A&A 649, A3 (2021).

Contents of the fifth addendum,

# version 4.2 (Chapters 54–56)

54 Addendum 5: version 4.2, Kepler mode
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*Pipeline: triceratops_validation.py **v4.2** · GUI: triceratops_gui.py **v1.2***
*Siegfried Vanaverbeke · siegfriedvanaverbeke@gmail.com · https://github.com/svaverbe/triceratops-gui*
*November 2026*

Chapter 54

# Addendum 5: version 4.2, Kepler mode

Version 4.2 extends `triceratops_validation.py` so the same analysis path can validate long-period **Kepler** candidates. The TESS path is unchanged when `--mission` is left at its default `tess`. This addendum documents the mission switch, the KIC→TIC resolution, the light-curve and time-system differences, the new CLI options, and the GUI/output consequences. It continues Addendum 3 (v4.0) and Addendum 4 (v4.1).

## 54.1 What changed

- **Mission flag.** `--mission tess|kepler` sets global `MISSION`, `TIME_ZERO` and `TIME_LABEL` (BTJD vs BKJD).
- **KIC input.** `--kic N` is required in Kepler mode. If `--tic` is omitted, the TIC is resolved from the KIC (Catalogue query, disposition flags, 4″ cone search).
- **Photometry.** Kepler long-cadence SAP/PDCSAP and target-pixel files *per quarter* replace TESScut sectors. Default `--lc-source` becomes `pdcsap` (no FFI cutouts).
- **Stellar parameters.** `--star-source tic|kepler` and `--stellar R M Teff [logg]`; optional `--host-tic` for a different TRICERATOPS host.
- **TRICERATOPS.** Called with `mission="Kepler"` (4″ pixels) and Kepler-band limb darkening.
- **Outputs.** Default folder `KIC<kic>/`.
- **GUI.** `triceratops_gui.py` v1.2 exposes the new parser fields on the Target & ephemeris tab (live `build_parser()`).

## 54.2 Design goals and non-goals

**Goals.** One codebase for TESS and Kepler; same diagnostic sequence (ephemeris check, per-transit depths, secondary search, ETV, difference imaging, ground clearance, pooled FPP); same observer-facing GUI; BKJD treated consistently in logs, plots and the transit-time tool.

**Non-goals.** Kepler short-cadence products; custom Kepler FFI cutouts; re-deriving the Giacalone et al. (2021) priors for the Kepler pixel scale beyond what the installed TRICERATOPS package already implements for `mission="Kepler"`.

## 54.3 Where it sits in the pipeline

After argument parsing, `set_mission(args.mission)` runs before any time arithmetic or catalogue lookup. Kepler-specific branches appear in: TIC resolution; light-curve search string (`KIC n` vs `TIC n`); default `lc_source`; limb-darkening coefficients; TRICERATOPS target construction; default `outdir`. Ground photometry (v4.1) and the robust FPP loop (v4.0) are mission-agnostic once the light curve and ephemeris are in hand.

Chapter 55

# Kepler data path

## 55.1 Mission switch and time system

Globals set by `set_mission(mission)`:

| Quantity | TESS | Kepler |
|---|---|---|
| MISSION | tess | kepler |
| TIME_ZERO | 2457000.0 | 2454833.0 |
| TIME_LABEL | BTJD | BKJD |
| Time in logs / ephemeris | BJD_TDB – 2457000 | BJD_TDB – 2454833 |

A full BJD_TDB given to `--ephemeris` (T0 > $2.4\times10^6$) is converted with the active `TIME_ZERO`. The GUI transit-time tool uses the same zero point when Mission = kepler.

## 55.2 Resolving TIC from KIC

`resolve_tic_from_kic(kic)`:

- Query TIC by KIC column; prefer rows not flagged ARTIFACT / DUPLICATE / SPLIT.
- If DUPLICATE, follow `duplicate_id` when that replacement is unflagged.
- Otherwise cone-search 4″ around the KIC position (Vizier fallback if needed) and pick the best unflagged star within 1″, else the nearest unflagged with a warning.
- All candidates are printed so the user can override with `--tic`.

*Old 2MASS-based TIC entries of close pairs are often SPLIT into Gaia-based stars; the logic above is intended to land on the star that is the KIC target rather than a companion.*

## 55.3 Light curves and pixel files

Quarters replace sectors. lightkurve searches use `KIC n`. When `--mission kepler` and `--lc-source tesscut` were requested, the script forces `pdcsap` (no Kepler FFI cutout path in this version). SAP remains available when the aperture-scale flux is required (notably with `--host-tic`, where PDCSAP crowding correction applies to the catalogue target, not the chosen host).

## 55.4 Stellar parameters and host star

`--star-source kepler` prefers Berger et al. (2020) Gaia-Kepler parameters, else Kepler DR25. `--stellar R M Teff [logg]` overrides both. `--host-tic` sets the star TRICERATOPS treats as host while the light curve and pixels stay those of `--tic` / `--kic`; dilution mode is forced to `none` and PDCSAP is rejected in favour of SAP.

## 55.5 TRICERATOPS call

The target is built with the Kepler mission flag expected by the installed package (pixel scale 4″). Limb darkening uses Kepler-band coefficients. Ground-light-curve clearance (v4.1) and TRICERATOPS+ multi-colour input remain available and are independent of mission once the folded light curve exists.

Chapter 56

# CLI, GUI, outputs and tests

## 56.1 New command-line options

| Option | Default | Role |
|---|---|---|
| --mission tess\|kepler | tess | Selects time system, LC products, TRICERATOPS mission. |
| --kic N | — | Required for kepler; TIC resolved if --tic omitted. |
| --host-tic TIC | — | TRICERATOPS host ≠ light-curve target. |
| --star-source tic\|kepler | tic | Catalogue for R, M, Teff, logg. |
| --stellar R M Teff [logg] | — | Manual host parameters (3 or 4 floats). |

Mutual exclusions enforced by the parser: `--kic` only with `--mission kepler`; `--host-tic` incompatible with `--dilution-corrected`; `--star-source kepler` needs `--kic`.

```
python triceratops_validation.py --mission kepler --kic 3755522 \
--ephemeris 10.5 245.3 1200 3.2 --ld-fit --outdir KIC3755522
```

## 56.2 GUI

`triceratops_gui.py` compiles only `Config` and `build_parser` from the validation script (AST), so the new options appear on Target & ephemeris without a GUI release cycle beyond shipping v1.2 labels. Predicted transit times and default output-folder logic follow `mission`.

## 56.3 Output layout

Default directory: `KIC<kic>/` when mission is kepler and `--outdir` is unset; otherwise `TIC<tic>/`. File naming (`summary_*.json`, `triceratops_output_TIC*.txt`, plot stems) still uses the numeric TIC id once resolved, so downstream tools (including `triceratops_audit.py`) keep a stable key.

## 56.4 Compatibility and tests

TESS-only regressions from v4.0/v4.1 remain the baseline (`pytest tests/`). Kepler mode additionally depends on MAST availability for long-cadence products and on correct TIC resolution for crowded KIC entries—always inspect the printed candidate list or pass `--tic` explicitly. NumPy 2 / TRICERATOPS+ shims from earlier versions are unchanged.

*Recommended checks after modifying Kepler branches: (1) full pytest suite; (2) one known Kepler ephemeris run with `--ld-fit` on a quiet KIC target; (3) confirm BKJD in the log header and in `summary_*.json` time fields; (4) GUI load of a Kepler profile and Tools → Check interpreter.*

---

*End of Addendum 5. Together with Addenda 3 (v4.0) and 4 (v4.1) this describes `triceratops_validation.py` through version 4.2. Author: Siegfried Vanaverbeke · siegfriedvanaverbeke@gmail.com · https://github.com/svaverbe/triceratops-gui*